MIL-HDBK-17B
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16 June 1989

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DEPARTMENT OF DEFENSE
WASHINGTON, D.C. 20025

MIL-HDBK-17B: Volume I
Polymer Matrix Composites

1. Military Handbook 17 (Part 1) provides guidelines and material properties for polymer (organic) matrix composites materials. This handbook encompasses, but is not limited to, polymeric composites intended for aircraft and aerospace vehicles and military combat vehicle applications. Structural sandwich composites are covered in MIL-HDBK-23. Metal matrix composites (MMC), ceramic matrix composites (CMC), and carbon/carbon composites (C/C) will be covered in separate military handbooks as developments occur.

2. This standardization handbook has been developed and is being maintained as a joint effort of the Department of Defense and the Federal Aviation Administration.

3. The information contained in this handbook was obtained from materials producers, industry, reports on Government-sponsored research, the open literature, and by contact with research laboratories and those who participate in the MIL-HDBK-17 coordination activity.

4. All information and data contained in this handbook have been coordinated with industry and the U.S. Army, Navy, Air Force, NASA, and Federal Aviation Administration prior to publication.

5. Every effort has been made to reflect the latest information on polymeric composites. The handbook is continually reviewed and revised to insure its completeness and currentness. Users of this document are encouraged to report any errors discovered and recommendations for changes or additions to Department of the Army, U.S. Army Laboratory Command, Materials Technology Laboratory, ATTN: SLCMT-OM, MIL-HDBK-17 Coordinator, Arsenal St., Watertown, MA 02172-0001. Documentation for the Secretariat should be directed to Materials Sciences Corporation, MIL-HDBK-17 Secretariat, Gwynedd Plaza II, Spring House, PA 19477.

6. Copies of this document and revisions thereto may be obtained from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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1.1 Introduction. The standardization of a statistically-based mechanical property data base, procedures used, and overall material guidelines for characterization of composite material systems is recognized as being beneficial to both manufacturers and governmental agencies. It is also recognized that a complete characterization of the capabilities of any engineering material system is primarily dependent on the inherent material physical and chemical composition which precede, and are independent of, specific applications. Therefore, at the material system characterization level, the data and guidelines contained in this handbook are applicable to military and commercial products and provide the technical basis for establishing statistically valid design values acceptable to certificating or procuring agencies.

This handbook specifically provides statistically-based mechanical property data on current and emerging polymer matrix composite materials, provides guidelines for the analysis and presentation of data, and provides fabrication and characterization documentation to ensure repeatability of results or reliable detection of differences. The primary focus of MIL-HDBK-17 in the overall characterization/design procedure as commonly applied to composites is shown in Figure 1.1.

The data contained herein, or appearing in approved items in the minutes of MIL-HDBK-17 coordination meetings are acceptable to the Army, the Navy, the Air Force, and the Federal Aviation Administration. Approval by the certificating or procuring agency must be obtained for the use of data or guidelines not contained herein.

This standardization handbook has been developed and is maintained as a joint effort of the Department of Defense and the Federal Aviation Administration. It is oriented toward the standardization of methods used to develop and analyze mechanical property data on current and emerging composite materials.

MIL-HDBK-17 is divided into three volumes. The first volume is oriented toward guidelines for data development and analyses. Chapter 2 provides guidelines for the generation of material properties, including the qualification of alternate materials and pooling of data from different sources. Chapters 3 through 5 define acceptable procedures for the evaluation of composite constituents including reinforcement fibers, resins, and prepreg materials. Chapter 6 addresses acceptable procedures for the evaluation of lamina and laminate materials. Procedures for analyzing composite structural details, and in particular, bolted joints are presented in Chapter 7. Chapter 8 of Volume I addresses the analysis and presentation of composite material property data. Important references are cited at the end of each chapter and appendix.

The second volume of MIL-HDBK-17 provides a compilation of statistically-based material properties for current and emerging composite materials. The first two chapters of this volume address matrix material and reinforcement fiber constituent properties. The remaining chapters address composite system properties and they are defined according to the primary reinforcement fiber types. Properties for composite materials reinforced with glass fibers, aramid fibers, carbon fibers, and so on are covered in separate chapters.

Volume III provides guidelines for the application of the data which are presented in Volume II. Chapter 2 addresses typical composite materials and

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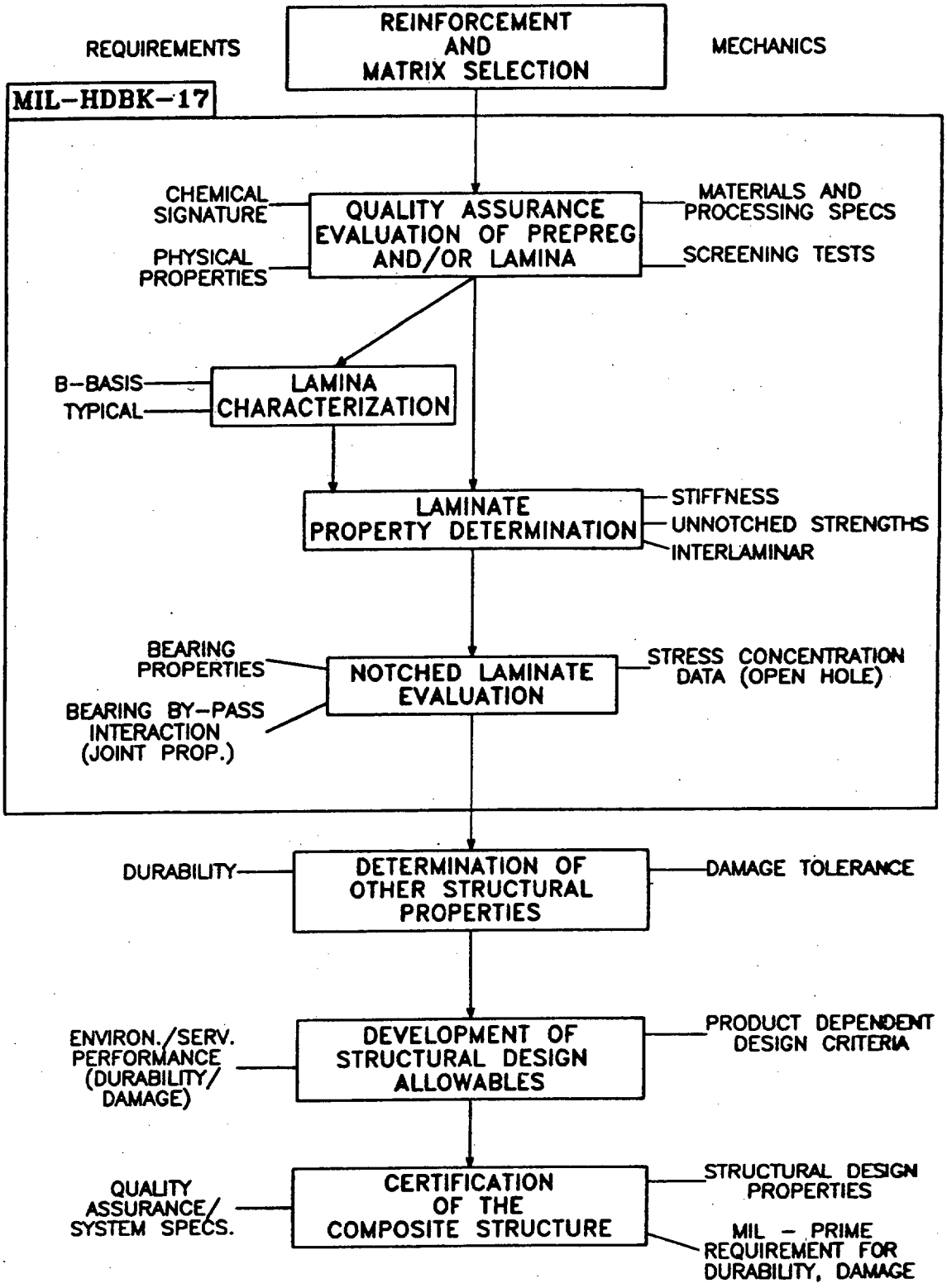


Figure 1.1 Focus of MIL-HDBK-17.

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processes. The matrix materials and reinforcement fibers of interest in MIL-HDBK-17 are reviewed, as are the typical product forms. Fabrication methods are also covered for information purposes. Chapter 3 reviews important issues related to quality control in the production of composite materials. Recommended manufacturing inspection procedures are reviewed, along with techniques for material property verification and statistical quality control. Chapter 4 addresses the design and analysis of composite material systems. It provides an overview of the current techniques and describes how the various constituent properties reported in MIL-HDBK-17 are used in the design and analysis of a composite system.

The remainder of Chapter 1 of this volume provides additional information of importance to any user of MIL-HDBK-17. The purpose and scope of the document are defined and some comments are provided on its use and limitations. The accepted system of symbols, abbreviations, and units are also reviewed, and a large collection of terms pertinent to this handbook are defined.

1.2 Purpose. The purpose of this handbook is to provide a standard source of statistically-based mechanical property data for current and emerging composite materials. In order to serve this purpose the handbook must provide specific guidelines on how the necessary data should be developed and analyzed. Documentation requirements on the fabrication and characterization of these composites must also be clearly defined.

Twice yearly MIL-HDBK-17 coordination meetings are held for the specific purpose of reviewing and approving new guidelines and data proposals. These meetings consist of representatives from the DOD, FAA, and industry. Materials which are approved and included in this handbook or the minutes of the MIL-HDBK-17 coordination meetings are acceptable to the Army, Navy, Air Force, and Federal Aviation Administration and considered effective on the date approved by MIL-HDBK-17 coordination committee. The use of data or guidelines that are not approved and therefore not included in this document must be approved by the appropriate certification or procurement agency.

1.3 Scope. MIL-HDBK-17 is published in three volumes, and serves as a source for the following:

Volume I - Provides guidelines for the characterization of composite material systems to be used in aerospace vehicles and structures. Composite material systems must normally be evaluated in accordance with these, or equivalent guidelines, in order to be considered acceptable by government certification and procuring agencies.

Volume II - Provides a compilation of statistically-based mechanical property data for current and emerging composite material systems used in the aerospace industry. B-basis strength and strain-to-failure values will be presented along with related data.

Volume III - Provides information regarding materials and fabrication procedures, quality control, and design and analysis.

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Specifically, Volume I provides guidelines for the physical, chemical, and mechanical characterization of composite materials. These guidelines address characterization of the fiber, matrix, and prepreg materials, which are the primary constituents in a typical composite system. The guidelines also address the characterization of the composite system, with particular emphasis on lamina properties, rather than laminate properties. Recommendations on the evaluation of bolted joints in composite materials are provided. Volume I also provides guidelines for the statistical analysis and presentation of data.

Volume II provides statistically-based mechanical property data for composite material systems used in the aerospace industry. Strength and strain-to-failure properties are reported either in terms of B-values or S-values (see Section 1.7). Stiffness properties are generally reported as typical values. The specific statistical significance of each of these quantities is defined in the first chapter of Volume II. Physical, chemical, and mechanical properties of the composite constituents - the fibers, matrix material and prepreg - are reported where applicable. Later chapters include data summaries for the various composite systems. Individual chapters focus on particular reinforcement fibers.

Volume III provides guidelines for the design and analysis of composite materials. The chapters in Volume III address related topics of 1) Materials and Processes, 2) Quality Control of Production Composites, and 3) Design and Analysis. This information is included primarily for background information and are not offered for regulatory purposes.

Statistically-based strength properties are defined for each composite material system over a range of potential usage conditions. The intent is to provide data at the upper and lower limits of the potential environmental conditions for a particular material, so that applications issues do not govern the mechanical property characterizations. If data are also available at intermediate environmental conditions, they are used to more exactly define the relationship between the mechanical properties and the effect of the environment on those properties. The statistically-based strength data which are available are tabulated in Volume II. These data are useful as a starting point for establishing structural design allowables when stress and strength analysis capabilities permit lamina level margin of safety checks. Depending on the application, some structural design allowables will have to be determined empirically at the laminate and composite level, since MIL-HDBK-17 does not provide these data.

Additional information and properties are added as they become available and are demonstrated to meet the guideline criteria. Typical property values, as well as S-values (see definitions) are included if they meet the approval of the MIL-HDBK-17 Coordination Group.

When the guidelines or data requirements of MIL-HDBK-17 cannot be followed, the certifying or procuring government agency should be contacted to determine data requirements and other documentation which may be necessary to justify data values proposed or used by the manufacturer.

1.4 Use of the document and limitations. The information contained in MIL-HDBK-17 is obtained from materials producers and fabricators, the aerospace industry, reports on government-sponsored research, the open literature, and by contact with research laboratories and those who participate in the MIL-HDBK-17

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coordination activity. All of the information and data contained in this document have been coordinated with representatives from industry and the Army, Navy, Air Force, and Federal Aviation Administration prior to publication. Every effort has been made to reflect the latest information on composite materials and structural details for aerospace vehicles and structures. The handbook is continually reviewed and revised to insure its completeness and to keep it as current as possible.

All data included herein are based on test specimens only. Test specimens dimensions conform with those specified for the particular test method which is used. Standard test methods are recommended where possible (ASTM standards are the primary source). The designer and all other users must be responsible for any translation of the data contained herein to other coupon dimensions, temperature, humidity, and other environmental conditions not covered in this document. Problems such as scale up effects and the influence of the test method selected on properties are also not addressed in this document. The manner in which S-basis values are used is also up to the discretion of the designer. In general, decisions concerning which properties to use for a specific application or design are the responsibility of the designer and are outside the scope of this handbook.

The data which are tabulated in this handbook are intended as an aid in assigning property values to a material. In specific cases where it is necessary or preferable to develop strength properties superior to those in this handbook acceptance of such values must be obtained from the appropriate procurement or certification agency. The applicability and interpretation of specific provisions of this handbook must also be defined by the appropriate procurement, regulatory, or certification agency.

Reference information which is cited in this handbook may not comply in every respect with the guidelines or other criteria specified in this document. References are provided at the end of each chapter primarily as a source of additional information in a given subject area.

The use of tradenames and proprietary product names does not constitute an endorsement of those products by the Government.

1.5 Approval procedures. The MIL-HDBK-17 Coordination Group is a joint government-industry activity that meets twice yearly. At each meeting, this group acts upon proposed changes or additions to the document that are submitted by any of the working groups. The agenda is usually mailed to attendees four weeks prior to the meeting date, and the minutes four weeks following the meeting. Requests for consideration of material for inclusion in the handbook should be submitted to the appropriate working group and the Secretariat well in advance of the mailing date.

Attachments containing proposed changes or additions to the document shall include specific notation of the changes or additions to be made and adequate documentation of supporting data and analytical procedures. Reproducible copies of drawings or photographs intended for inclusion in the document shall be furnished to the Secretariat.

Requests for inclusion of data in MIL-HDBK-17 should be submitted to the Secretariat with the documentation specified in Section 8.1.4. Following analysis

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and review of the data by the Secretariat, the data will be presented at the next meeting of the Coordination Group.

The choice of new materials to be included herein is governed by the MIL-HDBK-17 Coordination Group. Practical considerations preclude inclusion of all advanced composite materials that may be of interest. Reasonable attempts will be made to add new materials of interest in a timely manner.

1.6 Symbols, abbreviations, and systems of units. This section defines the symbols and abbreviations which are used within MIL-HDBK-17 and describes the system of units which is maintained. Common usage is maintained where possible. References 1.6(a), 1.6(b), and 1.6(c) served as primary sources for this information.

1.6.1 Symbols and abbreviations. The symbols and abbreviations used in this document are defined in this section with the exception of statistical symbols. These latter symbols are defined in Chapter 8.

- The symbols f and m, when used as either subscripts or superscripts, always denote fiber and matrix, respectively.
- The type of stress (e.g., cy - compression yield) is always used in the superscript position.
- Direction indicators (e.g., x, y, z, L, T, etc.) are always used in the subscript position.
- Ordinal indicators of laminae sequence (e.g., 1, 2, 3, etc.) are used in the superscript position and must be parenthesized to distinguish them from mathematical exponents.
- Other indicators may be used in either subscript or superscript position, as appropriate for clarity.
- Compound symbols (i.e., basic symbol plus indicators) which deviate from these rules are shown in their specific form in the following list.

The following general symbols and abbreviations are considered standard for use in MIL-HDBK-17. Where exceptions are made, they are noted in the text and tables.

A - (1) area (m^2,in^2)
- (2) ratio of alternating stress to mean stress
- (3) A-basis for mechanical property values

ASTM - American Society for Testing and Materials

a - (1) length dimension (mm,in)
- (2) acceleration (m/sec^2,ft/sec^2)
- (3) amplitude

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- (4) crack or flaw dimension (mm,in)

B - (1) B-basis for mechanical property values
- (2) biaxial ratio

Btu - British thermal unit(s)

b - width dimension (mm,in), e.g. the width of a bearing or compression panel normal to load, or breadth of beam cross-section

C - (1) specific heat (kJ/kg °C,BTU/lb °F)
- (2) Celsius

CF - centrifugal force (N,lbf)

CG - (1) center of mass, "center of gravity"
- (2) area or volume centroid

$\mathbb{C}_L$ - centerline

c - column buckling end-fixity coefficient

$\bar{c}$ - honeycomb sandwich core depth (mm,in)

cpm - cycles per minute

D - (1) diameter (mm,in)
- (2) hole or fastener diameter (mm,in)
- (3) plate stiffness (N-m,lbf-in)

d - mathematical operator denoting differential

E - modulus of elasticity in tension, average ratio of stress to strain for stress below proportional limit (GPa,Msi)

E' - storage modulus (GPa,Msi)

E" - loss modulus (GPa,Msi)

E^c - modulus of elasticity in compression, average ratio of stress to strain for stress below proportional limit (GPa,Msi)

E'_c - modulus of elasticity of honeycomb core normal to sandwich plane (GPa,Msi)

E^{sec} - secant modulus (GPa,Msi)

E^{tan} - tangent modulus (GPa,Msi)

e - minimum distance from a hole center to the edge of the sheet (mm,in)

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e/D - ratio of edge distance to hole diameter (bearing strength)

F - (1) stress (MPa,ksi)

- (2) Fahrenheit

F^b - bending stress (MPa,ksi)

F^{ccr} - crushing or crippling stress (upper limit of column stress for failure) (MPa,ksi)

F^{su} - ultimate stress in pure shear (this value represents the average shear stress over the cross-section) (MPa,ksi)

f - (1) internal (or calculated) stress (MPa,ksi)

- (2) stress applied to the gross flawed section (MPa,ksi)

- (3) creep stress (MPa,ksi)

f^c - internal (or calculated) compressive stress (MPa,ksi)

f_c - (1) maximum stress at fracture (MPa,ksi)

- (2) gross stress limit (for screening elastic fracture data (MPa,ksi)

ft - foot, feet

G - modulus of rigidity (shear modulus) (GPa,Msi)

GPa - gigapascal(s)

g - (1) gram(s)

- (2) acceleration due to gravity (m/s^2,ft/s^2)

H/C - honeycomb (sandwich)

h - height dimension (mm,in) e.g. the height of a beam cross-section

hr - hour(s)

I - area moment of inertia (mm^4,in^4)

i - slope (due to bending) of neutral plane in a beam, in radians

in. - inch(es)

J - (1) torsion constant (= I_p for round tubes) (m^4,in^4)

- (2) Joule

K - (1) Kelvin

- (2) stress intensity factor ($MPa\sqrt{m}$,$ksi\sqrt{in}$)

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- (3) coefficient of thermal conductivity (W/m °C, BTU/ft^2/hr/in/°F)
- (4) correction factor
- (5) dielectric constant

K_{app} - apparent plane strain fracture toughness or residual strength (MPa$\sqrt{m}$,ksi$\sqrt{in}$)

K_c - critical plane strain fracture toughness, a measure of fracture toughness at point of crack growth instability (MPa$\sqrt{m}$,ksi$\sqrt{in}$)

K_{Ic} - plane strain fracture toughness (MPa$\sqrt{m}$,ksi$\sqrt{in}$)

K_N - empirically calculated fatigue notch factor

K_s - plate or cylinder shear buckling coefficient

K_t - (1) theoretical elastic stress concentration factor

- (2) t_w/c ratio in H/C sandwich

K_x, K_y - plate or cylinder compression buckling coefficient

k - strain at unit stress (m/m,in/in)

L - cylinder, beam, or column length (mm,in)

L' - effective column length (mm,in)

lb - pound

M - applied moment or couple (N-m,in-lbf)

Mg - megagram(s)

MPa - megapascal(s)

MS - military standard

M.S. - margin of safety

MW - molecular weight

MWD - molecular weight distribution

m - (1) mass (kg,lb)

- (2) number of half wave lengths
- (3) metre
- (4) slope

N - (1) number of fatigue cycles to failure

- (2) number of laminae in a laminate

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- (3) distributed in-plane forces on a panel (lbf/in)
- (4) Newton
- (5) normalized

NA - neutral axis

NAS - National Aerospace Standard

n - (1) number of times in a set
- (2) number of half or total wavelengths
- (3) number of fatigue cycles endured

P - (1) applied load (N,lbf)
- (2) exposure parameter
- (3) probability

P^u - test ultimate load, (N,lb per fastener)

P^y - test yield load, (N,lb per fastener)

p - normal pressure (Pa,psi)

psi - pounds per square inch

Q - area static moment of a cross-section (mm^3,in^3)

q - shear flow (N/m,lbf/in)

R - (1) algebraic ratio of minimum load to maximum load in cyclic loading
- (2) reduced ratio

RA - reduction of area

R.H. - relative humidity

RMS - root-mean-square

RT - room temperature

r - (1) radius (mm,in)
- (2) root radius (mm,in)
- (3) reduced ratio (regression analysis)

S - (1) shear force (N,lbf)
- (2) nominal stress in fatigue (MPa,ksi)
- (3) S-basis for mechanical property values

S_a - stress amplitude in fatigue (MPa,ksi)

S_e - fatigue limit (MPa,ksi)

S_m - mean stress in fatigue (MPa,ksi)

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S_{max} - highest algebraic value of stress in the stress cycle (MPa,ksi)

S_{min} - lowest algebraic value of stress in the stress cycle (MPa,ksi)

S_R - algebraic difference between the minimum and maximum stresses in one cycle (MPa,ksi)

S.F. - safety factor

s - (1) arc length (mm,in)
- (2) H/C sandwich cell size (mm,in)

T - (1) temperature (°C,°F)
- (2) applied torsional moment (N-m,in-lbf)

T_d - thermal decomposition temperature (°C,°F)

T_F - exposure temperature (°C,°F)

T_g - glass transition temperature (°C,°F)

T_m - melting temperature (°C,°F)

t - (1) thickness (mm,in)
- (2) exposure time (s)
- (3) elapsed time (s)

V - (1) volume (mm^3,in^3)
- (2) shear force (N,lbf)

W - (1) weight (N,lbf)
- (2) width (mm,in)
- (3) Watt

x - distance along a coordinate axis

Y - nondimensional factor relating component geometry and flaw size

y - (1) deflection (due to bending) of elastic curve of a beam (mm,in)
- (2) distance from neutral axis to given point
- (3) distance along a coordinate axis

Z - section modulus, I/y (mm^3,in^3)

α - coefficient of thermal expansion (m/m/°C,in/in/°F)

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γ - shear strain (m/m,in/in)

Δ - difference (used as prefix to quantitative symbols)

δ - elongation or deflection (mm,in)

ϵ - strain (m/m,in/in)

ϵ^e - elastic strain (m/m,in/in)

ϵ^p - plastic strain (m/m,in/in)

η - plasticity reduction factor

$[\eta]$ - intrinsic viscosity

η^* - dynamic complex viscosity

ν - Poisson's ratio

ρ - radius of gyration (mm,in)

Σ - total, summation

σ - standard deviation

σ_{ij}, τ_{ij} - stress in j direction on surface whose outer normal is in i direction (i, j = 1, 2, 3 or x, y, z) (MPa,ksi)

T - applied shear stress (MPa,ksi)

ω - (1) density (kg/m^3,lb/in^3)
- (2) angular velocity (radians/s)

ω'_c - H/C sandwich core density (kg/m^3,lb/in^3)

∞ - infinity

1.6.1.1 Constituent properties. The following symbols apply specifically to the constituent properties of a typical composite material.

E^f - Young's modulus of filament material (MPa,ksi)

E^m - Young's modulus of matrix material (MPa,ksi)

E^g_L - Young's modulus of impregnated glass scrim cloth in the filament direction or in the warp direction of a fabric (MPa,ksi)

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E_T^g - Young's modulus of impregnated glass scrim cloth transverse to the filament direction or to the warp direction in a fabric (MPa,ksi)

G^f - shear modulus of filament material (MPa,ksi)

G^m - shear modulus of matrix (MPa,ksi)

G_{LT}^g - shear modulus of impregnated glass scrim cloth (MPa,ksi)

G'_{cx} - shear modulus of sandwich core along X axis (MPa,ksi)

G'_{cy} - shear modulus of sandwich core along Y axis (MPa,ksi)

ℓ - filament length (mm,in)

α^f - coefficient of thermal expansion for filament material (m/m/°C,in/in/°F)

α^m - coefficient of thermal expansion for matrix material (m/m/°C,in/in/°F)

α_L^g - coefficient of thermal expansion of impregnated glass scrim cloth in the filament direction or in the warp direction of a fabric (m/m/°C,in/in/°F)

α_T^g - coefficient of thermal expansion of impregnated glass scrim cloth transverse to the filament direction or to the warp direction in a fabric (m/m/°C,in/in/°F)

ν^f - Poisson's ratio of filament material

ν^m - Poisson's ratio of matrix material

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ν^{g}_{LT} - glass scrim cloth Poisson's ratio relating to contraction in the transverse (or fill) direction as a result of extension in the longitudinal (or warp) direction

ν^{g}_{TL} - glass scrim cloth Poisson's ratio relating to contraction in the longitudinal (or warp) direction as a result of extension in the transverse (or fill) direction

$\bar{\sigma}$ - applied axial stress at a point, as used in micromechanics analysis (MPa,ksi)

$\bar{\tau}$ - applied shear stress at a point, as used in micromechanics analysis (MPa,ksi)

1.6.1.2 Laminae and laminates. The following symbols, abbreviations, and notations apply to composite laminae and laminates. At the present time the focus in MIL-HDBK-17 is on laminae properties. However, commonly used nomenclature for both laminae and laminates are included here to avoid potential confusion.

A_{ij} (i,j = 1,2,6) - extensional rigidities (N/m,lbf/in)

B_{ij} (i,j = 1,2,6) - coupling matrix (N,lbf)

C_{ij} (i,j = 1,2,6) - elements of stiffness matrix (Pa,psi)

D_x, D_y - flexural rigidities (N-m,lbf-in)

D_{xy} - twisting rigidity (N-m,lbf-in)

D_{ij} (i,j = 1,2,6) - flexural rigidities (N-m,lbf-in)

E_L, E_α - Young's modulus of lamina parallel to filament or warp direction (GPa,Msi)

E_T, E_β - Young's modulus of lamina transverse to filament or warp direction (GPa,Msi)

E_x - Young's modulus of laminate along X reference axis (GPa,Msi)

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E_y - Young's modulus of laminate along Y reference axis (GPa,Msi)

G_{LT}, $G_{\alpha\beta}$ - shear modulus of lamina in LT or $\alpha\beta$ plane (GPa,Msi)

G_{xy} - shear modulus of laminate in XY reference plane (GPa,Msi)

h_i - thickness of ith ply or lamina (mm,in)

M_x, M_y, M_{xy} - bending and twisting moment components (N-m/m, in-lbf/in in plate and shell analysis)

n_f - number of filaments per unit length per lamina

Q_x, Q_y - shear force parallel to Z axis of sections of a plate perpendicular to X and Y axes, respectively (N/m,lbf/in)

Q_{ij} (i,j = 1,2,6) - reduced stiffness matrix (Pa,psi)

u, v, w - components of the displacement vector (mm,in)

u_o, v_o, w_o - components of the displacement vector at the laminate's midsurface (mm,in)

V_v - void content (% by volume)

V_f - filament content or fiber volume (% by volume)

V_g - glass scrim cloth content (% by volume)

V_m - matrix content (% by volume)

V_x, V_y - edge or support shear force (N/m,lbf/in)

W_f - filament content (% by weight)

W_g - glass scrim cloth content (% by weight)

W_m - matrix content (% by weight)

W_s - weight of laminate per unit surface area (N/m^2,lbf/in^2)

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α_L, α_α - lamina coefficient of thermal expansion along L or α axis (m/m/°C, in/in/°F)

α_T, α_β - lamina coefficient of thermal expansion along T or β axis (m/m/°C, in/in/°F)

α_x - laminate coefficient of thermal expansion along general reference X axis (m/m/°C, in/in/°F)

α_y - laminate coefficient of thermal expansion along general reference Y axis (m/m/°C, in/in/°F)

α_{xy} - laminate shear distortion coefficient of thermal expansion (m/m/°C, in/in/°F)

θ - angular orientation of a lamina in a laminate, i.e., angle between L and X axes (°)

λ_{xy} - product of ν_{xy} and ν_{yx}

ν_{LT}, $\nu_{\alpha\beta}$ - Poisson's ratio relating contraction in the T or β direction as a result of extension in the L or α direction.

ν_{TL}, $\nu_{\beta\alpha}$ - Poisson's ratio relating contraction in the L or α direction as a result of extension in the T or β direction.

ν_{xy} - Poisson's ratio relating contraction in the y direction as a result of extension in the x direction

ν_{yx} - Poisson's ratio relating contraction in the x direction as a result of extension in the y direction

ϕ - (1) general angular coordinate, (°)
- (2) angle between X and load axes in off-axis loading (°)

ω_c - density of a single lamina (kg/m^3, lb/in^3)

$\bar{\omega}_c$ - density of a laminate (kg/m^3, lb/in^3)

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1.6.1.3 Subscripts. The following subscript notations are considered standard in MIL-HDBK-17.

A	- axial
a	- (1) adhesive
	- (2) alternating
app	- apparent
c	- (1) composite system, specific filament/matrix composition. Composite as a whole, contrasted to individual constituents. Also, sandwich core when used in conjunction with prime (')
	- (2) critical
cf	- centrifugal force
e	- fatigue or endurance
eff	- effective
eq	- equivalent
f	- filament
g	- glass scrim cloth
H	- hoop
i	- i^{th} position in a sequence
L	- lateral
L, T, z	- laminae natural orthogonal coordinates (L is filament or warp direction)
m	- (1) matrix
	- (2) mean
max	- maximum
min	- minimum
n	- (1) n^{th} (last) position in a sequence
	- (2) normal
p	- polar
s	- symmetric
st	- stiffener
T	- transverse
t	- value of parameter at time t
x, y, z	- general coordinate system

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α, β, z	- laminae natural orthogonal coordinates (α is filament or warp direction)
Σ	- total, or summation
o	- initial or reference datum
$( )_T$	- format for indicating specific, temperature associated with term in parentheses. RT - room temperature (21°C,70°F); all other temperatures in °F unless specified.

1.6.1.4 Superscripts. The following superscript notations are considered standard in MIL-HDBK-17.

b	- bending
br	- bearing
c	- (1) compression
	- (2) creep
ccr	- compression buckling, crushing, or crippling
e	- elastic
f	- filament
g	- glass scrim cloth
is	- interlaminar shear
(i)	- i^{th} ply or lamina
lim	- limit, used to indicate limit loading
m	- matrix
p	- plastic
pl	- proportional limit
rup	- rupture
s	- shear
scr	- shear buckling
sec	- secant (modulus)
T	- temperature or thermal
t	- tension
tan	- tangent (modulus)
u	- ultimate
y	- yield
'	- secondary (modulus), or denotes properties of H/C core when used with subscript c

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1.6.2 System of units. To comply with Department of Defense Directive 4120.18, "Metric System of Measurement," dated January 28, 1980, the data in MIL-HDBK-17 are generally presented in both the International System of Units (SI units) and the U. S. Customary (English) system of units. ASTM E-380, Standard for Metric Practice, provides guidance for the application for SI units which are intended as a basis for worldwide standardization of measurement units. Further guidelines on the use of the SI system of units and conversion factors are contained in the following publications:

(1) DARCOM P 706-470, Engineering Design Handbook: Metric Conversion Guide, July 1976.

(2) NBS Special Publication 330, "The International System of Units (SI)," National Bureau of Standards, 1986 edition.

(3) NBS Letter Circular LC 1035, "Units and Systems of Weights and Measures, Their Origin, Development, and Present Status," National Bureau of Standards, November 1985.

(4) NASA Special Publication 7012, "The International System of Units Physical Constants and Conversion Factors" (1964).

English to SI conversion factors pertinent to MIL-HDBK-17 data are contained in Table 1.6.2.

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Table 1.6.2 English to SI conversion factors.

To convert from	to	Multiply by
Btu (thermochemical)/in^2-s	watt/$meter^2$ (W/m^2)	1.634 246 E+06
Btu-in/(s-ft^2-°F)	W/(m K)	5.192 204 E+02
degree Fahrenheit	degree Celsius (°C)	$T_C = (T_F - 32)/1.8$
degree Fahrenheit	kelvin (K)	$T_K = (T_F + 459.67)/1.8$
foot	meter (m)	3.048 000 E-01
ft^2	m^2	9.290 304 E-02
foot/second	meter/second (m/s)	3.048 000 E-01
ft/s^2	m/s^2	3.048 000 E-01
inch	meter (m)	2.540 000 E-02
$in.^2$	$meter^2$ (m^2)	6.451 600 E-04
$in.^3$	m^3	1.638 706 E-05
kilogram-force (kgf)	newton (N)	9.806 650 E+00
kgf/m^2	pascal (Pa)	9.806 650 E+00
kip (1000 lbf)	newton (N)	4.448 222 E+03
ksi (kip/in^2)	MPa	6.894 757 E+00
lbf-in	N-m	1.129 848 E-01
lbf-ft	N-m	1.355 818 E+00
lbf/in^2 (psi)	pascal (Pa)	6.894 757 E+03
lb/in^2	gm/m^2	2.926 397 E-01
lb/in^3	kg/m^3	2.767 990 E+04
Msi (10^6 psi)	GPa	6.894 757 E+00
pound-force (lbf)	newton (N)	4.488 222 E+00
pound-mass (lb avoirdupois)	kilogram (kg)	4.535 924 E-01
torr	pascal (Pa)	1.333 22 E+02

*The letter "E" following the conversion factor stands for exponent and the two digits after the letter "E" indicate the power of 10 by which the number is to be multiplied.

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1.7 Definitions. The following definitions are used within MIL-HDBK-17. This glossary of terms is not totally comprehensive but it does represent nearly all commonly used terms. Where exceptions are made, they are noted in the text and tables.

For ease of identification the definitions have been organized alphabetically.

A-Basis (or A-Value) -- The mechanical property value above which 99 percent of the population of values is expected to fall, with a confidence of 95 percent. Note - A-values are not currently presented in MIL-HDBK-17. This is a 95% lower confidence limit on the first percentile.

A-Stage -- An early stage in the reaction of thermosetting resins in which the material is still soluble in certain liquids and may be liquid or capable of becoming liquid upon heating. (Sometimes referred to as resol.)

Absorption -- A process in which one material (the absorbent) takes in or absorbs another (the absorbate).

Accelerator -- A material which, when mixed with a catalyzed resin, will speed up the chemical reaction between the catalyst and the resin.

Accuracy -- The degree of conformity of a measured or calculated value to some recognized standard or specified value. Accuracy involves the systematic error of an operation.

Addition Polymerization -- Polymerization in which monomers are linked together without the splitting off of water or other simple molecules.

Adhesion -- The state in which two surfaces are held together at an interface by forces or interlocking action or both.

Adhesive -- A substance capable of holding two materials together by surface attachment. In the handbook, the term is used specifically to designate structural adhesives, those which produce attachments capable of transmitting significant structural loads.

ADK -- Notation used for the k-sample Anderson-Darling statistic, which is used to test the hypothesis that k batches have the same distribution.

Aging -- The effect, on materials, of exposure to an environment for a period of time; the process of exposing materials to an environment for an interval of time.

Ambient -- The surrounding environmental conditions such as pressure or temperature.

Anelasticity -- A characteristic exhibited by certain materials in which strain is a function of both stress and time, such that, while no permanent deformations are involved, a finite time is required to establish equilibrium between stress and strain in both the loading and unloading directions.

Angleply -- Same as Crossply.

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Anisotropic -- Not isotropic; having mechanical and/or physical properties which vary with direction relative to natural reference axes inherent in the material.

Aramid -- A manufactured fiber in which the fiber-forming substance consisting of a long-chain synthetic aromatic polyamide in which at least 85% of the amide (-CONH-) linkages are attached directly to two aromatic rings.

Areal Weight of Fiber -- The weight of fiber per unit area of prepreg. This is often expressed as grams per square meter. See Table 1.6.2 for conversion factors.

Artificial Weathering -- Exposure to laboratory conditions which may be cyclic, involving changes in temperature, relative humidity, radiant energy and any other elements found in the atmosphere in various geographical areas.

Aspect Ratio -- In an essentially two-dimensional rectangular structure (e.g., a panel), the ratio of the long dimension to the short dimension. However, in compression loading, it is sometimes considered to be the ratio of the load direction dimension to the transverse dimension. Also, in fiber micro-mechanics, it is referred to as the ratio of length to diameter.

Autoclave -- A closed vessel for producing an environment of fluid pressure, with or without heat, to an enclosed object which is undergoing a chemical reaction or other operation.

Autoclave Molding -- A process similar to the pressure bag technique. The lay-up is covered by a pressure bag, and the entire assembly is placed in an autoclave capable of providing heat and pressure for curing the part. The pressure bag is normally vented to the outside.

B-Basis (or B-Value) -- The mechanical property value above which at least 90 percent of the population of values is expected to fall, with a confidence of 95 percent, specifically the same as A-Basis except on the tenth percentile. (See Section 8.5.1.1)

B-Stage -- An intermediate stage in the reaction of a thermosetting resin in which the material softens when heated and swells when in contact with certain liquids but does not entirely fuse or dissolve. Materials are usually precured to this stage to facilitate handling and processing prior to final cure. (Sometimes referred to as resitol.)

Bag Molding -- A method of molding or laminating which involves the application of fluid pressure to a flexible material which transmits the pressure to the material being molded or bonded. Fluid pressure usually is applied by means of air, steam, water or vacuum.

Balanced Laminate -- A composite laminate in which all laminae at angles other than 0 degrees and 90 degrees occur only in ± pairs (not necessarily adjacent).

Batch (or Lot) -- In general, a quantity of material formed during the same process and having identical characteristics throughout. As applied to the handbook, a batch of prepreg is defined as a quantity which is produced from a single

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batch of matrix material and fiber. The prepreg batch is produced at one time in the same equipment under identical conditions.

Bearing Area -- The product of the pin diameter and the specimen thickness.

Bearing Load -- A compressive load on an interface.

Bearing Yield Strength -- The bearing stress at which a material exhibits a specified limiting deviation from the proportionality of bearing stress to bearing strain.

Bend Test -- A test of ductility by bending or folding, usually with steadily applied forces. In some instances the test may involve blows to a specimen having a cross section that is essentially uniform over a length several times as great as the largest dimension of the cross section.

Binder -- A bonding resin used to hold strands together in a mat or preform during manufacture of a molded object.

Binomial Random Variable -- The number of successes in independent trials where the probability of success is the same for each trial.

Birefringence -- The difference between the two principal refractive indices (of a fiber) or the ratio between the retardation and thickness of a material at a given point.

Bleeder Cloth -- A nonstructural layer of material used in the manufacture of composite parts to allow the escape of excess gas and resin during cure. The bleeder cloth is removed after the curing process and is not part of the final composite.

Bond -- The adhesion of one surface to another, with or without the use of an adhesive as a bonding agent.

Broadgoods -- A term loosely applied to prepreg material greater than about 12 inches in width, usually furnished by suppliers in continuous rolls. The term is currently used to designate both collimated uniaxial tape and woven fabric prepregs.

Buckling (Composite) -- A mode of structural response characterized by an out-of-plane material deflection due to compressive action on the structural element involved. In advanced composites, buckling may take the form not only of conventional general instability and local instability but also a micro-instability of individual fibers.

Bundle -- A general term for a collection of essentially parallel filaments or fibers.

C-Stage -- The final stage of the curing reaction of a thermosetting resin in which the material has become practically infusable and insoluble. (Normally considered fully cured and sometimes referred to as resite.)

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Carbon Fibers -- Fibers produced by the pyrolysis of organic precursor fibers such as rayon, polyacrylonitrile (PAN), and pitch in an inert atmosphere. The term is often used interchangeably with "graphite"; however, carbon fibers and graphite fibers differ in the temperature at which the fibers are made and heat-treated, and the amount of carbon produced. Carbon fibers typically are carbonized at about 2400°F (1300°C)and assay at 93 to 95% carbon, while graphite fibers are graphitized at 3450 to 5450°F (1900 to 3000°C)and assay at more than 99% elemental carbon.

Caul Plates -- Smooth metal plates, free of surface defects, the same size and shape as a composite lay-up, used immediately in contact with the lay-up during the curing process to transmit normal pressure and to provide a smooth surface on the finished laminate.

Censoring -- Data is right (left) censored at M, if, whenever an observation is less than or equal to M (greater than or equal to M), the actual value of the observation is recorded. If the observation exceeds (is less than) M, the observation is recorded as M.

Chain-Growth Polymerization -- A chemical reaction in which polymer formation is initiated by a reactive species R* produced from some compound I termed an initiator. The reactive species may be a free radical, cation or anion. The reactive center, once produced, adds monomer units in a chain reaction and grows rapidly to a large size. High molecular weight polymer forms immediately with the molecular weight changing slightly, if at all, as the monomer concentration decreases steadily during the reaction.

Chromatogram -- A plot of detector response against peak volume of solution (eluate) emerging from the system for each of the constituents which have been separated.

Circuit -- One complete traverse of the fiber feed mechanism of a winding machine; one complete traverse of a winding band from one arbitrary point along the winding path to another point on a plane through the starting point and perpendicular to the axis.

Cocuring -- The act of curing a composite laminate and simultaneously bonding it to some other prepared surface during the same cure cycle (see Secondary Bonding).

Coefficient of Linear Thermal Expansion -- The change in length per unit length resulting from a one-degree rise in temperature.

Coefficient of Variation -- The ratio of the population (or sample) standard deviation to the population (or sample) mean.

Collimated -- Rendered parallel.

Compatible -- Descriptive term referring to different batches which may be treated as coming from the same population.

Composite Class -- As used in the handbook, a major subdivision of composite construction in which the class is defined by the fiber system and the matrix class, e.g., organic-matrix filamentary laminate.

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Composite Material -- Composites are considered to be combinations of materials differing in composition or form on a macroscale. The constituents retain their identities in the composite; that is, they do not dissolve or otherwise merge completely into each other although they act in concert. Normally, the components can be physically identified and exhibit an interface between one another.

Compound -- An intimate mixture of polymer or polymers with all the materials necessary for the finished product.

Condensation Polymerization -- This is a special type of step-growth polymerization characterized by the formation of water or other simple molecules during the stepwise addition of reactive groups.

Confidence Coefficient -- See confidence interval.

Confidence Interval -- A confidence interval is defined by a statement of one of the following forms:

$$(1)\ P\{a<\theta\} \geq 1-\alpha$$
$$(2)\ P\{\theta<b\} \geq 1-\alpha$$
$$(3)\ P\{a<\theta<b\} \geq 1-\alpha$$

where $1-\alpha$ is called the confidence coefficient. A statement of type (1) or (2) is called a one-sided confidence interval and a statement of type (3) is called a two-sided confidence interval. In (1) a is a lower confidence limit and in (2) b is an upper confidence limit. With probability at least $1-\alpha$, the confidence interval will contain the parameter θ.

Constituent -- In general, an element of a larger grouping. In advanced composites, the principal constituents are the fibers and the matrix.

Continuous Filament -- A yarn or strand in which the individual filaments are substantially the same length as the strand.

Coupling Agent -- Any chemical substance designed to react with both the reinforcement and matrix phases of a composite material to form or promote a stronger bond at the interface. Coupling agents are applied to the reinforcement phase from an aqueous or organic solution or from a gas phase, or added to the matrix as an integral blend.

Crazing -- Apparent fine cracks at or under the surface of an organic matrix.

Creep -- The time dependent part of strain resulting from an applied stress.

Creep, Rate Of -- The slope of the creep-time curve at a given time.

Critical Value(s) -- When testing a one-sided statistical hypothesis, a critical value is the value such that, if the test statistic is greater than (less than) the critical value, the hypothesis is rejected. When testing a two-sided statistical hypothesis, two critical values are determined. If the test statistic is either less than the smaller critical value or greater than the larger critical value, then the hypothesis is rejected. In both cases, the critical value chosen

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depends on the desired risk (often 0.05) of rejecting the hypothesis when it is true.

Crossply -- Any filamentary laminate which is not uniaxial. Same as Angleply. In some references, the term crossply is used to designate only those laminates in which the laminae are at right angles to one another, while the term angleply is used for all others. In the handbook, the two terms are used synonymously. The reservation of a separate terminology for only one of several basic orientations is unwarranted because a laminate orientation code is used.

Cumulative Distribution Function -- See Section 8.5.1.1.

Cure -- To change the properties of a thermosetting resin irreversibly by chemical reaction, i.e., condensation, ring closure, or addition. Cure may be accomplished by addition of curing (cross-linking) agents, with or without catalyst, and with or without heat. Cure may occur also by addition, such as occurs with anhydride cures for epoxy resin systems.

Cure Cycle -- The schedule of time periods at specified conditions to which a reacting thermosetting material is subjected in order to reach a specified property level.

Cure Stress -- A residual internal stress produced during the curing cycle of composite structures. Normally, these stresses originate when different components of a lay-up have different thermal coefficients of expansion.

Debond -- A deliberate separation of a bonded joint or interface, usually for repair or rework purposes. (see Disbond, Unbond).

Deformation -- The change in shape of a specimen caused by the application of a load or force.

Degradation -- A deleterious change in chemical structure, physical properties or appearance.

Delamination -- The separation of the layers of material in a laminate. This may be local or may cover a large area of the laminate. It may occur at any time in the cure or subsequent life of the laminate and may arise from a wide variety of causes.

Denier -- A direct numbering system for expressing linear density, equal to the mass in grams per 9000 meters of yarn, filament, fiber, or other textile strand.

Density -- The mass per unit volume.

Desorption -- A process in which an absorbed or adsorbed material is released from another material. Desorption is the reverse of absorption, adsorption, or both.

Deviation -- Variation from a specified dimension or requirement, usually defining the upper and lower limits.

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Dielectric Constant -- The ratio of the capacity of a condenser having a dielectric constant between the plates to that of the same condenser when the dielectric is replaced by a vacuum; a measure of the electrical charge stored per unit volume at unit potential.

Dielectric Strength -- The average potential per unit thickness at which failure of the dielectric material occurs.

Disbond -- An area within a bonded interface between two adherends in which an adhesion failure or separation has occurred. It may occur at any time during the life of the structure and may arise from a wide variety of causes. Also, colloquially, an area of separation between two laminae in the finished laminate (in this case the term "delamination" is normally preferred.) (See Debond, Unbond, Delamination.)

Distribution -- A formula which gives the probability that a value will fall within prescribed limits. (See normal, Weibull, and lognormal distributions, also Section 8.5.1.1).

Dry Fiber Area -- Area of fiber not totally encapsulated by resin.

Ductility -- The ability of a material to deform plastically before fracturing.

Elasticity -- The property of a material which allows it to recover its original size and shape immediately after removal of the force causing deformation.

Elongation -- The increase in gage length or extension of a specimen during a tension test, usually expressed as a percentage of the original gage length.

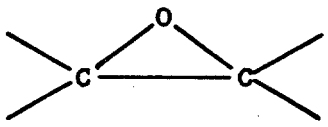
Eluate -- The liquid emerging from a column (in liquid chromatography).

Eluent -- The mobile phase used to sweep or elute the sample (solute) components into, through, and out of the column.

End -- A single fiber, strand, roving or yarn being or already incorporated into a product. An end may be an individual wrap yarn or cord in a woven fabric. In referring to aramid and glass fibers, an end is usually an untwisted bundle of continuous filaments.

Epoxy Equivalent Weight -- The number of grams of resin which contain one chemical equivalent of the epoxy group.

Epoxy Resin -- Resins which may be of widely different structures but are characterized by the presence of the epoxy group:



(The epoxy or epoxide group is usually present as a glycidyl ether, glycidyl amine, or as part of an aliphatic ring system. The aromatic type epoxy resins are normally used in composites.)
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Extensometer -- A device for measuring linear strain.

F-Distribution -- See Section 8.5.1.1.

Fabric, Nonwoven -- A textile structure produced by bonding or interlocking of fibers, or both, accomplished by mechanical, chemical, thermal, or solvent means, and combinations thereof.

Fabric, Woven -- A generic material construction consisting of interlaced yarns or fibers, usually a planar structure. Specifically, as used in this handbook, a cloth woven in an established weave pattern from advanced fiber yarns and used as the fibrous constituent in an advanced composite lamina. In a fabric lamina, the warp direction is considered the longitudinal direction, analogous to the filament direction in a filamentary lamina.

Failure Rate -- See Section 8.5.1.1.

Fiber -- A general term used to refer to filamentary materials. Often, fiber is used synonymously with filament. It is a general term for a filament of finite length.

Fiber Content -- The amount of fiber present in a composite. This is usually expressed as a percentage volume fraction or weight fraction of the composite.

Fiber Count -- The number of fibers per unit width of ply present in a specified section of a composite.

Fiber Direction -- The orientation or alignment of the longitudinal axis of the fiber with respect to a stated reference axis.

Fiber System -- The type and arrangement of fibrous material which comprises the fiber constituent of an advanced composite. Examples of fiber systems are collimated filaments or filament yarns, woven fabric, randomly oriented short-fiber ribbons, random fiber mats, whiskers, etc.

Filament -- The smallest unit of a fibrous material. The basic units formed during spinning and which are gathered into strands of fiber, (for use in composites). Filaments usually are of extreme length and of very small diameter. Filaments normally are not used individually. Some textile filaments can function as a yarn when they are of sufficient strength and flexibility.

Filamentary Composites -- A major form of advanced composites in which the fiber constituent consists of continuous filaments. Specifically, a filamentary composite is a laminate comprised of a number of laminae, each of which consists of a nonwoven, parallel, uniaxial, planar array of filaments (or filament yarns) embedded in the selected matrix material. Individual laminae are directionally oriented and combined into specific multiaxial laminates for application to specific envelopes of strength and stiffness requirements.

Filament Winding -- A reinforced-plastics process that employs a series of continuous, resin-impregnated fibers applied to a mandrel in a predetermined geometrical relationship under controlled tension.

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Filament Wound -- Pertaining to an object created by the filament winding method of fabrication.

Fill -- Yarn oriented at right angles to the warp in a woven fabric.

Filler -- A relatively inert substance added to a material to alter its physical, mechanical, thermal, electrical, and other properties or to lower cost. Sometimes the term is used specifically to mean particulate additives.

Finish (or Size System) -- A material, with which filaments are treated, which contains a coupling agent to improve the bond between the filament surface and the resin matrix in a composite material. In addition, finishes often contain ingredients which provide lubricity to the filament surface, preventing abrasive damage during handling, and a binder which promotes strand integrity and facilitates packing of the filaments.

Flash -- Excess material which forms at the parting line of a mold or die, or which is extruded from a closed mold.

Fracture Ductility -- The true plastic strain at fracture.

Gage Length -- the original length of that portion of the specimen over which strain or change of length is determined.

Gel -- The initial jelly-like solid phase that develops during formation of a resin from a liquid. Also, a semi-solid system consisting of a network of solid aggregates in which liquid is held.

Gel Coat -- A quick-setting resin used in molding processes to provide an improved surface for the composite; it is the first resin applied to the mold after the mold-release agent.

Gel Point -- The stage at which a liquid begins to exhibit pseudo-elastic properties. (This can be seen from the inflection point on a viscosity-time plot.)

Gel Time -- The period of time from a pre-determined starting point to the onset of gelation (gel point) as defined by a specific test method.

Glass -- An inorganic product of fusion which has cooled to a rigid condition without crystallizing. In the handbook, all reference to glass will be to the fibrous form as used in filaments, woven fabric, yarns, mats, chopped fibers, etc.

Glass Cloth -- Conventionally-woven glass fiber material (see Scrim).

Glass Fibers -- A fiber spun from an inorganic product of fusion which has cooled to a rigid condition without crystallizing.

Glass Transition -- The reversible change in an amorphous polymer or in amorphous regions of a partially crystalline polymer from (or to) a viscous or rubbery condition to (or from) a hard and relatively brittle one.

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Glass Transition Temperature -- The approximate midpoint of the temperature range over which the glass transition takes place.

Graphite Fibers -- See Carbon Fibers.

Greige -- Fabric that has received no finish.

Hand lay-up -- A process in which components are applied either to a mold or a working surface, and the successive plies are built up and worked by hand.

Hardness -- Resistance to deformation; usually measured by indention. Types of standard tests include Brinell, Rockwell, Knoop, and Vickers.

Hazard Rate -- See Section 8.5.1.1.

Heat Cleaned -- Glass or other fibers which have been exposed to elevated temperatures to remove preliminary sizings or binders which are not compatible with the resin system to be applied.

Heterogeneous -- Descriptive term for a material consisting of dissimilar constituents separately identifiable; a medium consisting of regions of unlike properties separated by internal boundaries. (Note that all nonhomogeneous materials are not necessarily heterogeneous).

Homogeneous -- Descriptive term for a material of uniform composition throughout; a medium which has no internal physical boundaries; a material whose properties are constant at every point, i.e., constant with respect to spatial coordinates (but not necessarily with respect to directional coordinates).

Horizontal Shear -- Sometimes used to indicate interlaminar shear. This is not an approved term for use in this handbook.

Humidity, Relative -- The ratio of the pressure of water vapor present to the pressure of saturated water vapor at the same temperature.

Hybrid -- A composite laminate comprised of laminae of two or more composite material systems. Or, a combination of two or more different fibers such as carbon and glass or carbon and aramid into a structure (tapes, fabrics and other forms may be combined).

Hygroscopic -- Capable of absorbing and retaining atmospheric moisture.

Hysteresis -- The energy absorbed in a complete cycle of loading and unloading.

Inclusion -- A physical and mechanical discontinuity occurring within a material or part, usually consisting of solid, encapsulated foreign material. Inclusions are often capable of transmitting some structural stresses and energy fields, but in a noticeably different manner from the parent material.

Integral Composite Structure -- Composite structure in which several structural elements, which would conventionally be assembled by bonding or with mechanical fasteners after separate fabrication, are instead laid up and cured as a single, complex, continuous structure; e.g., spars, ribs, and one stiffened cover of a

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wing box fabricated as a single integral part. The term is sometimes applied more loosely to any composite structure not assembled by mechanical fasteners.

Interface -- The boundary between the individual, physically distinguishable constituents of a composite.

Interlaminar -- Descriptive term pertaining to some object (e.g., voids), event (e.g., fracture), or potential field (e.g., shear stress) referenced as existing or occurring between two or more adjacent laminae.

Interlaminar Shear -- Shearing force tending to produce a relative displacement between two laminae in a laminate along the plane of their interface.

Intermediate Bearing Stress -- The bearing stress at the point on the bearing load-deformation curve where the tangent is equal to the bearing stress divided by a designated percentage (usually 4%) of the original hole diameter.

Intralaminar -- Descriptive term pertaining to some object (e.g., voids), event (e.g., fracture), or potential field (e.g., temperature gradient) existing entirely within a single lamina without reference to any adjacent laminae.

Isotropic -- Having uniform properties in all directions. The measured properties of an isotropic material are independent of the axis of testing.

Knuckle Area -- The area of transition between sections of different geometry in a filament wound part.

k-Sample Data -- A collection of data consisting of values observed when sampling from k batches.

Lamina -- A single ply or layer in a laminate made up of a series of layers.

Laminae -- Plural of lamina.

Laminate -- A product made by bonding together two or more layers or laminae of material or materials.

Laminate Orientation -- The configuration of a crossplied composite laminate with regard to the angles of crossplying, the number of laminae at each angle, and the exact sequence of the lamina lay-up.

Lattice Pattern -- A pattern of filament winding with a fixed arrangement of open voids.

Lay-up -- A process of fabrication involving the assembly of successive layers of resin-impregnated material.

Lognormal Distribution -- A probability distribution for which the probability that an observation selected at random from this population falls between a and b ($0 < a < b < \infty$) is given by the area under the normal distribution between $\ell og\ a$ and $\ell og\ b$. The common (base 10) or the natural (base e) logarithm may be used. (See Section 8.5.1.1.)

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Lower Confidence Bound -- See Confidence Interval.

Macro -- In relation to composites, denotes the gross properties of a composite as a structural element but does not consider the individual properties or identity of the constituents.

Macrostrain -- The mean strain over any finite gage length of measurement which is large in comparison to the material's interatomic distance.

Mandrel -- A form fixture or male mold used for the base in the production of a part by lay-up or filament winding.

Mat -- A fibrous material consisting of randomly oriented chopped or swirled filaments loosely held together with a binder.

Material System -- A specific composite material made from specifically identified constituents in specific geometric proportions and arrangements and possessed of numerically defined properties.

Material System Class -- As used in this handbook, a group consisting of material systems categorized by the same generic constituent materials, but without defining the constituents uniquely; e.g., the graphite/epoxy class.

Matrix -- The essentially homogeneous material in which the fiber system of a composite is embedded.

Mean -- See Sample Mean and Population Mean.

Mechanical Properties -- The properties of a material that are associated with elastic and inelastic reaction when force is applied, or the properties involving the relationship between stress and strain.

Median -- See Sample Median and Population Median.

Micro -- In relation to composites, denotes the properties of the constituents, i.e., matrix and reinforcement and interface only, as well as their effects on the composite properties.

Microstrain -- The strain over a gage length comparable to the material's interatomic distance.

Modulus, Initial -- The slope of the initial straight portion of a stress-strain or load-elongation curve.

Modulus, Secant -- The ratio of change in stress to change in strain between two points on a stress-strain curve, particularly the points of zero stress and stress at a particular strain.

Modulus, Tangent -- The ratio of change in stress to change in strain derived from the tangent to any point on a stress-strain curve.

Modulus, Young's -- The ratio of change in stress to change in strain below the elastic limit of a material. (Applicable to tension and compression).

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Modulus of Rigidity (also Shear Modulus or Torsional Modulus) -- The ratio of stress to strain below the proportional limit for shear or torsional stress.

Modulus of Rupture, in Bending -- The maximum tensile or compressive stress (whichever causes failure) value in the extreme fiber of a beam loaded to failure in bending. The value is computed from the flexure equation:

$$F^b = \frac{Mc}{I}. \qquad 1.7(a)$$

where M = maximum bending moment computed from the maximum load and the original moment arm,
c = initial distance from the neutral axis to the extreme fiber where failure occurs,
I = the initial moment of inertia of the cross section about its neutral axis.

Modulus of Rupture, in Torsion -- The maximum shear stress in the extreme fiber of a member of circular cross section loaded to failure in torsion calculated from the equation:

$$F^s = \frac{Tr}{J} \qquad 1.7(b)$$

where T = maximum twisting moment,
r = original outer radius,
J = polar moment of inertia of the original cross section.

Moisture Content -- The amount of moisture in a material determined under prescribed condition and expressed as a percentage of the mass of the moist specimen, i.e., the mass of the dry substance plus the moisture present.

Moisture Equilibrium -- The condition reached by a sample when it no longer takes up moisture from, or gives up moisture to, the surrounding environment.

Mold Release Agent -- A lubricant applied to mold surfaces to facilitate release of the molded article.

Molded Edge -- An edge which is not physically altered after molding for use in final form and particularly one which does not have fiber ends along its length.

Molding -- The forming of a polymer or composite into a solid mass of prescribed shape and size by the application of pressure and heat.

Monolayer -- The basic laminate unit from which crossplied or other laminates are constructed.

Monomer -- A compound consisting of molecules each of which can provide one or more constitutional units.

NDE -- Nondestructive evaluation. Broadly considered synonymous with NDI.

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NDI -- Nondestructive Inspection. A process or procedure for determining the quality or characteristics of a material, part, or assembly without permanently altering the subject or its properties.

NDT -- Nondestructive Testing. Broadly considered synonymous with NDI.

Necking -- A localized reduction in cross-sectional area which may occur in a material under tensile stress.

Negatively Skewed -- A distribution is said to be negatively skewed if the distribution is not symmetric and the longest tail is on the left.

Nominal Specimen Thickness -- The nominal ply thickness multiplied by the number of plies.

Nominal Value -- A value assigned for the purpose of a convenient designation. A nominal value exists in name only.

Normal Distribution -- A two parameter (μ,σ) family of probability distributions for which the probability that an observation will fall between a and b is given by the area under the curve

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \; e^{-(x-\mu)^2/2\sigma^2} \qquad 1.7(c)$$

between a and b. (See Section 8.5.1.1.)

Normalized Stress -- Stress calculated by multiplying the raw stress value by the ratio of measured fiber volume to the nominal fiber volume. This ratio is often approximated by the ratio of the measured specimen thickness to the nominal specimen thickness. Stresses for fiber-dominated failure modes are often normalized.

Observed Significance Level (OSL) -- The probability of observing a more extreme value of the test statistic when the null hypotheses is true.

Oligomer -- A polymer consisting of only a few monomer units such as a dimer, trimer, etc. or their mixtures.

One-Sided Tolerance Limit Factor -- See Tolerance Limit Factor.

Orthotropic -- Having three mutually perpendicular planes of elastic symmetry.

Oven Dry -- The condition of a material that has been heated under prescribed conditions of temperature and humidity until there is no further significant change in its mass.

PAN Fibers -- Polyacrylonitrile spun and stabilized fibers.

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Parallel Laminate -- A laminate of woven fabric in which the plies are aligned in the same position as originally aligned in the fabric roll.

Peel Ply -- A layer of resin free material used to protect a laminate for later secondary bonding.

pH -- A measure of acidity or alkalinity of a solution, with neutrality represented by a value of 7, with increasing acidity corresponding to progressively smaller values, and increasing alkalinity corresponding to progressively higher values.

Pick Count -- The number of filling yarns per inch of woven fabric.

Pitch Fibers -- Fibers derived from a special petroleum pitch.

Plastic -- A material that contains one or more organic polymers of large molecular weight, is solid in its finished state, and, at some state in its manufacture or processing into finished articles, can be shaped by flow.

Plasticizer -- A material of lower molecular weight added to a polymer to separate the molecular chains. This results in a depression of the glass-transition temperature, reduced stiffness and brittleness, and improved processability. (Note, many polymeric materials do not need a plasticizer.)

Poisson's Ratio -- The absolute value of the ratio of transverse strain to the corresponding axial strain resulting from uniformly distributed axial stress below the proportional limit of the material.

Polymer -- An organic material composed of molecules characterized by the repetition of one of more types of monomeric units.

Polymerization -- A chemical reaction in which the molecules of monomers are linked together to form polymers.

Population -- The set of measurements about which inferences are to be made or the totality of possible measurements which might be obtained in a given testing situation. For example, "all possible ultimate tensile strength measurements for Hexcel, conditioned at 95% relative humidity and room temperature". In order to make inferences about a population, it is often necessary to make assumptions about its distributional form. The assumed distributional form may also be referred to as the population. (See Section 8.5.1.1.)

Population Mean -- The average of all potential measurements in a given population weighted by their relative frequencies in the population. (See Section 8.5.1.1.)

Population Median -- That value in the population such that the probability of exceeding it is 0.5 and the probability of being less than it is 0.5. (See Section 8.5.1.1.)

Population Variance -- A measure of dispersion in the population.

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Porosity -- A condition of trapped pockets of air, gas, or vacuum within a solid material, usually expressed as a percentage of the total nonsolid volume to the total volume (solid plus nonsolid) of a unit quantity of material.

Positively Skewed -- A distribution is said to be positively skewed if the distribution is not symmetric and the longest tail is on the right.

Postcure -- Additional elevated temperature cure, usually without pressure, to improve final properties or complete the cure or both.

Pot Life -- The period of time during which a reacting thermosetting composition remains suitable for its intended processing after mixing with a reaction initiating agent.

Precision -- The degree of agreement within a set of observations or test results obtained. Precision involves repeatability and reproducibility.

Precursor (for Carbon or Graphite Fiber) -- Either the PAN or pitch fibers from which carbon and graphite fibers are derived.

Preply -- A composite material lamina in the raw material stage ready to be fabricated into a finished laminate. The lamina is usually combined with other raw laminae prior to fabrication. A preply includes all of the fiber system placed in position relative to all or part of the required matrix material that together will comprise the finished lamina. An organic matrix preply is called a prepreg.

Prepreg -- Ready to mold or cure material in sheet form which may be fiber, cloth, or mat impregnated with resin and stored for use. The resin is partially cured to a B-stage and supplied to the fabricator for lay-up and cure.

Pressure - The force or load per unit area.

Probability Density Function -- See Section 8.5.1.1.

Proportional Limit -- The maximum stress that a material is capable of sustaining without any deviation from the proportionality of stress to strain (also known as Hooke's law).

Quasi-Isotropic Laminate -- A laminate approximating isotropy by orientation of plies in several or more directions.

Reduction of Area -- The difference between the original cross sectional area of a tension test specimen and the area of its smallest cross section, usually expressed as a percentage of the original area.

Refractive Index - The ratio of the velocity of light (of specified wavelength) in air to its velocity in the substance under examination. Also defined as the sine of the angle of incidence divided by the sine of the angle of refraction as light passes from air into the substance.

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Reinforced Plastic -- A plastic with relatively high stiffness or very high strength fibers embedded in the composition. This improves some mechanical properties over that of the base resin.

Release Agent -- See Mold Release Agent.

Resilience -- A property of a material which is able to do work against restraining forces during return from a deformed condition.

Resin -- A solid or pseudo-solid organic material usually of high molecular weight which exhibits a tendency to flow when subjected to stress, usually has a softening or melting range, and fractures conchoidally.

Resin Content -- The amount of matrix present in a composite either by percent weight or percent volume.

Resin Starve Area -- Area of composite part where the resin has a non-continuous smooth coverage of the fiber.

Resin System -- A mixture of resin, with ingredients such as catalyst, initiator, diluents, etc. required for the intended processing and final product.

Roving -- A number of strands, tows, or ends collected into a parallel bundle with little or no twist.

S-Basis (or S-Value) -- The mechanical property value which is usually the specified minimum value of the appropriate government specification or SAE Aerospace Material Specification for this material.

Sample -- A small portion of a material or product intended to be representative of the whole. Statistically, a sample is the collection of measurements taken from a specified population. (See Section 8.5.1.1.)

Sample Mean -- The arithmetic average of the measurements in a sample. The sample mean is an estimator of the population mean. (See Section 8.5.1.1.)

Sample Median -- Order the observation from smallest to largest. Then the sample median is the value of the middle observation if the sample size is odd; the average of the two central observations if n is even. If the population is symmetric about its mean, the sample median is also an estimator of the population mean. (See Section 8.5.1.1.)

Sample Standard Deviation -- The square root of the sample variance. (See Section 8.5.1.1.)

Sample Variance -- The sum of the squared deviations from the sample mean, divided by n-1. (See Section 8.5.1.1.)

Sandwich Construction -- A structural panel concept consisting in its simplest form of two relatively thin, parallel sheets of structural material bonded to, and separated by, a relatively thick, light-weight core.

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Saturation -- An equilibrium condition in which the net rate of absorption under prescribed conditions falls essentially to zero.

Scrim (also called Glass Cloth, Carrier) -- A low cost fabric woven into an open mesh construction, used in the processing of tape or other B-stage material to facilitate handling.

Secondary Bonding -- The joining together, by the process of adhesive bonding, of two or more already-cured composite parts, during which the only chemical or thermal reaction occurring is the curing of the adhesive itself.

Selvedge -- The woven edge portion of a fabric parallel to the warp.

Set -- The strain remaining after complete release of the force producing the deformation.

Shear Fracture (for crystalline type materials) -- A mode of fracture resulting from translation along slip planes which are preferentially oriented in the direction of the shearing stress.

Shelf Life -- The length of time a material, substance, product, or reagent can be stored under specified environmental conditions and continue to meet all applicable specification requirements and/or remain suitable for its intended function.

Significant -- Statistically, the value of a test statistic is significant if the probability of a value at least as extreme is less than or equal to a predetermined number called the significance level of the test.

Significant Digit -- Any digit that is necessary to define a value or quantity.

Size System -- See Finish.

Skewness -- See Positively Skewed, Negatively Skewed.

Slenderness Ratio -- The unsupported effective length of a uniform column divided by the least radius of gyration of the cross-sectional area.

Sliver -- A continuous strand of loosely assembled fiber that is approximately uniform in cross-sectional area and has no twist.

Solute -- The dissolved material.

Specific Gravity -- The ratio of the weight of any volume of a substance to the weight of an equal volume of another substance taken as standard at a constant or stated temperature. Solids and liquids are usually compared with water at 4°C (39°F).

Specific Heat -- The quantity of heat required to raise the temperature of a unit mass of a substance one degree under specified conditions.

Specimen -- A piece or portion of a sample or other material taken to be tested. Specimens normally are prepared to conform with the applicable test method.

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Standard Deviation -- See Sample Standard Deviation.

Staple -- Either naturally occurring fibers or lengths cut from filaments.

Step-Growth Polymerization -- A chemical reaction in which polymers are formed by the stepwise intermolecular addition of molecules through reactive groups. Any two molecular species present can react. Monomers disappear early in the reaction and polymer molecular weight rises steadily throughout the reaction.

Strain -- the per unit change, due to force, in the size or shape of a body referred to its original size or shape. Strain is a nondimensional quantity, but it is frequently expressed in inches per inch, metres per metre, or percent.

Strand -- Normally an untwisted bundle or assembly of continuous filaments used as a unit, including slivers, tow, ends, yarn, etc. Sometimes a single fiber or filament is called a strand.

Strength -- the maximum stress which a material is capable of sustaining.

Stress -- The intensity at a point in a body of the forces or components of forces that act on a given plane through the point. Stress is expressed in force per unit area (pounds-force per square inch, megapascals, etc.).

Stress Relaxation -- The time dependent decrease in stress in a solid under given constraint conditions.

Stress-Strain Curve (Diagram) -- A graphical representation showing the relationship between the change in dimension of the specimen in the direction of the externally applied stress and the magnitude of the applied stress. Values of stress usually are plotted as ordinates (vertically) and strain values as abscissa (horizontally).

Surfacing Mat -- A thin mat of fine fibers used primarily to produce a smooth surface on an organic matrix composite.

Symmetrical Laminate -- A composite laminate in which the sequence of plies below the laminate midplane is a mirror image of the stacking sequence above the midplane.

Tack -- Stickiness of the prepreg.

Tape -- Prepreg fabricated in widths up to 12 inches wide for carbon and 3 inches for boron. Cross stitched carbon tapes up to 60 inches wide are available commercially in some cases.

Tenacity -- The tensile stress expressed as force per unit linear density of the unstrained specimen i.e., grams-force per denier or grams-force per tex.

Tex -- A unit for expressing linear density equal to the mass or weight in grams of 1000 meters of filament, fiber, yarn or other textile strand.

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Thermal Conductivity -- Ability of a material to conduct heat. The physical constant for quantity of heat that passes through unit cube of a substance in unit time when the difference in temperature of two faces is one degree.

Thermoplastic -- A plastic that repeatedly can be softened by heating and hardened by cooling through a temperature range characteristic of the plastic, and when in the softened stage, can be shaped by flow into articles by molding or extrusion.

Thermoset -- A plastic that is substantially infusible and insoluble after having been cured by heat or other means.

Tolerance -- The total amount by which a quantity is allowed to vary.

Tolerance Limit -- A lower (upper) confidence limit on a specified percentile of a distribution. For example, the B-basis value is a 95% lower confidence limit on the tenth percentile of a distribution.

Tolerance Limit Factor -- The factor which is multiplied by the estimate of variability in computing the tolerance limit.

Toughness -- A measure of a material's ability to absorb work, or the actual work per unit volume or unit mass of material that is required to rupture it. Toughness is proportional to the area under the load-elongation curve from the origin to the breaking point.

Tow -- An untwisted bundle of continuous filaments. Commonly used in referring to man-made fibers, particularly carbon and graphite fibers, in the composites industry.

Transformation -- A transformation of data values is a change in the units of measurement accomplished by applying a mathematical function to all data values. For example, if the data is given by x, then $y = x + 1$, x^2, $1/x$, $\log x$, and $\cos x$ are transformations.

Transition, First Order -- A change of state associated with crystallization or melting in a polymer.

Transversely Isotropic -- Descriptive term for a material exhibiting a special case of orthotropy in which properties are identical in two orthotropic dimensions, but not the third; having identical properties in both transverse directions but not the longitudinal direction.

Twist -- The number of turns about its axis per unit of length in a yarn or other textile strand. It may be expressed as turns per inch (tpi).

Twist, Direction of -- The direction of twist in yarns and other textile strands is indicated by the capital letters S and Z. Yarn has S twist if when held in a vertical position the visible spirals or helices around its central axis are in the direction of slope of the central portion of the letter S and Z twist is in the other direction.

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Typical Basis -- A typical property value is a sample mean. Note that the typical value is defined as the simple arithmetic mean which has a statistical connotation of 50% reliability with a 50% confidence.

Unbond -- An area within a bonded interface between two adherends in which the intended bonding action failed to take place. Also used to denote specific areas deliberately prevented from bonding in order to simulate a defective bond, such as in the generation of quality standards specimens. (See Disbond, Debond).

Unidirectional Laminate -- A laminate with nonwoven reinforcements and all layers layed up in the same direction.

Upper Confidence Limit -- See Confidence Interval.

Vacuum Bag Molding -- A process in which the lay-up is cured under pressure generated by drawing a vacuum in the space between the lay-up and a flexible sheet placed over it and sealed at the edges.

Variance -- See Sample Variance.

Viscosity -- The property of resistance to flow exhibited within the body of a material.

Void -- A physical and mechanical discontinuity occurring within a material or part which may be two-dimensional (e.g., disbonds, delaminations) or three-dimensional (e.g., vacuum-, air-, or gas-filled pockets). Porosity is an aggregation of micro-voids. Voids are essentially incapable of transmitting structural stresses or nonradiative energy fields. (See Inclusion).

Warp -- The longitudinally oriented yarn in a woven fabric (see Fill); a group of yarns in long lengths and approximately parallel.

Weibull Distribution (Two - Parameter) -- A probability distribution for which the probability that a randomly selected observation from this population lies between a and b ($0 < a < b < \infty$) is given by

$$e^{-(a/\alpha)^{\beta}} - e^{-(b/\alpha)^{\beta}} \qquad 1.7(d)$$

where α is called the scale parameter and β is called the shape parameter. (See Section 8.5.1.1.)

Wet Lay-up -- A method of making a reinforced product by applying a liquid resin system while the reinforcement is put in place.

Wet Strength -- The strength of an organic matrix composite when the matrix resin is saturated with absorbed moisture. (See Saturation).

Wet Winding -- A method of filament winding in which the fiber reinforcement is coated with the resin system as a liquid just prior to wrapping on a mandrel.

Whisker -- A short single crystal fiber or filament. Whisker diameters range from 1 to 25 microns, with aspect ratios between 100 and 15,000.

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Work Life -- The period during which a compound, after mixing with a catalyst, solvent, or other compounding ingredient, remains suitable for its intended use.

Woven Fabric Composite -- A major form of advanced composites in which the fiber constituent consists of woven fabric. A woven fabric composite normally is a laminate comprised of a number of laminae, each of which consists of one layer of fabric embedded in the selected matrix material. Individual fabric laminae are directionally oriented and combined into specific multiaxial laminates for application to specific envelopes of strength and stiffness requirements.

Yarn -- A generic term for strands or bundles of continuous filaments or fibers, usually twisted and suitable for making textile fabric.

Yarn, Plied -- Yarns made by collecting two or more single yarns together. Normally, the yarns are twisted together though sometimes they are collected without twist.

Yield Point -- The first stress in a material at which an increase in strain occurs without an increase in stress. (The stress is less than the maximum attainable). It should be noted that only materials that exhibit the unique phenomenon of yielding have a "yield point".

Yield Strength -- The stress at which a material exhibits a specified limiting deviation from the proportionality of stress to strain. (The deviation is expressed in terms of strain such as 0.2 percent for the Offset Method or 0.5 percent for the Total Extension Under Load Method).

X-Axis -- In composite laminates, an axis in the plane of the laminate which is used as the 0 degree reference for designating the angle of a lamina.

X-Y Plane -- In composite laminates, the reference plane parallel to the plane of the laminate.

Y-Axis -- In composite laminates, the axis in the plane of the laminate which is perpendicular to the x-axis.

Z-Axis -- In composite laminates, the reference axis normal to the plane of the laminate.

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1.6(a) Military Standardization Handbook, "Metallic Materials and Elements for Aerospace Vehicle Structures", MIL-HDBK-5D, Change Notice 2, (May, 1985).

1.6(b) DOD/NASA Advanced Composites Design Guide, Air Force Wright Aeronautical Laboratories, Dayton, OH, prepared by Rockwell International Corporation (1983) (distribution limited).

1.6(c) ASTM E206, "Definitions of Terms Relating to Fatigue Testing and the Statistical Analysis of Fatigue Data," 1984 Annual Book of ASTM Standards, 3.01, ASTM, Philadelphia, PA (1984).

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2. OBJECTIVES IN GENERATING PROPERTY DATA

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2.1 Introduction. The purpose of Chapter 2 is to provide guidelines for the characterization of composite material properties. While current procedures emphasize the development of a lamina-level database, this does not preclude higher level testing. Further, it is possible that qualification can be achieved at the application level of testing.

This chapter also provides guidelines for the screening of new material systems. Mechanical properties testing is proposed which provides a preliminary, typical value, evaluation of of key stiffness and strength properties. The intent of this screening test program is to provide an initial evaluation of new material systems under worst-case environmental and loading test conditions. The screening includes both lamina (ply) and laminae (application) level testing and should be sufficient to eliminate deficient material systems and to reveal promising new material systems for planning of in-depth evaluations.

2.2 Recommendations for the generation of physical and mechanical properties.

2.2.1 General guidelines. Generation of design allowables for advanced composite materials requires physical and chemical property characterization of the prepreg material as well as physical and mechanical property tests on cured lamina and laminates. These guidelines contain recommendations for the number and type of tests sufficient to establish B-basis mechanical properties for advanced composite unidirectional tape and woven fabric materials. Recommendations are also provided for vendor acceptance tests by the manufacturer of the prepreg materials.

These guidelines present the methods recommended by MIL-HDBK-17 for generating statistically-based material properties for lamina coupons. Other methods involving statistical analysis of mechanical property tests on oriented laminates are equally acceptable when agreed to by the contractor and certifying agency.

2.2.2 Environmental sensitivity.

2.2.2.1 Moisture effects. Organic matrix composites absorb small amounts of moisture. The absorbed water causes a dimensional change (swelling), lowers the glass transition temperature of the resin, and results in a reduction of the matrix and matrix/fiber interface dependent mechanical properties of the composite, particularly at elevated temperatures. Except for the aramid fiber composites, it is usually assumed that the moisture absorption is limited to the organic matrix.

The degradation of composite properties as a result of exposure to high temperatures is nonlinear. A drastic reduction in properties occurs at a characteristic temperature level, defined here as the material operational limit (MOL). For composites, this behavior is more complicated because the introduction of moisture will result in more than one critical temperature, as illustrated in Figure 2.2.2.1(a). The various moisture contents are represented schematically by saturation at different relative humidity levels. There are no standard criteria for the determination of the MOL. One method (References 2.2.2.1(b) - (d)), illustrated in Figure 2.2.2.1(b), utilizes the glass transition temperature (T_g) with some safety factor K as the basis for the MOL. For epoxy composites, 50 F° (28 C°) has been proposed for the value of the safety factor. If service usage at

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high temperatures is limited to occasional excursions, it can be argued that lower safety factors are acceptable.

The elevated temperature wet testing required by MIL-HDBK-17 should be at the MOL temperature, corresponding to the highest practical relative humidity. For aircraft, the worst-case relative humidity is 85%. As can be seen from Figure 2.2.2.1(a), the effect of moisture is small for matrix-dependent properties at temperatures below room temperature (RT). However, if Figure 2.2.2.1(a) were produced for the fiber dependent properties, degradation in properties would be observed below room temperature. Therefore, MIL-HDBK-17 requires testing at cold temperatures. Since the changes in properties at these temperatures are not drastic, the lowest service temperature will suffice and there is no need to determine a MOL.

The equilibrium level of weight gain due to moisture absorption is generally considered to be a function of relative humidity (RH) only. The relationship is not necessarily linear as shown by the test data fitted curves for epoxy composites in Figure 2.2.2.1(c). The rate of moisture absorption is very strongly dependent on temperature as shown in Figure 2.2.2.1(d); a decrease of 60 F° (33 C°) increases the time to absorb 1% moisture by 5 times. The relationship between the diffusion coefficient and absolute temperature is of the Arrhenius form

$$D = D_o \exp(-E_d/RT) \qquad 2.2.2.1(a)$$

where D_o and E_d are material constants and R is the universal gas constant. The moisture absorption and desorption is mathematically modeled using a one-dimensional Fickian equation

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} \qquad 2.2.2.1(b)$$

where c is moisture concentration. The boundary conditions are:

$$c = c_i \text{ at } t = 0$$

$$c = c_a \text{ at } t \gg 0$$

and x is the thickness parameter varying between 0 and h. This equation assumes that the moisture diffusion and temperature are constant inside the material. Because the rate of temperature diffusion is about 10^6 times faster than the moisture diffusion, and because diffusivity changes very little with moisture content, these assumptions are quite valid. If a further simplification of constant boundary conditions with time is made, solution to Equation 2.2.2.1(b) can be approximated (Reference 2.2.2.1(a)) as

$$M_{avg} = G(M_m - M_i) + M_i \qquad 2.2.2.1(c)$$

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where

M_{avg} = moisture content of slab in percent by weight

M_m = moisture content at equilibrium for the boundary relative humidity

M_i = initial uniform moisture content (assumed to be zero)

G = a time-dependent parameter defined below

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The specimens or travellers should also be weighed just before and after mechanical testing, especially for elevated temperatures. At these temperatures, it is also important not to soak the specimens longer than needed so as not to drive-out moisture. A typical soak time after the thermocouples on the specimens reach the desired temperature is half a minute for carbon/epoxy for temperatures up to 350°F (177°C). Because thick laminates would require unacceptably long times to reach equilibrium moisture levels, thin laminates are usually used in moisture conditioning testing. The diffusion properties and equilibrium moisture content obtained from thin laminates are assumed to be applicable to thick laminates unless test data or calculations show otherwise.

2.2.3 Material acquisition and prepreg physical property characterization. For material property characterization suitable for inclusion in MIL-HDBK-17, five prepreg batches of each material type shall be prepared by the material supplier using prepreg production facilities. The five prepreg batches shall be made with five different fiber lots impregnated with five different production resin batches. The contractor shall prepare and coordinate a preliminary material specification with the supplier prior to manufacture of the prepreg batches. The preliminary material specification shall define agreed-to laminate and specimen fabrication procedures, test procedures, and physical and mechanical property requirements.

The five-batch requirement only applies to material properties that are to be incorporated in MIL-HDBK-17. An alternate number of tests and batches may be employed upon approval of the certifying agency. However, mechanical property test data should be examined by the recommended statistical methods in this document (MIL-HDBK-17) to ascertain that they are statistically acceptable.

Recommended tests to be performed on the prepreg materials are shown in Table 2.2.3. When applicable, the tests should be standard ASTM procedures.

2.2.4 Lamina physical and mechanical property tests. A minimum of two test panels for each prepreg batch shall be cured in two separate autoclave runs to be used in fabricating test coupons. The prepreg material shall be layed up, bagged, and cured in accordance with a process specification prepared by the contractor. Cured test panels shall be non-destructively evaluated (NDE) using ultrasonic inspection. Imperfections in the test panels shall be reported with questionable areas marked on the panels. Test coupons shall not be extracted from these areas.

Suggested physical and mechanical property tests are shown in Tables 2.2.4(a) and 2.2.4(b). A test procedures document should be prepared by the contractor prior to the start of testing. It is recommended that, whenever possible, the tests should be ASTM standard methods.

The test matrix shown in Table 2.2.4(b) is based on thirty tests per data point (six tests per batch) to provide for parametric/non-parametric analysis in determining B-basis properties. Fewer tests per data point may be acceptable for non-critical properties as agreed to by the prime contractor and the certifying agency. For generating B-basis properties, 0° and 90° tension, 0° and 90° compression, and in-plane shear tests are required. Short beam shear tests are required for determining process control test minimum values.

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Extensometers or strain gages should be used to obtain strain measurements for each specimen. Two of each group of six compression specimens shall be strain gaged back-to-back with axial gages to assess possible premature buckling failure.

TABLE 2.2.3 - Recommended physical and chemical property tests to be performed by material supplier and prime contractor.

Test Property	Suggested Test Procedure*	Number of Tests per Batch**	Total Number of Tests
Resin Content, %	D3529	3	15
Volatile Content, %	D3530	3	15
Gel Time, min.	D3532	3	15
Resin Flow, %	D3531	3	15
Fiber Areal Wt., gm/M^2	†	3	15
Moisture Content, %	†	3	15
Tack	†	3	15
HPLC (High Performance Liquid Chromatography)	†	3	15
IR (Infrared Spectroscopy)	†	3	15
DMA (Dynamic Mechanical Analysis, neat resin only)	†	3	15
DSC (Differential Scanning Calorimetry)	†	3	15
RDS (Rheological Dynamic Spectroscopy)	†	3	15

* Test procedures shall be coordinated and agreed to prior to manufacture of prepreg material.

** Tests shall be performed on each of the five batches of prepreg material.

† Test procedures to be described at a later date.

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2.2.5 Statistical development of mechanical properties. Data interpretation and statistical analysis procedures are defined in Chapter 8.

2.2.6 Data pooling requirements. The capability of pooling data may be desirable in order to obtain sufficient data to calculate B-basis values for material properties. Data to be pooled may be available for materials from different composite component manufacturers, different locations of one manufacturer, or slightly different processes from the same manufacturer. Data for the same material system and equivalent processing procedures may be pooled if the statistical batch-to-batch variation test criteria of Section 8.5.3.2 are satisfied. These criteria should be satisfied for all properties measured in order to pool data for any of the properties. For example, if tensile data meet the batch-to-batch variation criteria but shear data for the same material do not meet these criteria, shear data and tensile data should not be pooled.

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2.3 Other useful test matrices.

2.3.1 Material system screening. The objective of this screening process is to reveal key mechanical property attributes and/or inadequacies in new material system candidates through use of a reduced test matrix. The screening process will identify the critical test and environmental conditions, and any special considerations for a particular composite material system. Proper test selection and environmental conditioning will enable comparisons with current productions systems.

Table 2.3.1 shows a typical mechanical property screening test matrix primarily intended for epoxy resin systems. The test matrix will vary depending on the purpose of the investigation. The 0° axial tensile tests examine fiber dominant properties, and 0° axial compression tests monitor matrix/fiber interactions. The 0° axial tensile tests may reveal matrix-fiber interaction in some materials at high strain rates. These tests will yield static strength and stiffness properties. To test matrix characteristics, ±45° tensile tests will be used. The ±45° tensile tests permit actual shear modulus and effective shear strength values to be obtained. Finally, damage tolerance will be evaluated using impact testing.

Testing will be conducted under three environmental conditions: cold temperature dry (CTD), room temperature ambient (RTA), and elevated temperature wet (ETW). These conditions are selected based on results for the epoxy resin systems currently in use which show that the CTD condition is a fiber critical environmental state and the ETW condition is the most severe test for the matrix. Moisture conditioning may be accomplished by placing the dried test specimens in an environmental chamber until an equilibrium moisture weight gain at the desired relative humidity is obtained.

The screening process presented allows key mechanical properties and environmental conditions to be evaluated. Dependent upon application, sensitivity to exposures to operational fluids and other special issues, additional tests may also be included in the screening evaluation. However, a reduced test matrix for mean property evaluations should permit preliminary efficient and economical assessment of new composite material systems.

The approach to screening is the performance of key static tests to provide sufficient data for mean property evaluations of stiffness and strength. The recommended specimens provide data at the lamina (ply) and laminate (application) levels. The lamina level tests provide key generic material stiffness and strength properties commonly used in classical lamination plate point stress analysis including tension, compression, and shear loadings. The laminate level tests provide screening strength data on application issues relating to effects of stress concentrations, manufactured stress risers (fastener holes), bolt bearing, or impact damage. Additional laminate level tests provide screening stiffness data to verify the use of lamina data with classical lamination plate theory for laminate stiffness predictions. The damage tolerance screening is patterned after NASA Reference Publication 1092 (Reference 2.3.1).

All laminate level tests suggested are performed at the RTD test condition. For screening purposes, it is assumed the lamina (ply) level unnotched specimens will conservatively provide mean data to assess environmental effects on strength and

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calorimetry (DSC), thermal mechanical analysis (TMA), dynamic mechanical analysis (DMA), and torsional braid analysis (TBA) are not strictly chemical analysis techniques; however, they provide useful information relating to the composition and processability of resins. Similarly, rheological and dielectric techniques are used frequently to evaluate the chemoviscosity properties of thermoset resins during cure, and there is increasing interest in applying such techniques for process monitoring and process control of both thermoset and thermoplastic resins.

5.2.1 High performance liquid chromatography. High performance liquid chromatography (HPLC) is a versatile and economically viable quality assurance technique for soluble resin materials (References 5.2.1(a) - 5.2.1(g)). HPLC involves the liquid-phase separation and monitoring of separated resin components. Dilute solutions of resin samples are prepared and injected into a liquid mobile phase which is pumped through column(s) packed with a stationary phase to facilitate separation and then into a detector. The detector monitors concentrations of the separated components; its signal response, recorded as a function of time after injection, provides a "fingerprint" of the sample's chemical composition. Quantitative information may be obtained if the sample components are known and sufficiently well-resolved, and if standards for the components are available. Size exclusion chromatography (SEC), an HPLC technique, is particularly useful in determining the average molecular weights and molecular weight distributions of thermoplastic resins (Reference 5.2.1(g)). Recent advances have resulted in improved and automated HPLC instrumentation that is relatively low cost and simple to operate and maintain.

5.2.2 Infrared spectroscopy. Infrared (IR) spectroscopy is sensitive to changes in the dipole moments of vibrating groups in molecules and, accordingly, yields useful information for the identification of resin components. IR spectroscopy provides a fingerprint of the resin composition and is not limited by the solubility of resin components (References 5.2.2(a) - 5.2.2(c)). Indeed, gases, liquids and solids may be analyzed by IR spectroscopy. Advances in technology have led to the development of Fourier transform infrared spectroscopy (FTIR) - a computer-supported IR technique for rapidly scanning and storing infrared spectra. Multiple scans and Fourier transformation of the infrared spectra enhance the signal-to-noise ratio and provide improved spectra for interpretation. In addition, the FTIR attenuated total reflection (ATR) technique may be applied for quality assurance of thermoset composite materials to assess their state of cure; i.e., residual epoxide concentration. (See Section 5.6.2.)

Other chromatographic and spectroscopic techniques have also been considered (References 5.2.2(a), 5.2.2(d) - 5.2.2(h)). Gas chromatography (GC), GC headspace analysis, and GC-mass spectroscopy are useful for analyzing residual solvents and some of the more volatile resin components. Combined thermal analysis - GC-mass spectroscopy can be used to identify volatile reaction products during cure (References 5.2.2(i) and 5.2.2(j)). Elemental analysis, titration, and wet chemical analysis for specific functional groups are useful techniques for characterizing individual epoxy components but have limited application and may provide misleading results when complex resin formulations are analyzed. When necessary, ion chromatography, atomic absorption (AA), x-ray fluorescence, or emission spectroscopy are applied to analyze specific elements, such as boron or fluorine. X-ray diffraction may be used to identify crystalline components, such as fillers, and determine the relative percent crystallinity for certain resins. (See Section 5.6.3.)

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5.2.3 Thermal analysis. Thermal gravimetric analysis (TGA) monitors the weight changes in a sample as a function of temperature. Although primarily used for studying the degradation processes, TGA can also be applied as a quality assurance technique to provide information about the volatiles, resin, fiber, and inorganic residue content of prepreg materials Reference (5.2.3(a)). Since dissimilar materials often degrade and volatilize at different temperature and rates, compositional differences may be reflected by differences in their TGA thermograms. Thermal oxidative degradation rates determined by TGA are useful for estimating the life cycles of resin materials (Reference 5.2.3(b)).

Differential scanning calorimetry (DSC) and differential thermal analysis (DTA) techniques are frequently employed for characterizing resins and composite materials (References 5.2.1(g), 5.2.2(a), 5.2.3(c), and 5.2.3(d)). Both DSC and DTA monitor enthalpy changes in materials as a function of temperature (DSC directly and DTA indirectly) and thereby provide similar information useful for quality assurance of prepreg materials. DTA measures the temperature difference (ΔT) between the epoxy resin specimen and a reference material; whereas DSC measures the rate of heat evolution (dH/dt) or enthalpy absorption of the specimen relative to a reference. DTA and DSC measure thermal changes (1) as a function of time with both the specimen and reference material held at the same temperature (isothermal), or (2) as a function of temperature with both the specimen and reference material heated at the same heating rate (dynamic). For quality assurance applications, DTA and DSC are usually run in the dynamic mode with the weighed specimen in an aluminum specimen holder and an empty holder used as the reference. Dynamic DTA and DSC measure the glass transition temperature T_g and heat of reaction ΔH of the prepreg resin but do not directly provide information about chemical composition. By monitoring the fraction of heat evolved as a function of temperature or time, information relating to the extent of cure and curing kinetics can be obtained. DSC and DTA may also be applied to evaluate the melting temperature T_m and to estimate the degree of crystallinity of thermoplastic resins and composites. Since the average specimen size used in DSC is only about 10 mg (0.00002 lb), special care must be taken in obtaining representative materials. Multiple specimens runs are advisable.

Thermal mechanical analysis (TMA) is used in conjunction with DTA and DSC to study the thermal transition behavior (e.g., T_g) of prepreg resins and cured laminates. TMA simulates a linear dilatometer to measure the thermal expansion and contraction of specimens under dynamic or isothermal heating conditions. Adjustable loads are applied via a specially designed probe resting upon the specimen surface. Sensitive displacement devices are employed to monitor the "nominal" thermal response of a material. Since thermal transition behavior is related to the chemical composition and extent of cure of a prepreg resin, TMA can be applied as a quality assurance technique.

5.2.4 Rheological analysis. Dynamic mechanical analysis (DMA), torsional braid analysis (TBA), and various mechanical spectrometers measure the rheological response of resins and composites as a function of frequency, temperature and/or state of cure. Both DMA and TBA provide information relating to the storage modulus (E') and the loss modulus (E'') of the resin during cure and to the T_g of the cured resin (references 5.2.3(c), 5.2.4(a) - 5.2.4(C)). The dynamic complex viscosity (η^*) and tan delta parameters can also be determined from rheological measurements of thermoset resins during cure. Rheological techniques are most often used to optimize processing parameters and evaluate the onset of gelation

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and vitrification during cure. However, since rheological properties are related to resin composition and morphology, rheological techniques may also be applied for the quality assurance of prepregs. Rheological data are often less reproducible when taken from fiber reinforced samples. Whenever possible, neat resin or extruded film samples of the resin lot should be procured along with the prepreg for rheological testing.

Dynamic dielectric analysis (DDA) techniques involve the use of electrical measurements to monitor changes in the dielectric constant, the dissipation factor, capacitance, and/or conductance of the prepreg resin during processing as a function of frequency, time, and temperature. Measured electrical parameters are highly responsive to changes in resin viscosity and, similar to the rheological techniques, are often employed to investigate and optimize prepreg processing parameters such as resin flow and gelation time/temperature. Since chemical composition affects the electrical properties and curing behavior of prepregs, DDA techniques may also be applied for their quality assurance (References 5.2.4(c) - 5.2.4(j)). (See Section 5.6.5.)

Whenever possible, complementary techniques should be used for the chemical quality assurance of resins and composites. Techniques, such as HPLC and IR spectroscopy, are fundamentally different from one another and provide direct, but different, information about a resin's composition. If appropriate test methods are applied, HPLC and IR spectroscopy are usually powerful enough to detect differences or changes in the chemical compositions of resins. DTA and DSC complement HPLC and IR spectroscopy by providing information relating to the handleability (i.e., the T_g and extent of reaction of the resin) and the processability of the prepreg. TGA and GC head-space analysis techniques for volatile components are secondary, but important, techniques. Special techniques for analyzing specific components or elements should be used if knowledge of the concentrations of the components is critical for processing the resin or if their presence could adversely effect the performance and durability of the cured composite. The information provided by mechanical, rheological, and dielectric analysis techniques is related to the chemical composition of the prepreg resin and thereby complements the more direct chemical techniques. However, caution is recommended in applying non-chemical techniques since the information obtained is complex and frequently ambiguous when attempts are made to relate measured parameters to chemical composition. (See Section 5.6.6.)

5.3 Sampling, handling and storage. Prepregs are specified by the manufacturer's trade name, resin type (e.g., 250°F) and lot number, fiber type and form (tape, fabric, roving, etc.), prepreg lot and roll numbers, and date of manufacture. The shipping date and expected shelf life are also usually designated along with recommended processing conditions. Generally, prepregs are shipped as rolls of impregnated woven fabric or unidirectional tape. (Standard widths are 38" (~97 cm) for woven aramid and glass, 42" (~107 cm) for woven graphite, and 12" (~30 cm) for tapes). The prepreg layers are separated by thin, removable plastic or coated paper films which are removed when the prepregs are analyzed.

To check uniformity, it is recommended that approximately 6 × 6 inch (15 x 15 cm) sections be cut from the center and each side of the front-end (first off) of each prepreg roll. The amount and number of samples required for a particular analysis or test depends upon the homogeneity of the resin and the uniformity of

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the prepreg fibers. For some techniques, such as HPLC, 0.5 to 2.0 grams of prepreg may be needed to provide a representative sample. Other techniques (e.g., DSC) which utilize relatively small specimens (10 to 20 milligrams) may demand multiple specimens to provide an "average" value or test result.

Care must be taken not to contaminate or in any way alter samples during handling and storage. The samples should be placed in clean, dry, sealable containers and be carefully labeled. The containers must not react with the samples and precautions must be taken not to expose the prepregs to moisture nor allow them to stand unrefrigerated for long periods of time. For reactive prepreg resins such as epoxies, it is recommended that the prepregs be stored in hermetically sealed containers at -10 to -15°F (-23 to -26 C). Upon removing the containers from cold storage, they should be allowed to achieve room temperature before opening to prevent condensation of moisture on the samples.

5.4 General characteristics of prepregs and composites.

5.4.1 Physical description of reinforcement. The physical description of the reinforcement used in a composite shall be described using the standard definitions of ASTM D3878 (Reference 5.4.1). The filaments in the prepreg should be uniformly wetted by the resin. No particulates should be observable upon visual examination.

5.4.1.1 Alignment. In unidirectional prepregs, the filament bundles must be parallel to the longitudinal direction of the prepreg within an angle of 0.5° when examined visually using appropriate aids to measure angular alignment.

5.4.1.2 Gaps. Any gap within or between filament bundles in unidirectional prepregs generally should comply with the following:

a. No gap shall exceed 0.060 inch (1.54 mm) in width.

b. The length of any portion of the gap with an average width of 0.040 inch (1.02) shall not exceed 24 inches.

c. Gaps in line with each other and no more than one inch (25 mm) apart shall be considered as one gap, regardless of number.

d. Gaps with excessive width or length shall be considered defective and will be the basis for flagging or replacing the prepreg.

5.4.1.3 Width. Width tolerance for unidirectional prepreg tape shall be ± 0.030 inch (0.76 mm) or as specified.

5.4.1.4 Length. The length of each roll of prepreg shall be provided together with sequence in production and batch identification as supporting data to prepreg certification. The maximum length of prepreg on any single roll shall be specified. Alternatively, suppliers and users may agree to specify the maximum prepreg weight (lbs, kg) or area (ft^2, m^2) per roll.

5.4.1.5 Edges. Maximum acceptable waviness of any 12-inch (30-cm) length of tape shall be 0.030 inch (0.76 mm) from the edge when measured with an appropriate straight edge.

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5.4.9 ASTM D 3531, "Test Methods for Resin Flow of Carbon Fiber-Epoxy Prepeg"; 1984 Annual Book of ASTM Standards, 15.03, ASTM, Philadelphia, PA (1984).

5.4.10 ASTM D 3532, "Test Method for Gel Time of Carbon Fiber-Epoxy Prepeg", 1984 Annual Book of ASTM Standards, 15.03, ASTM, Philadelphia, PA (1984).

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6. LAMINA AND LAMINATE CHARACTERIZATION

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6.1 Introduction. The use of composite materials continues to increase as new performance, reliability, and durability requirements drive hardware designs to higher levels of structural efficiency. Additionally, government requirements are becoming more stringent to assume proper levels of structural integrity are maintained. These design drivers, among others, have resulted in a growing recognition that certification or qualification of aerospace structure requires an extensive combination of analysis, testing, and documentation.

Further, because of the large number of design variables inherent to composite structure, analytic models are even more necessary than for metallic structure to assume completeness of the hardware qualification process. Inherent in all structural analysis models are material, physical, and mechanical property characterization data. Ideally, these analytic models would permit analysts to predict full-scale structural response (e.g. stability, deflections, strength, life) directly from a generic (lamina) material database. In truth, test data is required at design development (element, subcomponent, component) and full-scale article test levels as well as the generic (coupon) levels of evaluation.

The purpose of Chapter 6 is to provide guidelines on testing procedures for characterization of lamina (ply) physical and mechanical properties. While current procedures emphasize development of a lamina-level database, this does not preclude higher-level testing.

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6.2 Chemical property tests.

6.3 Physical property tests.

6.3.1 Glass transition temperature.

6.3.2 Density.

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6.6 Mechanical property tests.

6.6.1 General. Section 6.6 contains test methods for determining material property data for composite materials. The purpose of this section is to provide for uniformity in the use of standard test methods and, ultimately, to provide for combining the experimental data. The reader is referred to Chapter 8 for reporting of data to MIL-HDBK-17. The test methods are representative of procedures used in the composite materials industry and were selected after review of user material specifications. Specific standards are included to allow the user to perform tests consistent with industry practice; however, inclusion of these standards should not be considered an endorsement of any standard or organization by MIL-HDBK-17.

These discussions reflect the current dynamic state of test methods development for composite materials. The intent is to inform the reader of the availability of test standards, and to alert the reader to the limitations of these test methods. New test methods, or modifications to a test standard that require a dissimilar test fixture or deviate from the recommended specimen configuration, are not included in MIL-HDBK-17 unless there is a general consensus that mention of the new technique or modification will lessen confusion. By restricting the discussion to test methods which are standards, or to test which are being evaluated for standardization, MIL-HDBK-17 seeks to avoid an endorsement of a technique which may be popular, but which has not undergone the careful scientific scrutiny that insures a thorough and comprehensive evaluation of the precision, accuracy, and validation of the test.

It should be noted that procedures which may enhance the preparation of the test specimen may be given. These statements are deliberately general in terms of applicability and are based upon the experience of the composite materials industry. The reader should further investigate any enhancements of a standard before proceeding with a test program.

6.6.2 Tensile tests.

6.6.2.1 General considerations. Tensile testing of laminates will generally be accomplished in accordance with ASTM D3039. This test employs a straight-sided specimen with bonded end tabs for gripping, and is applicable to 0° and 90° unidirectional constructions as well as bidirectional and fabric laminates. Details for specific orientations are discussed in Sections 6.6.2.2, 6.6.2.3, and 6.6.2.4. The test yields tensile ultimate strength, tensile modulus, and the major Poisson's ratio.

Experience within the aerospace industry has indicated that the critical factors in obtaining consistent tensile test results are tab design, tab application, and specimen machining quality. With respect to machining, it is important that specimen edges be wet ground, cut, or polished to a fine surface finish to preclude premature failure.

The purpose of tabbing is to introduce uniform loading into the specimen. The degree to which this is realized depends upon a number of factors including tab bevel angle, and the mechanical properties of the tab material relative to the laminate being tested. In general, the tab material should be less stiff than the test laminate.

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Surface preparation prior to tab bonding may include light sanding or grit blasting to promote adhesion. Care must be taken not to abrade any part of the laminate gage section (any area not covered by the tab), as this damage could cause premature failure.

Thermal considerations are also important. If the coefficient of thermal expansion of the tab material is significantly different from that of the laminate, bond failure could occur either during bond cycle cool-down, or during elevated temperature testing. In addition, bond cycles that will advance the cure state of the laminate cannot be used, as this will invalidate the data. For this reason, many companies use 250°F (120°C) curing adhesives when tabbing 350°F (180°C) cured laminates. This is of less concern when postcured laminates are being tested.

6.6.2.2 0° Unidirectional laminae. When preparing 0° specimens, care must be taken to insure that the fibers are aligned parallel to the coupon load axis. A misalignment of only 0.5° has been shown to affect ultimate strength by as much as 5%. To accomodate capacities of standard composite test machines, the width of 0° unidirectional specimens is usually specified as 0.500 inches (12.7 mm).

6.6.2.3 90° Unidirectional laminae. As specified in ASTM D3039, the 90° tensile specimen is wider and shorter than the 0° coupon. Since this test is totally matrix dependent, the handling and precise alignment of the specimen are critical. Special alignment fixtures are often used in conjunction with hydraulic or pneumatic test grips to prevent introduction of out-of-plane forces.

As an alternate, a sandwich beam specimen in accordance with ASTM C393 may be used for 90° testing. This specimen is less susceptible to handling damage, but panel warpage is a problem.

6.6.2.4 Bidirectional and fabric laminates. The specimen for these laminates is wider than the 0° unidirectional coupon. For fabrics it is mandatory to orient the warp and fill for each ply in the same direction, and to report the direction tested. A modified ASTM D-3039 dog-bone shaped specimen is frequently utilized for testing fabrics. As larger tow, fewer ends per inch fabrics have become more common, it has become practical to increase the gage width to 1.000 inch (25.4 mm) to reduce scatter of test results.

6.6.3 Compression tests. Compression test methods for composite materials generally fall into three categories: unsupported coupon, supported coupon, and sandwich beam. ASTM D 3410, Standard Test Method for Compressive Properties of Unidirectional or Cross-ply Fiber-Resin Composites, describes the specimens and fixtures for the unsupported coupon and sandwich beam tests. The unsupported coupon has a recommended gage length of 0.500 inch (12.7 mm) and can be placed in one of the two compression fixtures given in ASTM D 3410. The two fixtures have split collet-type grips at both ends but they differ in the way load is introduced and in the gripping arrangement. The sandwich beam is a rectangular bonded beam with a 6 ply, all 0° composite test skin. The skin is bonded to a high density core. It should be noted that the sandwich beam specimen is generally expensive to test and unsuitable for environmental testing. There are also questions concerning the influence of honeycomb on the test results.

ASTM D 695, Standard Test Method for Compressive Properties of Rigid Plastics, was not designed for composite materials, but nonetheless it has been modified by

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the aerospace industry for use with composites. References will often be made to a "modified ASTM D 695 test", but each company has developed its own modification and there is no one accepted test fixture, gage length, or test procedure. In general, a modified ASTM D 695 test supports the face of the specimen and uses a gage section ranging from 0.1 to 0.2 inches (2.5 to 5.1 mm) in length. The supported specimen face of this test method, in addition to the smaller gage section, generally results in higher compressive strength and moduli values than ASTM D 3410. These higher values make it the method of choice for many designers.

There is considerable debate as to whether a modified ASTM D 695 test results in a true compressive failure. The constrained specimen may not be representative of a compressive failure in composite materials. ASTM D 3410 may produce a more accurate mechanistic compressive failure in composites, but the design values are generally lower and may not be representative of a compressive failure mode in an aircraft structure.

Regardless of choice, ASTM D-3410 or modified ASTM D-695, it has become more practical to incorporate 1.000 inch (25.4 mm) gage width laminate specimens for fabrics with low end count per inch weaves (such as 6000 filament graphite tow fabrics).

6.6.4 Flexure tests. There is not a recommended test method for determining the flexural properties of composite laminates. Even though there are approved flexure test methods, there is some debate as to the validity of the results.

Within the aerospace industry, flexure testing is primarily used for quality control. ASTM D790, Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials, was originally written for plastics but has since been modified and approved for composites. In some cases, ASTM C393, Flexure Test of Flat Sandwich Constructions, has been adapted for use with composite laminates.

6.6.5 Shear tests. The use of continuous fiber reinforced composites as high performance structural materials in modern aerospace vehicles necessitates the accurate measurement of the material response to mechanical and thermal loads. Most of the current composite material shear test methods were originally developed for metals, wood, and adhesives. However, coupling effects, nonlinear behavior of the matrix or the fiber-matrix interface, and the presence of normal stresses combine to make the present shear test methods of questionable value for composite materials (References 6.6.5(a) - (d)).

The shear test methods most widely used in the aerospace industry to measure the interlaminar and in-plane shear moduli and shear strengths of composite materials are subject to the above-mentioned limitations. For instance, in the Short Beam Shear test (ASTM D2344), the shear stress distribution depends upon the beam length-to-thickness ratio and eccentricities of the rollers applying the loads (Reference 6.6.5(a)). Stinchcomb et al. (Reference 6.6.5(b)) found that a good quality specimen failed in microbuckling or a combination of shear and microbuckling and that a shear failure only occurred in poor quality specimens. The Short Beam Shear test should not be utilized for determining the interlaminar shear strength. Short Beam Shear is useful for quality control and perhaps for selecting candidate materials, but it does not provide an accurate measurement of interlaminar shear strength.

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Similar problems exist for the Rail Shear test (ASTM D4255), which is one of the most analyzed methods for determining the in-plane shear modulus of composite materials. Stress concentrations at the corners and significant normal stresses are produced in the test section depending upon the method of load introduction, stiffness of rails, and properties of the composite (Reference 6.6.5(c)).

The torsional shear of a thin-walled circular tube (ASTM E143) is considered the most desirable method to obtain both the shear strength and the shear modulus of a material. Though from the applied mechanics viewpoint this is the ideal shear test, the torsional tube presents many problems to the experimentalist. Cost of fabricating tubular specimens can be prohibitive unless the manufacturer is involved in filament winding. Preparation of a tube is time-intensive and requires more material than a flat specimen. Composite tubes can also be extremely fragile and difficult to handle. Further, the specimen must be mounted concentrically in the test apparatus to prevent the introduction of bending moments, and the tube must be free to move axially to avoid introducing axial forces. Buckling of the tube must also be prevented. This requires special test equipment which can be expensive to manufacture. Yet, for all its limitations, when shear properties from this test are available, these values are the standard by which results from other shear tests are judged.

The ±45 Off-Axis Tensile Shear Test (ASTM D3518) consists of loading a ±45-degree symmetric laminate uniaxially in tension. This test has the advantage of being economical with material and time and it involves a simple test procedure (Reference 6.6.5(d)). Researchers have reported good correlation with unidirectional laminates between the results of the ±45-degree shear test and other shear test methods (References 6.6.5(d-g)), including the torsion tube. However, like all other shear test methods, this shear test is not without its problems. The stress/strain response tends toward ductile characteristics as a result of interply effects (Reference 6.6.5(f)). The stiffness is considered reliable up to 1.3% strain, but at higher stress levels the stiffness is underestimated due to edge effects (Reference 6.6.5(d)). Strength is also governed by these edge effects and is considered a conservative, lower-bound design value. Note that ±45 off-axis shear test results with fabric materials become more difficult to interpret due to the nature of the product form, mechanical testing, etc.

However, for determining the initial in-plane shear modulus, the ±45 Off-Axis Tensile Shear Test (ASTM D3518) may provide a value which reflects the actual stress state in a laminated structure. The specimen's ductile stress-strain response resulting from interply effects is indicative of the fact that a "pure" shear stress does not exist in the specimen, yet it may mimic the actual stress state that may occur in a laminate and may be representative of the interaction of one lamina upon adjoining laminae. The resulting value is an "effective" shear modulus, and may be a more realistic determination for the designer.

6.6.6 Fatigue. Static testing of unidirectional composite coupons is useful for material characterization, comparison of materials, and for prediction of application laminate properties through the use of lamination plate theory. In the area of fatigue, however, no generalized methodology has yet been devised to predict laminate behavior from unidirectional coupon data. Hence, the development of fatigue design values becomes a unique problem for each application lay-up.

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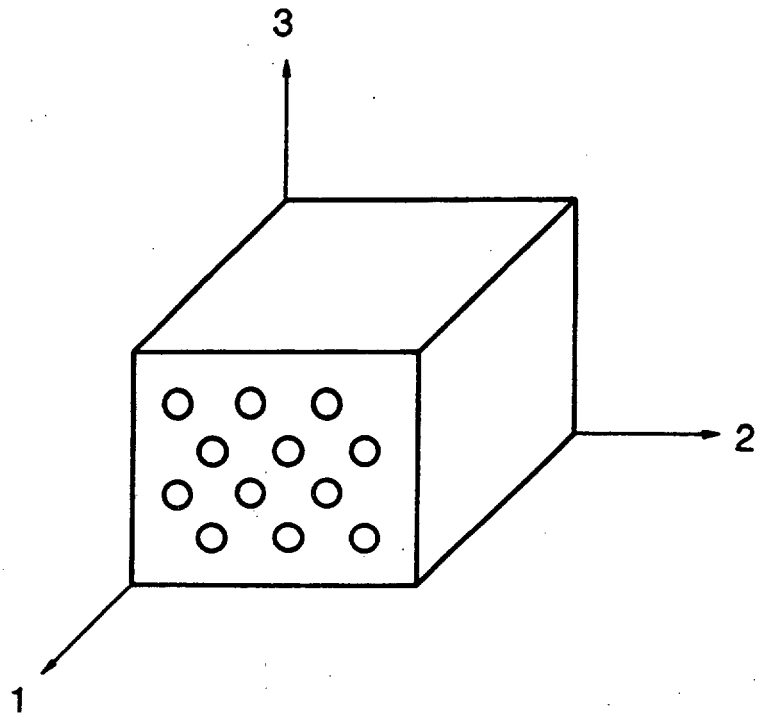
Many studies have been undertaken, and much has been written concerning life prediction for specific laminates under cyclic loading spectra. Even at this level, empirical methods have been favored due to the inadequacy of results obtained from cumulative damage models, fracture mechanics analysis, and other theoretical approaches (References 6.6.6(a) and (b)).

ASTM D 3479, Tension - Tension Fatigue of Oriented Fiber, Resin Matrix Composites, is a generalized coupon testing method. However, because composite fatigue is so application dependent, it is important that the laminates represent the application and that the laminates be tested accounting for the usage load

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TABLE 6.6.9 JANNAF interim standard test methods.

0° Tension:	Pressurized NOL Ring, Pressurized Tube (90° Wind)	E_{11}, ν_{12}, σ_{11}, ϵ_{11}
90° Tension:	Tube (90° Wind)	E_{22}, ν_{21}, σ_{22}, ϵ_{22}
0° Compression:	Flat Laminate (0°)	E_{11}, ν_{12}, σ_{11}, ϵ_{11}
90° Compression:	Tube (90° Wind)	E_{22}, ν_{21}, σ_{22}, ϵ_{22}
In-Plane Shear:	Torsion Tube (90° Wind)	G_{12}, τ_{12}, γ_{12}
Transverse Shear	Iosipescu	G_{23}, τ_{23}, γ_{23}



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7. STRUCTURAL ELEMENT CHARACTERIZATION

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7.1 Introduction.

7.1.1 Background. Testing and analysis of composite joints are essential for maintaining the structural integrity of composite structures and to ensure their reliability. Any joint in a composite structure is a potential failure site. Without proper design, a joint can act as a damage initiation point, which can lead to a loss in structural stability and eventual failure of a component. Two types of joints are in common use, namely (1) mechanically-fastened joints and (2) adhesively-bonded joints. Only mechanically-fastened joints are considered in MIL-HDBK-17 at this time. These guidelines define the recommended approach for testing bolted joints to determine bearing strength properties. A detailed analysis of the stress distribution around a fastener hole is not presented here. Discussion on both theoretical and empirical approaches to the stress analysis of bolted joints in composite materials can be found in Reference 7.1.1.

An important consideration in joint testing and analysis is the selection of the type of test method with due attention to the failure mode which is likely to result with a specific joint design in a particular composite system. A brief discussion on various failure modes is provided in the next section.

7.1.2 Failure modes. The occurrence of a particular failure mode is dependent primarily on joint geometry. Composite bolted joints may fail in various modes as shown in Figure 7.1.2. The likelihood of a particular failure mode is influenced by bolt diameter (D), laminate width (b), edge distance (e), and thickness (t).

Net section tension failures occur when the bolt diameter is a sufficiently large fraction of the strip width. This fraction is about one-quarter or more for near-isotropic lay-ups in graphite-epoxy systems. Bearing failures typically occur when the bolt diameter is a small fraction of the strip width. Shear-out failures are essentially a special case of bearing failure. Quite often a shear-out failure is the result of a bearing failure with short edge distance (e). For highly orthotropic laminates, shear-out failures may occur at very large edge distances.

Cleavage failures occur because of the proximity of the end of the specimen. A cleavage failure can be triggered from a net-section tension failure. This type of failure often initiates at the end of the specimen rather than adjacent to the fastener.

In some instances the bolt head may be pulled out through the laminate after the bolt is bent and deformed. This mode is frequently associated with countersunk fasteners.

Finally, it is important to note that for any given geometry, the failure mode may vary as a function of fiber pattern and lay-up sequence.

Within MIL-HDBK-17, only the bearing strength test method is discussed in detail to allow proper measurement and computation of joint strength. Strength measurements based on other failure modes may be included in MIL-HDBK-17 in the future.

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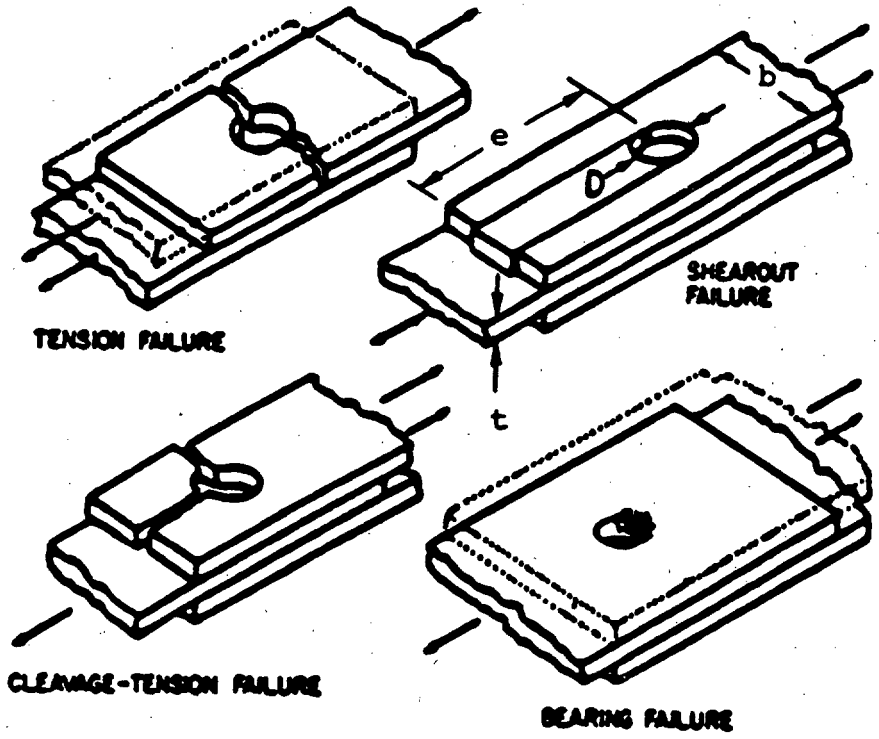


Figure 7.1.2 Typical failure modes for bolted joints in advanced composites.

7.2 Bearing strength characterization. Determination of bearing strength of continuous fiber-reinforced advanced composites is discussed in this section.

7.2.1 Significance. Bearing strength, as measured by the proposed test, is considered a structural property for relative evaluation and design. In an actual structural application, factors like laminate lay-up and load eccentricity will significantly influence the realizable fraction of the bearing strength measured in the proposed test.

The proposed test introduces the bearing load in a double shear configuration. In actual applications, load transfer in a single shear configuration is more commonplace, resulting in larger stress concentrations in the thickness direction, and lowering the realizable bearing strength.

The proposed test specimen will induce pure bearing failures because it contains a low percentage of 0° plies. If the structural application calls for over forty percent of 0° plies in the laminate, the failure mode will generally not be pure bearing, and a lowering of the bearing stress at failure will result. A high percentage of 0° plies will normally result in a shear-out mode of failure. In

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this case, the bearing stress at failure will be influenced by the shear-out area, which will be dependent on the edge distance.

Only a tensile loading condition is proposed for evaluating bearing failures; under compression, the larger edge distance (e >> 3D) should only influence the bearing stress at failure minimally unless a shear-out mode of failure is possible (e.g., a laminate with a large percent of 0° plies).

In design, not only the final bearing failure is usable information, but bearing deformation is equally important. Therefore, it is recommended that the bearing stress variation as a function of hole deformation be documented, and that bearing stress values corresponding to the proportional limit, yield bearing (see definition below), and ultimate failure be recorded.

7.2.2 Definitions. The following definitions are relevant to this chapter.

Edge Distance Ratio -- The distance from the center of the bearing hole to the edge of the specimen in the direction of the applied load, divided by the diameter of the hole.

Bearing Area -- The diameter of the hole multiplied by the thickness of the specimen.

Bearing Load -- A compressive load on an interface.

Bearing Strain -- The ratio of the deformation of the bearing hole in the direction of the applied force, to the pin diameter.

Bearing Strength -- The maximum bearing stress which can be sustained.

Bearing Stress -- The applied load divided by the bearing area.

Proportional Limit Bearing Stress -- The bearing stress value corresponding to the deviation from linearity of the bearing stress versus hole elongation curve.

Yield Bearing Strength -- The bearing stress at the intersection of the bearing load-deformation curve with the tangent modulus drawn from the 4% elongation value of the original hole diameter.

Ultimate Bearing Strength -- The bearing stress corresponding to total failure of the test specimen.

7.2.3 Specimen design and testing. Using standard test equipment, the specimen shall be tested in the double shear arrangement shown in Figure 7.2.3. Test specimens shall conform to the dimensions and tolerances shown. Recommended b/D and e/D values are 6 and 3, respectively, and D = 1/4 inch (6.4 mm). A $[\pm 45/0/90]_{3s}$ lay-up, where n is selected so that the total thickness is approximately 0.125 inch (3.2 mm) is also recommended. The bearing hole shall be located as shown in Figure 7.2.3(b'). The hole shall be drilled undersized and reamed to size. The bolt shall be made of hardened steel (Rockwell Hardness of C60 - C64) and shall be 0.249 (+0.0000/-0.0010). Thus a close tolerance fit between the specimen and pin is required since a loose fit will tend to give lower

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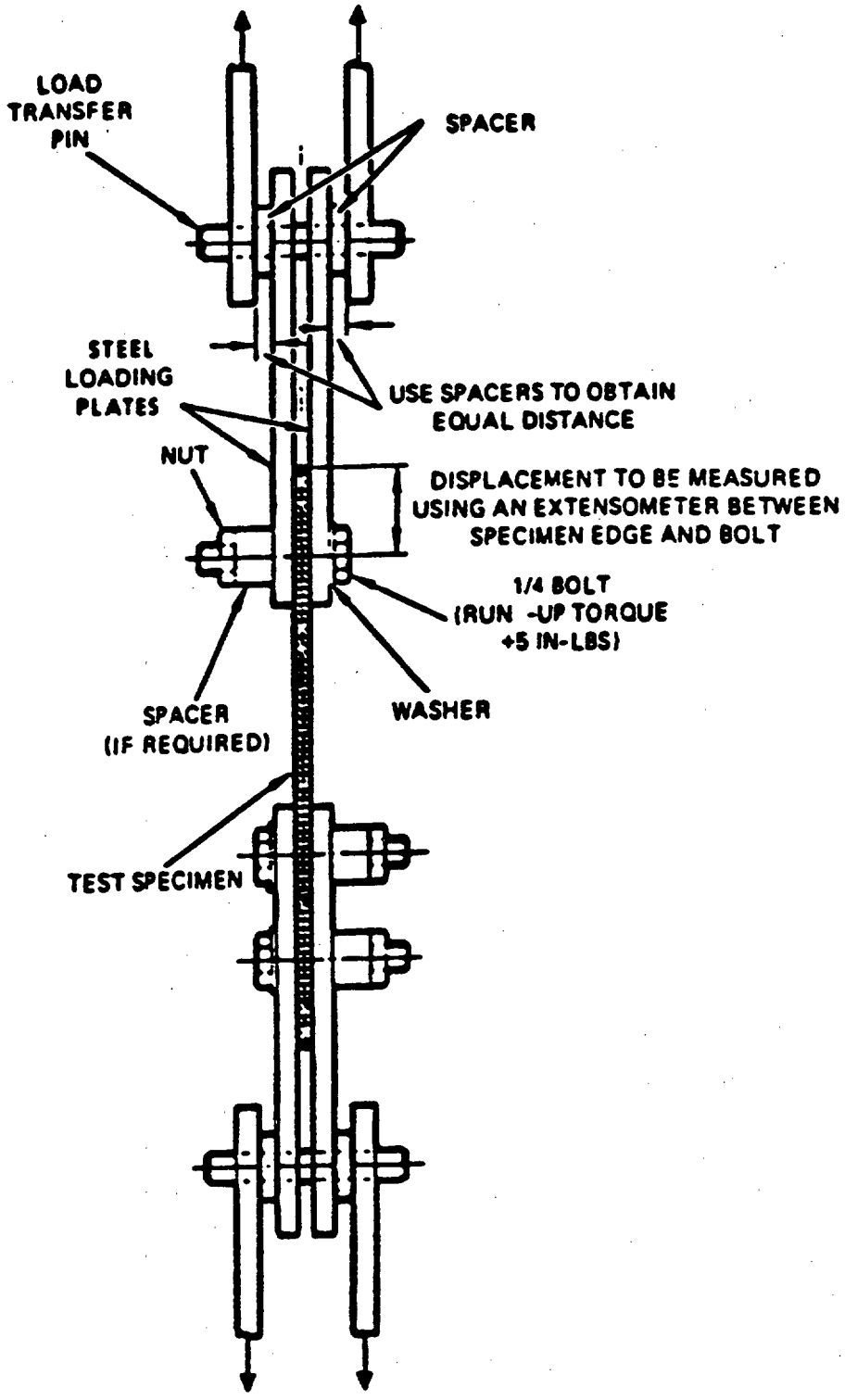


Figure 7.2.3. Test arrangement for bearing strength measurement.

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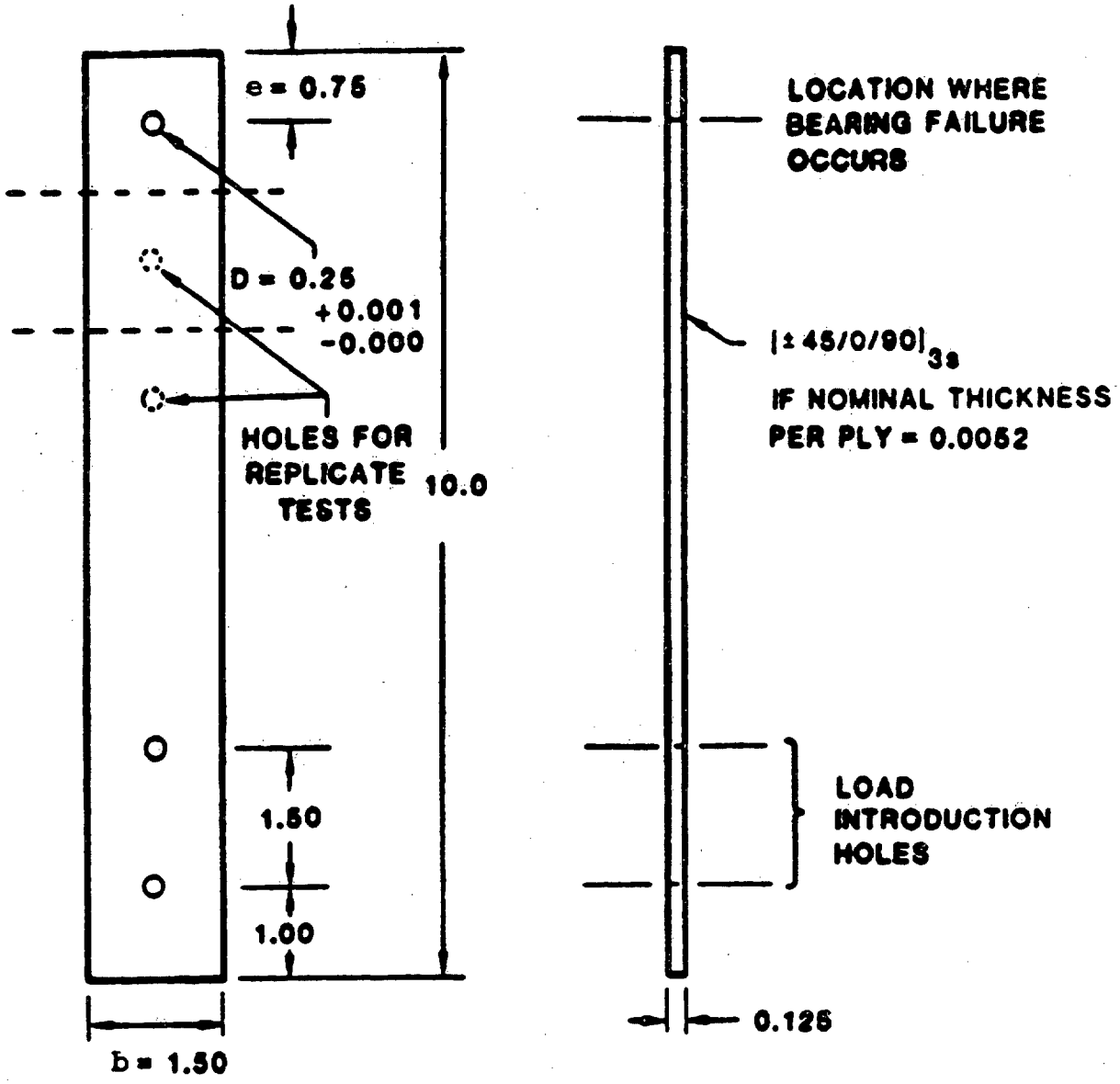


Figure 7.2.4. Test specimen geometry.

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results. Prior to assembly, clean the specimen, bolt, and adjacent areas of all foreign matter and contamination, especially lubricants.

7.2.4 Replication requirements. Five replicates shall be tested as a minimum. If three tests are conducted on one specimen, as shown in Figure 7.2.4, a total of six tests on two specimens will suffice.

7.2.5 Test conditions. Tests shall be conducted under as-fabricated or ambient dry or room temperature dry (RTD) conditions, and two hot, wet conditions. The hot, wet tests shall be conducted on specimens after they are preconditioned to near saturation level moisture contents, at temperatures that are dependent on the wet glass transition temperature for the material (wet T_g). Recommended test temperatures for hot, wet tests are T_g - 50F° (T_g - 28C°) and T_g - 20F° (T_g - 11C°).

Measurement of specimen dimensions, test procedures, etc., shall follow ASTM D 953 recommendations (Reference 7.2.5). The determination of yield bearing strength is shown for a typical curve in Figure 7.2.5.

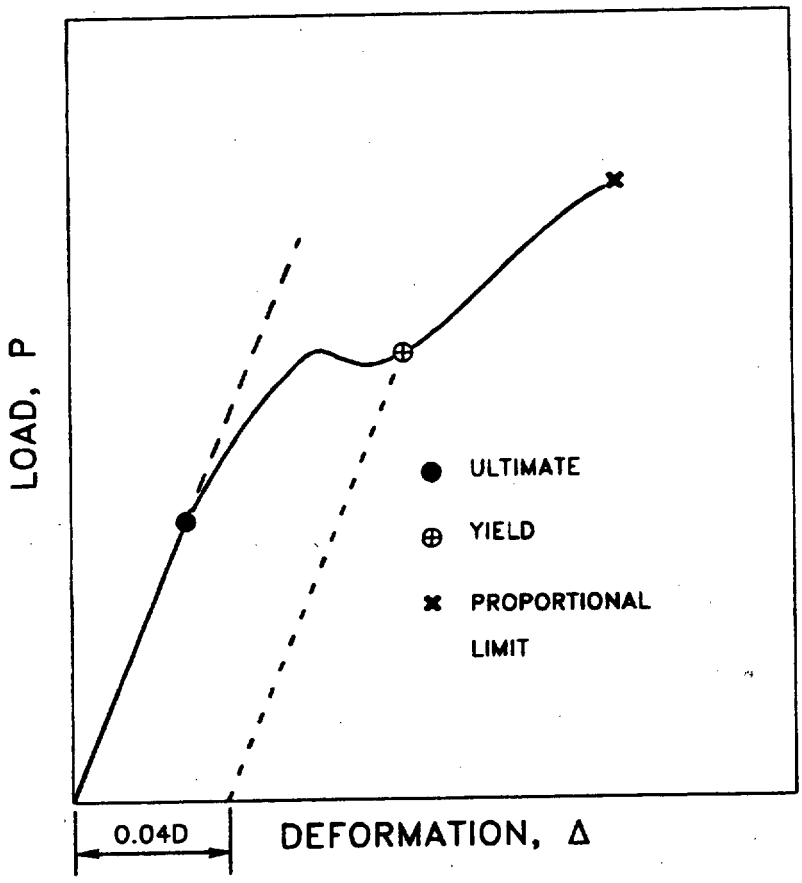


Figure 7.2.5 Illustration of a typical load-deformation curve.

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7.2.6 Bearing strength calculations. Bearing strength should be calculated using the following equation:

$$F^{br} = P/tD \qquad 7.2.6$$

where

F^{br} = bearing strength, psi (Pa)
P = bearing load, lb_f (N)
D = bearing hole diameter, in. (m)
t = specimen thickness, in. (m)

Superscripts bry and bru are commonly used to differentiate between yield and ultimate bearing strengths.

Bearing strength values are reported in MIL-HDBK-17 as typical or average values. Therefore, bearing strength values which are available for each specific condition should be analyzed to produce typical property values as described in Chapter 8.

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8.1 General. This section of Chapter 8 covers general information applicable to the sections that follow. Information specific to individual properties is to be found in the pertinent sections.

8.1.1 Introduction. The mechanical properties in MIL-HDBK-17 are used in the design of aerospace structures and elements. Thus it is exceedingly important that the values presented in MIL-HDBK-17 reflect as accurately as possible the actual properties of the materials covered.

The statistical procedures used in determining the materials properties values presented in this handbook are described in Sections 8.5 through 8.8. Section 8.2 presents the definitions for typical and B-basis values and defines the minimum number of batches and specimens to obtain a B-basis property value. Guidelines for the computation of individual mechanical properties are presented in Section 8.3. Section 8.4 describes the procedures and formats for presenting the material property data in Volume II.

8.1.2 Data documentation requirements. The purpose of this section is to outline data documentation requirements necessary to establish the validity of a manufacturer's material system physical, chemical, and mechanical property database. These requirements are necessary to establish justification for the inclusion of data in MIL-HDBK-17. The essence of documentation requirements is complete traceability and control of the the database development process from material production, through procurement, fabrication, machining, gaging, environmental conditioning, testing, data acquisition, data normalization, and final statistical interpretation.

The following information should be documented in any material property determination. The items checked should be included with data submitted to the Secretariat. All other information should be traceable and available to the Secretariat for validation of outliers. This list is based on the information necessary for lamina/laminate level mechanical property testing. The information required for other tests or material levels is similar. For instance, prepreg property testing would require the prepreg, reinforcement, and matrix material analysis information, as well as appropriate information on specimen preparation and testing procedure. For testing on laminae made from prepreg, redundant information on the matrix and reinforcement would not be required.

Material identification

X material identification

material procurement specification

Prepreg analysis

X ply manufacturer

X date of manufacturer

X material lot number

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X	commercial designation
X	material form
X	reinforcement areal weight
X	resin content
	moisture content
	high performance liquid chromatography results
	infrared spectroscopy results
	differential scanning calorimetry results
	rheological dynamic spectroscopy results

Reinforcement analysis

X	specification
	precursor type
X	commercial designation
X	manufacturer
X	date of manufacturer
X	lot number
X	surface treatment identification
X	specific gravity, measured
X	nominal filament count
X	fiber areal weight
X	twist

Matrix material analysis

X	specification
X	material chemical family
X	commercial designation

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X	manufacturer
X	lot number
	gel time and test conditions
X	specific gravity, measured
	dynamic mechanical analysis results for neat resin
	high performance liquid chromatography results
	infrared spectroscopy results
	differential scanning calorimetry results

Processing information

X	process specification
X	panel manufacturer
X	date of manufacture
	panel identification number
X	process cure cycle

Lamina analysis

X	form (panel, tube, etc.)
X	ply count
X	layup code
X	dimensions
	NDT conclusions
X	fiber volume
X	resin content (weight or volume)
X	density
	glass transition temperature (wet and dry, batch basis) and test method

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Specimen preparation

specimen lay-out data and numbering system

X specimen drawing

specimen acceptance criteria (machining specifications, etc.)

Mechanical testing

X test matrix performed

X test procedure (including deviations from standard procedures)

X specimen dimensions for each specimen

X environmental conditioning (drying) procedures

X moisture absorption data to equilibrium

X fastener type and torque-up conditions (if any)

X test temperature and moisture content at failure, environment

X continuous stress-strain data to failure

X failure mode identification and location

X all non-normalized (raw) data

Data analysis

(1) normalized data and procedures used

statistical procedures used

statistically-reduced data and parameters (by batch and pooled)

statistical hypothesis checks

---

(1) optional for submission to Secretariat

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8.1.3 Normalization. The values of fiber-dominated strength and stiffness properties (i.e., 0° tension, 0° compression) are dependent upon the volume fraction of fiber present in the laminate. In typical applications, the strength and stiffness have been found to vary linearly with percent fiber. Properties other than 0° tension and 0° compression strengths and stiffnesses may also be dependent on fiber volume fraction, but not in a well-defined way. For example, interlaminar shear properties are frequently found to remain fairly constant over a given fiber fraction range, but become lower when the fiber fraction falls above or below this range. Normalization is not considered appropriate for such properties.

In order to compare the properties of laminates of different fiber contents, it has become common practice to report values normalized to a given fiber volume fraction. This is accomplished by determining the fiber volume fraction in the cured test laminates by an appropriate method, and then multiplying the raw property values by the factor: (nominal fiber volume fraction)/(determined fiber volume fraction).

It may be shown that, for fiber/fabric of a given areal weight, the per ply thickness depends entirely on the fiber volume fraction. Hence, normalization may also be performed on a cured ply thickness basis. Although the true relationship is of the form, cured ply thickness = K/fiber fraction (where K is a constant), the departure from linearity in the 0.45 to 0.65 fiber volume fraction range is less than 0.1%. The normal thickness is chosen to correspond to a given fiber volume fraction or to the expected thickness under given process conditions.

The thickness method may be particularly useful for fiber reinforcements such as Kevlar(TM), where direct, repeatable methods for determining fiber volume of cured laminates are not well established or easily applied on a routine basis. The thickness method has the added advantage of allowing normalization of each individual test specimen, rather than applying a common fraction to all specimens from a given panel (based on panel fiber volume fraction). The latter approach might be less accurate since the thickness (and hence fiber fraction) will vary

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throughout the panel. The per ply thickness approach is not considered appropriate for test panels wound from rovings, however, since the per ply thickness is influenced greatly by the wind spacing and roving bandwidth, and may not consistently correspond to a fiber volume fraction.

The recommended data practices are as follows:

1. For 0° tension and 0° compression strengths and stiffnesses, normalized values shall be reported. Normalization may be by fiber volume fraction or thickness as appropriate for the material. The method of normalization shall be stated, and the normalizing factor shall be within the material process specification range. Raw data before normalization shall also be reported.

2. For properties other than those in (1.) above, raw data without normalization shall be reported. The fiber volume fraction and thickness for panels used shall be reported for information.

8.1.4 Symbols. The symbols which are used in Chapter 8, particularly those that are used in Sections 8.5 through 8.8, and not commonly used throughout this handbook are listed below, each with its definition and the section in which it is first used. These are primarily used to denote statistical variables.

SYMBOL	DEFINITION	SECTION
A_i	population	8.6.3.1
ADK	k-sample Anderson-Darling statistic	8.6.3.1
B	B-basis value	8.5.1.1
C.V.	critical value	8.6.2.1
e	error, residual	8.6.4.4
EV	equality of variances test statistic	8.6.3.2
F	F statistic	8.5.8
$F_{0.2}$	0.20 quantile of the F-distribution	8.5.4
IQ	informative quantile function	8.6.5.2
k	number of batches	8.5.4
k^*	total number of observations/root-mean-square number of observations per sample	8.5.4
k_B	one-sided tolerance limit factor	8.5.5.1
MNR	maximum normed residual test statistic	8.6.2.1

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8.4 Presentation of data. All data are presented in Volume II of MIL-HDBK-17B. This section describes how the data are presented and organized in that volume.

8.4.1 Properties and definitions. The properties and their definitions are found in the appropriate chapters of Volume I. Fiber properties and methods for obtaining them are discussed in Chapter 3. Resin properties are presented in Chapter 4. Methods for characterizing prepreg materials are discussed in Chapter 5 and properties and definitions for laminae and laminates are presented in Chapter 6. The statistical methods used in determining these properties are discussed in this chapter. Material system codes and laminate orientation codes are defined in Chapter I of Volume II.

8.4.2 Organization of data in handbook. The data in Volume II is divided into chapters of fiber properties, resin properties, and composite properties organized by fiber and then resin.

8.4.2.1 Fiber properties. Chapter 2 in Volume II provides data for fiber properties. Sections are included for different types of fiber, e.g. glass fibers and carbon fibers. In each section, the general characteristics of the type of fiber are given, as well as an index of suppliers, designations, and abbreviations. For each specific fiber, data are organized in the following manner. The X's in the subsection number are determined by the type of fiber and the specific fiber described.

2.X.X.1	Supplier and product data
2.X.X.2	Chemical and physical properties
2.X.X.2.1	Typical range of chemical constituents
2.X.X.2.2	Expected bound in physical properties
2.X.X.3	Thermal-mechanical properties
2.X.X.3.1	Stress-strain curves
2.X.X.3.2	Environmental effects

8.4.2.2 Matrix properties. Matrix or resin properties are included in Chapter 3 which is divided into sections according to the type of resin. For example, section 3.2 gives data for epoxies and section 3.3 provides data for polyester resins. The subsections for each specific resin are the same as those in Chapter 2 given above.

8.4.2.3 Composite properties. The remaining chapters of Volume II provide data for prepreg, lamina, laminate, and joint properties. There are individual chapters for each family of composites based on fiber type. For example, Chapter 4 describes glass fiber composites. Within each chapter, there is an index of suppliers, designations, and abbreviations. Sections are included based on the resin type used with the fiber described in the chapter, e.g. section 4.3 provides properties for epoxy-glass composites.

Properties are organized in the following manner for each specific composite:

X.X.X.1	Supplier and product data
.2	Prepreg chemical and physical properties
.2.1	Physical description
.2.2	Resin content
.2.3	Fiber content

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X.X.X.2.4 Volatiles content
.2.5 Moisture content
.2.6 Inorganic fillers and additives content
.2.7 Areal weight
.2.8 Tack and drape
.2.9 Resin flow
.2.10 Gel time
.3 Lamina chemical properties
.4 Lamina physical properties
.5 Lamina mechanical properties
.5.1 Data summaries
.5.2 Typical stress-strain curves
.6 Thermal properties
.7 Electrical properties
.8 Laminate thermal-mechanical properties
.8.1 Index of properties by lay-up
.8.2 Strength properties
a. Lay-up No. 1
b. Lay-up No. 2
.8.3 Thermal properties
.8.4 Electrical properties
.9 Joint thermal-mechanical properties
.9.1 Index of properties by joint and composite system
.9.2 Bearing strength
a. System No.1
b. System No. 2

8.4.3 Sample summary tables.

8.4.4 Sample graphs.

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Dispersion statistics:

Sample variance. - The sum of the squared deviations from the sample mean, divided by n - 1, where n denotes the sample size. The sample variance is expressed as:

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n - 1} \qquad 8.5.1.1(c)$$

or

$$S^2 = \frac{n \sum_{i=1}^{n} (X_i)^2 - \left(\sum_{i=1}^{n} X_i \right)^2}{n (n - 1)} \qquad 8.5.1.1(d)$$

Sample standard deviation. - The square root of the sample variance. The sample standard deviation is denoted by S.

Distribution terms:

Distribution. - A formula which gives the probability that a value will fall within prescribed limits.

Normal Distribution - A two parameter (μ,σ) family of probability distributions for which the probability that an observation will fall between a and b is given by the area under the curve

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \, e^{-(x-\mu)^2/2\sigma^2} \qquad 8.5.1.1(e)$$

between a and b. A normal distribution with parameters (μ,σ) has population mean μ and variance σ^2.

Lognormal Distribution - A probability distribution for which the probability that an observation selected at random from this population falls between a and b $(0 < a < b < \infty)$ is given by the area under the normal distribution between ℓog a and ℓog b.

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Two-Parameter Weibull Distribution - A probability distribution for which the probability that a randomly selected observation from this population lies between a and b ($0 < a < b < \infty$) is given by

$$e^{-(a/\alpha)^{\beta}} - e^{-(b/\alpha)^{\beta}} \qquad 8.5.1.1(f)$$

where α is called the scale parameter and β is called the shape parameter.

Probability function terms:

Cumulative Distribution Function - A function, usually denoted by F(x), which gives the probability that a random variable lies between any prescribed pair of numbers, that is

$$P\{ a < x \leq b \} = F(b) - F(a) \qquad 8.5.1.1(g)$$

Such functions are non-decreasing and satisfy

$$\lim_{x \to -\infty} F(x) = 0 \quad \text{and} \quad \lim_{x \to \infty} F(x) = 1$$

The cumulative distribution function is related to the probability density function by

$$f(x) = \frac{d}{dx} F(x) \qquad 8.5.1.1(h)$$

provided that F(x) is differentiable.

F-distribution - A probability distribution which is employed in the analysis of variance, regression analysis, and tests for equality of variance. Tables of this distribution are readily available.

Failure Rate - see Hazard Rate.

Hazard Rate - If F(x) is the cumulative distribution function, and f(x) is the corresponding probability density function, then the hazard rate, h(x), is defined by

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$$h(x) = \frac{f(x)}{1 - F(x)} \qquad 8.5.1.1(i)$$

The above formula has the following intuitive interpretation. h(x) is proportional to the probability of "instantaneous failure" at time x, given that a failure has not occurred by time x, where F(x) is the probability that a failure occurs on or before time x.

Probability Density Function - A function $f(x) \geq 0$ for all x with

$$\int_{-\infty}^{\infty} f(x)\, dx = 1 \qquad 8.5.1.1(j)$$

The probability density function determines the cumulative distribution function F(x) by

$$F(x) = \int_{-\infty}^{x} f(t)\, dt \qquad 8.5.1.1(k)$$

Note that the limits $(-\infty, \infty)$ may be conventional; that is, a random variable with a restricted range, for example, the exponential random variable which assumes only positive values satisfies the definition by defining its probability density function as

$$f(x) = \begin{cases} 0 & x \leq 0 \\ e^{-x} & x > 0 \end{cases} \qquad 8.5.1.1(l)$$

The probability density function is used to calculate probabilities as follows:

$$P\{ a \leq x \leq b \} = \int_{a}^{b} f(x)\, dx \qquad 8.5.1.1(m)$$

8.5.1.2 Sample size requirements. The sample size requirements for publishing a B-basis value in MIL-HDBK-17 depend upon the status of material and process specifications for the subject composite material system. If no standard specifications have been developed for the material system, a minimum of five batches of material and six specimens per batch are required for each fabricator that wishes to publish a B-basis value in MIL-HDBK-17.

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In the event that standard material and process specifications have been developed for a composite material system, several fabricators may jointly develop data to publish a single B-basis value in MIL-HDBK-17. In this case, a minimum of three fabricators, three batches of material per fabricator, and six specimens per batch are required.

The sample specimens should be obtained from randomly selected areas of randomly selected sheets of material fabricated to the material specification for which the B-basis value is desired. The number of specimens from each batch should be as nearly equal as possible with the largest batch size being no more than one and a half times the smallest batch size.

It should be noted that the magnitude of a B-basis value is a function of the amount of data obtained, the number of batches represented, and the uniformity of the batches produced. In general, B-basis values will tend to increase if the number of batches is increased or the number of specimens per batch is increased. B-basis values will also tend to increase if the batch-to-batch variability or the within-batch variability is reduced.

If the minimum data requirements specified above cannot be met, these guidelines may be used to calculate "preliminary" B-basis values. These data will be included as interim data in MIL-HDBK-17 but the B-basis values will not be reported in the handbook until the minimum data requirements are met.

8.5.2 Definition of computational procedures. The procedure used to determine a B-basis value depends on the characteristics of the data. The step-by-step procedure for selecting the appropriate computational method is illustrated by the flowchart in Figure 8.5.2. Details for the specific computational methods are provided in later sections.

If the sample is made up of data from several different batches, each of the batches should be screened for outliers by the method described in Section 8.5.3.1. One should then determine whether or not the data from the different batches should be analyzed as one sample by employing the k-sample Anderson-Darling test as specified in Section 8.5.3.2. If the k-sample test does not

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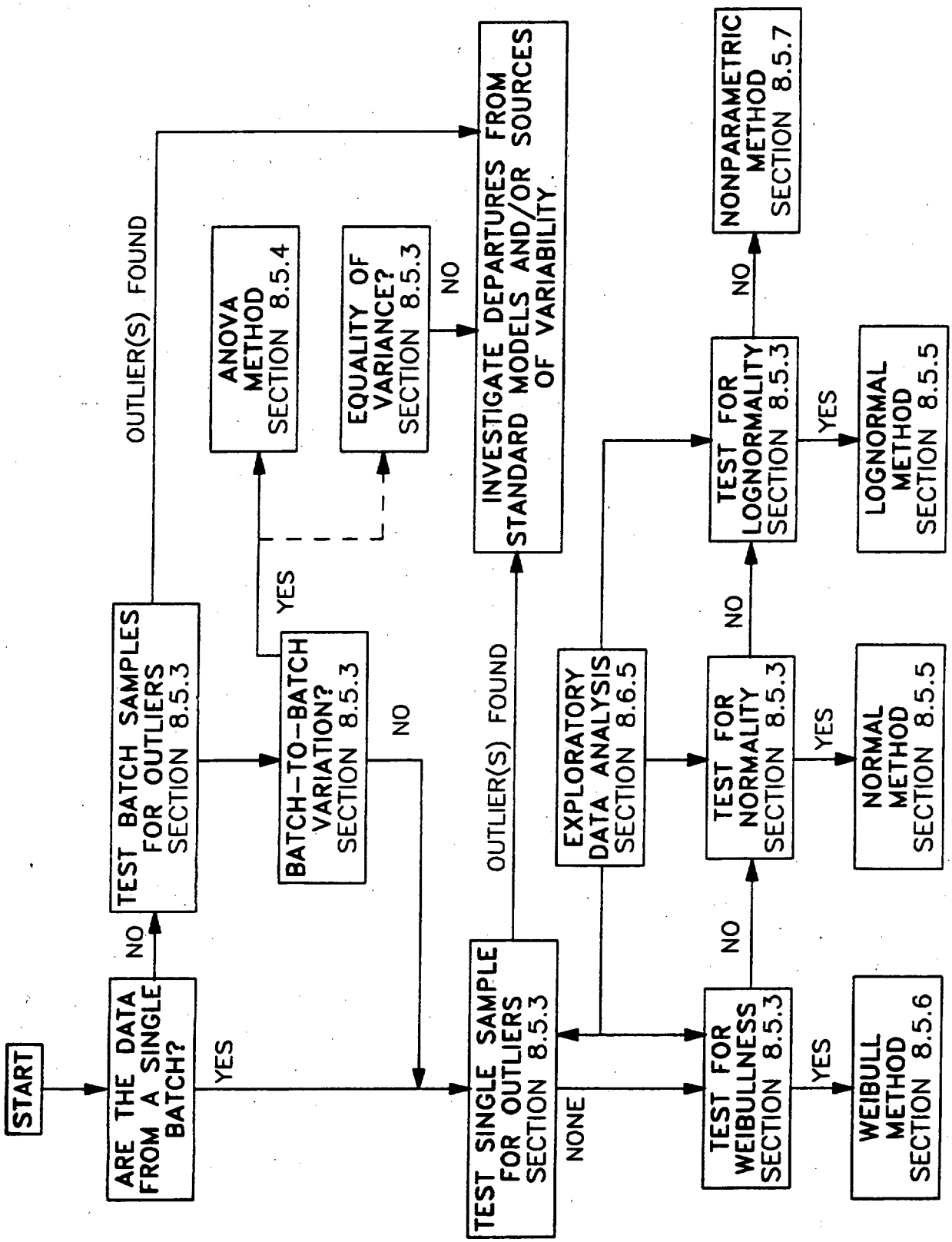


Figure 8.5.2 Flow chart illustrating computational procedures for B-basis material property values.

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batches should be analyzed as one sample by employing the k-sample Anderson-Darling test as specified in Section 8.5.3.2. If the k-sample test does not reject the hypothesis that the batches are from the same population, the batches should be combined and the data should be analyzed as a single sample.

If the hypothesis that the batches are from the same population is rejected, indicating significant batch-to-batch variability, then the B-basis value should be computed by the ANOVA method of Section 8.5.4. A diagnostic test for the equality of variance assumption is discussed in Section 8.5.3.3.

If the sample represents a single batch of material, or several batches which are to be analyzed as a single sample, the single sample of data should be screened for outliers as discussed in Section 8.5.3.1. Note that if only a single batch is represented, then the B-basis value should only be considered a "preliminary" value. One should then determine which, if any, distributional form will be assumed in calculating the B-basis value. The use of goodness-of-fit tests for the two-parameter Weibull, normal, and lognormal distributions is described in Section 8.5.3.4. These tests establish the degree to which the underlying population may be fitted by one of these distributions. The two-parameter Weibull distribution is tested first. If it does not adequately fit the data, then the normal distribution is tested. If neither of these provide an adequate fit, then the lognormal distribution is tested. Exploratory data analysis (EDA) techniques, described in Section 8.6.5, can provide graphical illustrations of the distribution of the sample in support of the goodness-of-fit tests.

If one of the distributions mentioned above provides a good fit to the data, the B-basis value is computed by the corresponding two-parameter Weibull, normal, or lognormal method as described in Sections 8.5.5 or 8.5.6. If none of the distributions adequately fit the data, and the sample contains 29 or more observations, the nonparametric method described in Section 8.5.7.1 is used to compute the B-basis value. If none of the distributions fit the data adequately, and the sample size is less than 29, the Hanson-Koopmans method, described in Section 8.5.7.2, may be employed. This method, however, does not produce an

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approved B-basis value due to the small sample size. These nonparametric procedures should only be used when batch-to-batch variation is negligible.

For strength measurements measured in thousands of pounds per square inch (ksi), data are generally measured to the nearest tenth of a ksi. B-basis values are generally rounded to ksi units by the following rule. Values with a fractional part less than 0.75 are rounded down to the next lowest integer. B-basis values with a fractional part greater than or equal to 0.75 are rounded up to the next highest integer.

8.5.3 Detecting outliers and characterizing the population. Before a B-basis value is computed, the data must be screened for erroneous values and the structure of the population from which the data were sampled must be investigated in order to determine the appropriate method. The methods discussed in this section address these two issues. These methods include the maximum normed residual outlier test (Section 8.5.3.1), the k-sample Anderson-Darling test (Section 8.5.3.2), an equality of variance test (Section 8.5.3.3), and goodness-of-fit tests for the two-parameter Weibull, normal, and lognormal distributions (Section 8.5.3.4).

8.5.3.1 Detecting outliers. An observation is said to be an outlier if it is an observation that has been recorded in error. For example, an erroneous observation could be the result of clerical error or the incorrect setting of environmental conditions during testing. Since outliers may have a substantial influence on the statistical analysis of the data, it is mandatory that the data be screened for outliers prior to the calculation of a B-basis value.

Initially, the data is subjected to the statistical outlier procedure discussed in Section 8.6.2, using a significance level of 0.05, to identify potential outliers. The test screens for outliers in a single sample. When different batches are represented, the data from each batch should be tested separately. Since the outlier detection procedure may miss erroneous observations, the sample should also be visually inspected for observations suspected of being outliers.

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If the statistical outlier procedure or the visual inspection identifies an observation as a potential outlier, the observation will be investigated to ascertain if a cause for the outlying value can be ascribed. If it can be corrected, as is often the case for clerical errors, this should be done. If it is determined to be an erroneous observation and it cannot be corrected, as might be the case with improper setting of environmental conditions during testing, then it is discarded. If no cause can be determined for the outlying value, it must be retained in the data set.

When errors in data collection and recording are discovered, it should be determined whether similar errors occurred for other data points that were not identified as potential outliers and these values should also be corrected or discarded. If any observations are corrected or discarded, both the statistical outlier procedure and the visual inspection should be repeated.

8.5.3.2 The k-sample Anderson-Darling test. The k-sample Anderson-Darling test is used to test the hypothesis that the mechanical property data from different batches are independent random samples from the same population. The procedure for performing this test is presented in Section 8.6.3.1. If the test statistic is less than the critical value from Equation 8.6.3.1(j), then the batches should be treated as a single sample and one should proceed to the goodness-of-fit tests for investigating the form of the distribution discussed in Section 8.5.3.4.

If the test statistic is greater than or equal to the critical value, it is concluded that the batches are not identically distributed and the test for equality of variance should be performed. This test is discussed in the next section.

8.5.3.3 Equality of variance test. The ANOVA method (Section 8.5.4) is derived under the assumption that the variances within each batch are equal. The recommended procedure for testing this assumption is presented in Section 8.6.3.2. If the test does not reject the hypothesis that the variances are equal, then the ANOVA method should be used to compute the B-basis value.

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If the test rejects the equality of variance assumption, then there is currently no approved method for computing a B-basis value. In this case, it is recommended that an investigation of the reason for the non-equal variances be carried out. This may reveal problems in the generation of the data or in the fabrication of the material.

8.5.3.4 Investigating the form of the distribution. The method employed in calculating the B-basis value for a single sample depends on the distributional form which is assumed for the data. The most frequently used parametric procedures involve the two-parameter Weibull, normal, or lognormal distributions. Section 8.6.4 contains procedures for performing a goodness-of-fit test for each of these distributions. These goodness-of-fit tests yield an observed significance level (OSL) for each of the distributions listed above. The OSL from each of the tests measures the probability of observing an Anderson-Darling statistic as extreme as the value calculated assuming that the given distribution is the correct one.

As shown in Figure 8.5.2, the test for the Weibull model (Section 8.6.4.3) is to be performed first. If the Weibull model cannot be demonstrated to adequately fit the data, then the normal and lognormal (Section 8.6.4.2) tests are performed in succession. This preference for the Weibull distribution is based on several factors:

1) Theoretical considerations suggest that the Weibull distribution is appropriate for the strength distribution of brittle materials such as composite fibers. For an elementary account of this result, see Reference 8.5.3.4(a).

2) The "Chain-of-Bundles" model for the strength of two- and three-dimensional unidirectional composites suggests that the Weibull model is appropriate for the strength distribution of such composites. This result is stated in References 8.5.3.4(b) and (c).

3) Empirical support for the Weibull model is provided by the model's good fit to data.

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To determine which method should be used in calculating the B-basis value, the goodness-of-fit tests should be employed in the following manner. The two-parameter Weibull distribution is tested first. If the OSL is larger than 0.05, the Weibull distribution method of Section 8.5.6 should be used to compute the B-basis value. If the OSL is less than or equal to 0.05, the hypothesis that the data is a sample from a normal distribution is tested next. If this test for normality gives an OSL larger than 0.05, the normal distribution method of Section 8.5.5 should be used to compute the B-basis value. If the normal OSL is less than or equal to 0.05, the lognormal distribution is tested next. If this results in an OSL greater than 0.05, the lognormal distribution method of Section 8.5.5 should be used to compute the B-basis value. If this OSL is less than or equal to 0.05, it is concluded that the data is not a sample from any of these families of parametric distributions, and the nonparametric procedure of Section 8.5.7.1 should be used to compute the B-basis value, provided that the sample size is greater than or equal to 29. If the sample size is less than 29, then the Hanson-Koopmans method, described in Section 8.5.7.2, may be employed.

The exploratory data analysis (EDA) techniques of Section 8.6.5 may be used to graphically display the distribution of measurements in the sample and to provide graphical evidence for the conclusions obtained through the goodness-of-fit tests.

8.5.4 Normal analysis procedure in the presence of batch-to-batch variation. When failure data from different batches are to be analyzed, it is possible that the variation from batch to batch will be significant. This will prevent pooling the data and using the methods of Section 8.5.5, 8.5.6 or 8.5.7 to compute B-basis values from a single sample. This section contains a method, referred to as the ANOVA method, for computing B-basis values when there is significant batch-to-batch variation. It is a modification of a method suggested in Reference 8.5.4(a). The original paper on this topic (Reference 8.5.4(b)) is of some historical interest but should not be used for computation.

The method is based on the one-way analysis of variance (ANOVA) random effects model discussed in Section 8.6.7. The assumptions are that

(1) the data from each batch are normally distributed,
(2) the within-batch variance is the same from batch to batch, and

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(3) the batch means are normally distributed.

There is currently no test available for the first assumption. Simulation studies, however, suggest that moderate violation of this assumption does not have an adverse effect on the properties of the ANOVA model. The second assumption should be validated by performing the test described in Section 8.6.3.2. This test is currently recommended as a diagnostic, since extensive simulation suggests that violation of this assumption will likely result in conservatism, although non-conservatism can arise in some situations. The third assumption is not testable except in the case of many (20 or more) batches.

In the analysis, all batches are treated the same, and no distinction is made between batches from different fabricators. For this reason, the sample size requirements in Section 8.5.2 require that data from at least three fabricators be included in the sample, with at least three batches from each fabricator. This attempts to insure that any variability among fabricators, while not explicitly estimated, will be included in the analysis via its contribution to the batch-to-batch variability.

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In what follows, X_{ij} denotes the jth data value in the ith batch, n_i denotes the sample size of batch i, k is the number of batches, and N is the total number of data values ($N = n_1 + n_2 + \ldots + n_k$). If the entire set of data values is not available, means, standard deviations, and sample sizes for the batches are sufficient to perform the calculations.

Computational Procedure. The first step in computing the B-basis value is to compute estimates of the between-batch and within-batch variances (S_b^2 and S_e^2, respectively). Computation of the intermediate values MSB and MSE, is discussed in Section 8.6.7.2. The within-batch variance (σ_e^2) is estimated by

$$S_e^2 = MSE \qquad 8.5.4(a)$$

The between-batch variance (σ_b^2) is estimated by

$$S_b^2 = (MSB - MSE)/n' \qquad 8.5.4(b)$$

where

$$n' = (N-n^*)/(k-1) \qquad 8.5.4(c)$$

and

$$n^* = \sum_{i=1}^{k} \frac{n_i^2}{N} \qquad 8.5.4(d)$$

If (MSB-MSE)/n' is negative, then set S_b^2 equal to 0. For more information on estimation procedures for the one-way random effects model, see Reference 8.5.4(c).

The quantity n' may be thought of as the "effective sample size" when the number of specimens in each batch is not the same. When the number of specimens per batch is the same, n' is equal to the common sample size, n.

The ratio of the between-batch to within-batch variances ($R = \sigma_b^2/\sigma_e^2$) must also be estimated. An upper bound for this ratio is estimated by the quantity $\hat{R}$, which is computed as

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$$\hat{R} = [(MSB/MSE)/F_{0.2} - 1]/n' \qquad 8.5.4(e)$$

where $F_{0.2}$ is the 0.20 quantile of an F distribution with k - 1 numerator degrees of freedom and N - k denominator degrees of freedom (Reference 8.5.4(d)). $\hat{R}$ is an upper 80 percent confidence bound on the true ratio. If $\hat{R}$ is negative, then set $\hat{R}$ equal to zero. Table 8.8.4 contains the $F_{0.2}$ values required in Equation 8.5.4(e).

The formula for the B-basis value is then

$$B = \bar{X} - T(S_b^2 + S_e^2)^{0.5} \qquad 8.5.4(f)$$

where $\bar{X}$ is the overall mean (average of all N measurements) defined in Equations 8.6.7.1.1(a) and 8.6.7.1.2(a), and T is a tolerance limit factor defined below and tabulated in Tables 8.8.6.

The tolerance limit factor T is defined in statistical terms as

$$T = \{(n^*\hat{R}+1)/[N(\hat{R}+1)]\}^{0.5}\, t_{\gamma,0.95}(\delta) \qquad 8.5.4(g)$$

where $t_{\gamma,0.95}(\delta)$ is the 0.95 quantile of the non-central t distribution with noncentrality parameter

$$\delta = 1.282\,[N(\hat{R} + 1)/(n^*\hat{R} + 1)]^{0.5} \qquad 8.5.4(h)$$

and degrees of freedom approximated by

$$\gamma = \frac{(\hat{R} + 1)^2}{\frac{(\hat{R} + \frac{1}{n^*})^2}{k^* - 1} + \frac{(\frac{n^*-1}{n^*})^2}{k^*(n^* - 1)}} \qquad 8.5.4(i)$$

where $k^* = N/n^*$. T may be computed from the following approximate formula:

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$$T = 1.282 + 1.282\,(t_{\gamma,0.95}/\delta) + 1.80\,(1/\gamma) - 1.85\,(1/\gamma^2) + 0.567\,(\delta/\gamma) + 5.24\,(\delta/\gamma^2) - 1.08\,(\delta^2/\gamma^2) + 0.0166\,(\delta^3/\gamma^2) + 7.79\,(1/\gamma^4) \qquad 8.5.4(j)$$

(Note: $t_{\gamma,0.95}$ is the 0.95 quantile of the central t-distribution with γ degrees of freedom.) Tolerance limit factors are tabulated in Table 8.8.6 for various values of n^*, k^* and $\hat{R}$.

Equations 8.5.4(e) through 8.5.4(i) were developed under the assumption of equal sample sizes. The performance of the method is not affected substantially by unequal sample sizes as long as the largest batch size is no more than one and one half times the smallest batch size. If sample sizes are equal, then $k^* = k$ and $n^* = n$.

8.5.5 Normal analysis procedure for a single population. This procedure should be used when a B-basis value is to be computed from a single sample of failure data that is assumed to be normally distributed. See Section 8.5.1.1 for a definition of the normal distribution. Section 8.5.5.1 contains the computational procedure and Section 8.5.5.2 describes how this procedure is applied to lognormal data. Further information on this procedure may be found in Reference 8.5.5.

8.5.5.1 Computational procedure. In order to compute B-basis values for a normal population, it is first necessary to obtain estimates of the population mean and standard deviation. The sample mean and standard deviation, $\bar{X}$ and S, are computed from the sample of available measurements using the equations presented in Section 8.5.1.1.

The B-basis value is then calculated by use of the following formula:

$$B = \bar{X} - k_B S \qquad 8.5.5.1$$

where

$\bar{X}$ = the sample mean based on n observations
S = the sample standard deviation
k_B = the one-sided tolerance-limit factor from Table 8.8.1.

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A numerical approximation to the k_B values is given in Equation 8.8.1.

8.5.5.2 Lognormal procedure. If the sample is assumed to follow a lognormal distribution, then the computational procedure presented in Section 8.5.5.1 is used to calculate the B-basis value. However, the calculations are performed using the logarithms of the data rather than the original observations. Either the natural or the common logarithm may be used. The computed B-basis value must then be transformed back to the original units by applying the inverse of the log transformation which was used. See Section 8.5.1.1 for definition of the lognormal distribution.

If, for example, the natural logarithm is used to transform the data, the B-basis value would be calculated as follows:

$$B = \exp\left(\bar{X}_L - k_B S_L\right) \qquad 8.5.5.2(a)$$

where

$$\bar{X}_L = \frac{\sum_{i=1}^{n} \ell n(X_i)}{n} \qquad 8.5.5.2(b)$$

$$S_L = \left[\frac{\sum_{i=1}^{n} [\ell n(X_i) - \bar{X}_L]^2}{n - 1}\right]^{0.5} \qquad 8.5.5.2(c)$$

and k_B is the one-sided tolerance-limit factor from Table 8.8.1.

8.5.6 Two-parameter Weibull analysis procedure for a single population. This procedure should be used when a B-basis value is to be computed from a single sample of failure data which is assumed to follow a two-parameter Weibull distribution. See Section 8.5.1.1 for a definition of the two-parameter Weibull distribution.

In order to compute a B-basis value for a two-parameter Weibull population, it is first necessary to obtain estimates of the population shape and scale

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parameters. Section 8.5.6.1 contains a step-by-step procedure for calculating maximum likelihood estimates of these parameters. The computational procedure for calculating B-basis values using these estimates and the tolerance limit factors in Table 8.8.2 is outlined in Section 8.5.6.2. For further information on these procedures, see Reference 8.5.6.

8.5.6.1 Estimating the shape and scale parameters of a Weibull distribution. In order to calculate the B-basis value for a random sample from the two-parameter Weibull distribution, estimates of the shape and scale parameters are needed. The procedure described below is the maximum likelihood method, and the estimates obtained are known as the maximum likelihood estimates. The shape parameter estimate is denoted by $\hat{\beta}$, and the scale parameter by $\hat{\alpha}$. The estimates are obtained by solving the pair of likelihood equations given below for $\hat{\beta}$ and $\hat{\alpha}$.

$$\hat{\alpha}\,\hat{\beta}\,n - \frac{\hat{\beta}}{\hat{\alpha}^{\hat{\beta}-1}} \sum_{i=1}^{n} X_i^{\hat{\beta}} = 0 \qquad 8.5.6.1(a)$$

$$\frac{n}{\hat{\beta}} - n\,\ell n\,\hat{\alpha} + \sum_{i=1}^{n} \ell n\,X_i - \sum_{i=1}^{n} \left(\frac{X_i}{\hat{\alpha}} \right)^{\hat{\beta}} (\ell n\,X_i - \ell n\,\hat{\alpha}) = 0 \qquad 8.5.6.1(b)$$

Equation 8.5.6.1(a) can be rewritten as

$$\hat{\alpha} = \left(\frac{\sum_{i=1}^{n} X_i^{\hat{\beta}}}{n} \right)^{1/\hat{\beta}} \qquad 8.5.6.1(c)$$

By substituting Equation 8.5.6.1(c) into Equation 8.5.6.1(b), the following equation is obtained.

$$\frac{n}{\hat{\beta}} + \sum_{i=1}^{n} \ell n\,X_i - \frac{n}{\sum_{i=1}^{n} X_i^{\hat{\beta}}} \sum_{i=1}^{n} X_i^{\hat{\beta}}\,\ell n\,X_i = 0 \qquad 8.5.6.1(d)$$

Since Equation 8.5.6.1(d) depends only on the data $X_1, X_2, \ldots, X_n$, it can be solved numerically, and the solution of $\hat{\beta}$ that is obtained is substituted into Equation 8.5.6.1(c) to obtain $\hat{\alpha}$, the estimate of α.

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Figure 8.5.6 shows FORTRAN source code for three routines which compute the estimates of $\hat{\alpha}$ and $\hat{\beta}$ by the method described above. WBLEST is a subroutine which returns the estimates of the parameters, $\hat{\beta}$ and $\hat{\alpha}$. FNALPH is a function which calculates the estimate of the scale parameter, $\hat{\alpha}$. GFUNCT is a function which evaluates Equation 8.5.6.1(d). Arguments to WBLEST are

X = a vector of length NOBS containing the data (input),
NOBS = the number of data values, N (input),
BETA = estimate of the shape parameter (output),
ALPHA = estimate of the scale parameter (output).

The algorithm by which the FORTRAN code computes the estimates is described in the following paragraph.

Equation 8.5.6.1(d) is a monotonically increasing continuous function of $\hat{\beta}$. Designate the left-hand side of Equation 8.5.6.1(d) divided by n as $G(\hat{\beta})$ and obtain a solution for $\hat{\beta}$ by the following iterative procedure. Let S_y denote the standard deviation of $Y_1,\ldots,Y_n$ where $Y_i = \ell n(X_i)$ for $i = 1,\ldots,n$. Calculate $I = 1.28/S_y$ as an initial guess at the solution and calculate G(I). If G(I) > 0, then find the smallest positive integer k such that $G(I/2^k) < 0$ and let $L = I/2^k$ and $H = I/2^{k-1}$. If G(I) < 0, then find the smallest positive integer k such that $G(2^k I) > 0$ and let $L = 2^{k-1} I$ and $H = 2^k I$. In either case, the interval (L,H) contains the solution to $G(\hat{\beta}) = 0$. Now calculate G(M) where M = (L + H)/2. If G(M) = 0, then the solution is $\hat{\beta} = M$. If G(M) > 0, then let H = M. If G(M) < 0 then let L = M. The new interval (L,H) still contains the solution to $G(\hat{\beta}) = 0$ but is only half as long as the old interval. Calculate a new M-value and begin the process of interval halving again. The process is repeated until $H-L < 2I/10^6$. The solution to $G(\hat{\beta}) = 0$ is then taken to be M = (L + H)/2. The solution is in error by at most $I/10^6$.

8.5.6.2 Computational procedure. With a suitable sample of size n and the population parameter estimates discussed above at hand, the computation of a B-basis value is carried out by use of the formula

$$B = \hat{Q} \exp\left(-V/(\hat{\beta}\sqrt{n})\right) \qquad 8.5.6.2(a)$$

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where

$$\hat{Q} = \hat{\alpha}\ (0.10536)^{1/\hat{\beta}} \qquad 8.5.6.2(b)$$

and V is the value in Table 8.8.2 corresponding to a sample of size n. A numerical approximation to the V values is given in Equation 8.8.2(b).

8.5.7 Nonparametric analysis procedure for a single population. This procedure should be used when a material property value is to be computed from failure data and the form of the distribution of the population is unknown. The distribution should be considered unknown if tests indicate significant departure from the two-parameter Weibull, normal, and lognormal distributions.

8.5.7.1 Sample size greater than 28. First, order the sample so that the observations are in increasing order. Then, using Table 8.8.3, determine the r value corresponding to the sample size n. For sample sizes lying between tabulated values, use the r value associated with the largest tabulated sample size which is smaller than the actual n. The B-basis value is the rth lowest observation in the ordered sample. For example, in a sample of size n = 30, the lowest (r = 1) observation is the B-basis value. As another example, in a sample of size n = 600, the 48th ordered observation (r = 48) is the B-basis value. For sample sizes less than 29, a B-basis value acceptable for MIL-HDBK-17 cannot be calculated with this procedure; the Hanson-Koopmans procedure can provide a B-basis value for these small sample sizes.

The value of r for any sample size n can be computed directly by using the following relationships. If $q_{0.10}$ is the 0.10 quantile of the underlying population distribution, then the number of observations in the sample less than $q_{0.10}$ is a binomial random variable with n trials and probability of success p = 0.10. Letting $X_{(r)}$ denote the rth ordered sample value, the statement $X_{(r)} \leq q_{0.10}$ is true if and only if there are r or more failure values in the sample which are less than or equal to $q_{0.10}$. Using this relationship, the B-basis value may be found as the rth ordered sample failure value, $X_{(r)}$, where $r \geq 1$ is the largest integer solution to

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$$\sum_{w=r}^{n} \binom{n}{w} (0.10)^{w} (0.90)^{n-w} \geq 0.95 \qquad 8.5.7.1(a)$$

where

$$\binom{n}{w} = \frac{n!}{w!(n-w)!} \qquad 8.5.7.1(b)$$

Table 8.8.3 was generated from this equation. A numerical approximation to the tabulated r values as a function of n is also given in Section 8.8. Further information on this procedure may be found in Reference 8.5.5.

8.5.7.2 The Hanson-Koopmans method. Although it can be demonstrated that for $n < 29$ there is no useful nonparametric method, the following procedure, due to D. L. Hanson and L. H. Koopmans (Reference 8.5.7.2), may be a useful method for obtaining a B-basis value for sample sizes not exceeding 28. This procedure requires the assumption that the observations are a random sample from a probability density function with a log-concave cumulative distribution function. That is, if the probability function is f(x), and the cumulative distribution function is F(x), then

$$h(x) = - \log F(x) \qquad 8.5.7.2(a)$$

is a concave function of x, for $x \leq 0$. This assumption is satisfied by a large class of probability distribution functions. Consequently the Hanson-Koopmans method does not involve strong parametric assumptions, though it is not completely nonparametric.

There is substantial empirical evidence that suggests the composite strength data that will be encountered by users of this handbook satisfies the assumptions underlying the Hanson-Koopmans method and, consequently, this procedure can be recommended for use when n is less than 29. However, in view of the required assumption, this is not an unconditional recommendation.

The B-basis value is calculated as follows. Order the observations from smallest to largest. Let $X_{(1)}$ be the smallest observation, $X_{(2)}$ the second smallest,

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and so on. In general, we denote the t-th smallest value by $X_{(t)}$. Then the B-basis value is given by

$$B = X_{(t)} - k(X_{(t)} - X_{(1)}) \qquad 8.5.7.2(b)$$

where t and k depend on n and are determined from Table 8.8.11. This equation for the B-basis value should not be employed if $X_{(t)} = X_{(1)}$.

Example. Assume that 25 observations have been obtained. Order these in increasing value. From the table, for n = 25, t = 11; therefore, we record the first and eleventh values. For purposes of illustration, let the smallest value be 36.64 and the eleventh value be 47.50. Then the B-basis value is

$$B = 47.50 - 1.087(47.50 - 36.64) = 35.70$$

8.5.8 Linear regression analysis procedure. In order to determine that the average of a property for which a B-basis value is being calculated varies linearly with some specimen property such as thickness, number of plies, or percent fiber volume, the linear regression analysis procedure may be appropriate. Linear regression analysis assumes that the property to be regressed is normally distributed about the true regression line. The steps for obtaining B-basis values using linear regression analysis are as follows.

(1) Fit a regression equation of the form

$$y = a + bx \qquad 8.5.8$$

to the data as described in Section 8.6.6 where y is the material property to be regressed and x is the specimen property such as thickness, number of plies, or percent fiber volume. Obtain the estimates a and b and the root mean square error of the regression (s_y). Also obtain the F statistic for testing the significance of the regression (F).

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(2) If F indicates that the regression is insignificant, one of the other appropriate analysis techniques, as described in Sections 8.5.5, 8.5.6, or 8.5.7, should be used. Otherwise, proceed to step 3.

(3) Evaluate the validity of the normality assumption for y by performing the Anderson-Darling test presented in Section 8.6.4.4. If the hypothesis of normality is rejected, there is currently no approved method for computing B-basis values using linear regression analysis. Otherwise, proceed to step 4.

(4) At selected values of x, determine the B-basis value for the property according to the following procedure.

8.5.8.1 Computational procedure. To compute a B-basis value at $x = x_o$, use Equation 8.5.8.1(a).

$$B = (a+bx_o) - k'_B s_y \qquad 8.5.8.1(a)$$

where s_y is the root mean square error for the regression. The tolerance limit factor is

$$k'_B = [(1+\Delta)/n]^{0.5}\, t_{\gamma,0.95}(\delta) \qquad 8.5.8.1(b)$$

where $t_{\gamma,0.95}(\delta)$ is the 0.95 percentile of the noncentral t-distribution with $\gamma = n - 2$ degrees of freedom, noncentrality parameter

$$\delta = 1.282/[(1+\Delta)/n]^{0.5} \qquad 8.5.8.1(c)$$

where

$$\Delta = \frac{(x_o - \bar{x})^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2 / n} \qquad 8.5.8.1(d)$$

and

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$$\bar{x} = \sum_{i=1}^{k} \frac{x_i}{n} \qquad 8.5.8.1(e)$$

is the average of the x values in the sample. The quantity represents the relative distance of x_o from the average x value.

The following approximation to k_B may be used when n is greater than or equal to 10 and $0 \leq \Delta \leq 10$.

$$k'_B \approx 1.282 + \exp\{0.595 - 0.508\ \ell n(n) + 4.62/n + (0.486 - 0.986/n)\ \ell n(1.82 + \Delta)\} \qquad 8.5.8.1(f)$$

Under the conditions stated above, the approximation is accurate to within ±1.0% (relative magnitude of error, see Section 8.8).

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8.6 Statistical properties.

8.6.1 Introduction. This section includes a number of statistical tools for use in the analysis of data for MIL-HDBK-17. These tools are intended to supplement the basic procedures described in Section 8.5.

Various statistical techniques are employed in the analysis of material-property data. This section presents brief descriptions of procedures that are used most frequently in this application. More detailed descriptions of these and other statistical techniques and tables can be found in a number of workbooks and texts, some of which are given as references.

When procedures other than those described below are employed in the preparation of data proposals, they should be described adequately in the proposal to allow a proper evaluation of their validity and pertinence.

8.6.2 Outlier detection procedure. An observation is said to be an outlier if it is an observation that has been recorded in error. Since outliers may have a substantial influence on the statistical analysis of a set of data, it is often desirable that a data set be screened for outliers prior to data analysis. The maximum normed residual method for detecting potential outliers is described below. For more information on this test procedure, see References 8.6.2(a) and 8.6.2(b).

8.6.2.1 The maximum normed residual method. The maximum normed residual (MNR) test is a screening procedure for identifying outliers in a single set of data. It involves an examination of the residuals (signed distances from the mean divided by the standard deviation) to determine whether or not they are unusually large. The test assumes that the non-erroneous data values are independently and identically normally distributed. Also, the outlier procedure searches for one outlier at a time and, therefore, the significance level pertains to a single decision.

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The first step is to compute the normed residuals and the MNR statistic. If X_1, X_2, ... X_n denote the data values in the sample of size n, the normed residuals are defined as:

$$r_i = \frac{X_i - \bar{X}}{S}, \quad i = 1, 2, \ldots, n \qquad 8.6.2.1(a)$$

where $\bar{X}$ and S are the sample mean and sample standard deviation, respectively, defined in Section 8.5.1.1. The MNR statistic is the maximum of the absolute values of the normed residuals:

$$MNR = \max_i (| r_i |) \qquad 8.6.2.1(b)$$

Next, the MNR statistic is compared to the critical value corresponding to the sample size n from either Table 8.8.9 or 8.8.10. The critical values for this test are computed from the following formula:

$$C.V. = \frac{n - 1}{\sqrt{n}} \left(\frac{t^2}{n - 2 + t^2} \right)^{0.5} \qquad 8.6.2.1(c)$$

where t is the $[1 - \alpha/(2n)]$ quantile of the t distribution with $n - 2$ degrees of freedom and α represents the significance level of the test. Critical values for sample sizes 3 through 200 and a significance level of $\alpha = 0.01$ are tabulated in Table 8.8.9. Critical values for the same sample sizes and a significance level of $\alpha = 0.05$ are presented in Table 8.8.10. Values for other sample sizes or other significance levels may be computed using the above formula.

If MNR is smaller than the critical value, then no outliers are detected in the sample. If MNR is larger than the critical value, then the data value associated with the largest absolute normed residual is declared to be a potential outlier.

If an outlier is detected, the procedure is repeated on the reduced set of data (eliminating the potential outlier). This process is repeated until no potential outliers are detected. Note that in the second iteration, the mean, standard deviation, and critical value are computed using a sample size of n - 1. In the third iteration, the sample size will be n - 2, and so on.

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8.6.3 Subpopulation compatibility tests. This section contains tests for determining the compatibility of two or more populations. The k-sample Anderson-Darling test in Section 8.6.3.1 is used to test the hypothesis that the populations from which two or more independent samples were taken are identically distributed. The equality of variance test in Section 8.6.3.2 is used to test the hypothesis that the variances of the populations from which two or more independent random samples were taken are equal.

8.6.3.1 The k-sample Anderson-Darling test. The k-sample Anderson-Darling statistic tests the hypothesis that the populations from which two or more independent random samples were drawn are identical. The test can be applied to determine whether two or more products differ with regard to their strength distributions. The test is a nonparametric statistical procedure and only requires the assumption that the samples are independent random samples from their respective populations.

Consider the populations A_1, A_2,..., A_k. Let X_{11}, X_{12}, ..., X_{1n_1}, denote a sample of n_1 data points from population A_1, let X_{21} X_{22}, ..., X_{2n_2}, denote a sample of the n_2 data points from population A_2, and so forth. Furthermore, let $N = n_1 + n_2 + \ldots + n_k$ represent the total number of data points in the combined samples.

Let L denote the total number of distinct data points in the combined samples and $Z_{(1)}$, $Z_{(2)}$,..., $Z_{(L)}$ denote the distinct values in the combined data set ordered from least to greatest. The k-sample Anderson-Darling statistic is defined by

$$ADK = \frac{N-1}{N^2(k-1)} \sum_{i=1}^{k} \left[\frac{1}{n_i} \sum_{j=1}^{L} h_j \frac{(NF_{ij} - n_i H_j)^2}{H_j (N-H_j) - Nh_j/4} \right] \qquad 8.6.3.1(a)$$

where h_j = the number of values in the combined samples equal to $Z_{(j)}$

H_j = the number of values in the combined samples less than $Z_{(j)}$ plus one half the number of values in the combined samples equal to $Z_{(j)}$, and

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F_{ij} = the number of values in sample corresponding to population A_i which are less than $Z_{(j)}$ plus one half the number of values in the sample corresponding to population A_i which are equal to $Z_{(j)}$.

Under the hypothesis of no difference in the sampled populations, the mean and variance of ADK are approximately 1 and

$$\sigma_N^2 = \mathrm{Var(ADK)} = \frac{aN^3 + bN^2 + cN + d}{(N-1)\ (N-2)\ (N-3)\ (k-1)^2} \qquad 8.6.3.1(b)$$

with

$$a = (4g-6)(k-1) + (10-6g)S \qquad 8.6.3.1(c)$$

$$b = (2g-4)k^2 + 8Tk + (2g-14T-4)S - 8T + 4g - 6 \qquad 8.6.3.1(d)$$

$$c = (6T+2g-2)k^2 + (4T-4g+6)k + (2T-6)S + 4T \qquad 8.6.3.1(e)$$

$$d = (2T+6)k^2 - 4Tk \qquad 8.6.3.1(f)$$

where

$$S = \sum_{i=1}^{k} \frac{1}{n_i} \qquad 8.6.3.1(g)$$

$$T = \sum_{i=1}^{N-1} \frac{1}{i} \qquad 8.6.3.1(h)$$

and

$$g = \sum_{i=1}^{N-2} \sum_{j=i+1}^{N-1} \frac{1}{(N-i)j} \qquad 8.6.3.1(i)$$

If

$$\mathrm{ADK} \geq 1 + \sigma_N \left[1.645 + \frac{0.678}{(k-1)^{0.5}} - \frac{0.362}{k-1} \right] \qquad 8.6.3.1(j)$$

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one may conclude (with a five percent risk of being in error) that the samples were drawn from different populations. Otherwise, the hypothesis that the samples were selected from identical populations is not rejected. Table 8.8.7 contains the critical values (Equation 8.6.3.1(j)) for the case of equal sample sizes.

For more information on the k-sample Anderson-Darling test, see Reference 8.6.3.1.

8.6.3.2 Test for equality of variances. This section describes a test suggested by Lehmann (Reference 8.6.3.2) for determining whether two or more estimates of variance differ significantly. With k independent random samples, the sample variance for the ith sample is denoted by S_i^2 and has $\gamma_i = n_i - 1$ degrees of freedom, where n_i is the size of the ith sample.

Letting $Z_i = \ell n\ (S_i^2)$, the test statistic is computed as

$$EV = \frac{1}{2} \sum_{i=1}^{k} \gamma_i Z_i^2 - \frac{1}{2} (N-k)\ \bar{Z}^2 \qquad 8.6.3.2(a)$$

where

$$\bar{Z} = \sum_{i=1}^{k} \gamma_i\ Z_i/(N-k) \qquad 8.6.3.2(b)$$

is the weighted average of the Z_i values, and N is the total sample size. This statistic is compared to the 0.95th quantile of a chi-square distribution with k - 1 degrees of freedom, values of which are tabulated in Table 8.8.8. (A numerical approximation to the tabulated values is given in Section 8.8.8.) If the test statistic is greater than or equal to the tabulated value, then the variances are declared to be significantly different. If the statistic is less than the tabulated value, then the hypothesis of equality of variance is not rejected.

8.6.4 Goodness-of-fit tests. The computational procedure selected to establish B-basis values by statistical techniques is dependent upon the underlying distribution of strength measurements. The most frequently used parametric procedures involve the use of the normal, lognormal, or two-parameter Weibull

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distributions. This section contains methods that are used to establish the degree to which a population may be fitted by one of these distributions.

Four goodness-of-fit test procedures are described below. The purpose of each is to indicate whether an initial distributional assumption should be rejected. The methods presented are based on the Anderson-Darling goodness-of-fit test statistic, which is particularly sensitive to discrepancies in the tail regions. The tests of fit to the normal, lognormal, and Weibull distributions for a single sample are presented in Sections 8.6.4.1, 8.6.4.2 and 8.6.4.3, respectively. A test for normality in a regression setting is discussed in Section 8.6.4.4.

An observed significance level (OSL) may be computed for each test. The OSL is the probability of obtaining a value of the test statistic at least as large as that obtained if the hypothesis that the data are actually from the distribution being tested is true. If the OSL is less than or equal to 0.05, the hypothesis is rejected (with at most a five percent risk of being in error) and one proceeds as if the data are not from the distribution being tested.

In what follows, unless otherwise noted, the sample size is denoted by n, the sample observations by $X_1, \ldots, X_n$, and the sample observations ordered from least to greatest by $X_{(1)}, \ldots, X_{(n)}$.

8.6.4.1 Anderson-Darling test for normality. The Anderson-Darling test for normality is used to test the hypothesis that the population from which a given sample of data was drawn is normally distributed. See Section 8.5.1.1 for a definition of the normal distribution.

The test compares the cumulative normal distribution function that fits the observed data best with the cumulative distribution function of the observed data. Let

$$Z_{(i)} = (X_{(i)} - \bar{X}) / S \qquad i = 1, \ldots, n \qquad 8.6.4.1(a)$$

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where $X_{(i)}$ is the ith smallest sample observation, $\bar{X}$ is the sample average, and S is the sample standard deviation. Equations for computing these sample statistics are presented in Section 8.5.1.1.

The Anderson-Darling test statistic is

$$AD = \sum_{i=1}^{n} \frac{1-2i}{n} \{\ell n[F_o(Z_{(i)})] + \ell n [1-F_o(Z_{(n+1-i)})]\} - n \qquad 8.6.4.1(b)$$

where F_o is the standard normal distribution function. The standard normal distribution function $F_o(x)$ represents the area to the left of the value x under the standard normal density curve. (See Equation 8.5.1.1(e).)

The observed significance level (OSL) is calculated as

$$OSL = 1/\{1+\exp[-0.48 + 0.78\ \ell n(AD^*) + 4.58(AD^*)]\} \qquad 8.6.4.1(c)$$

where

$$AD^* = (1 + 4/n - 25/n^2)AD \qquad 8.6.4.1(d)$$

The OSL is a measure of the goodness-of-fit of a normal distribution to the data. Specifically, the OSL measures the probability of observing an Anderson-Darling statistic at least as extreme as the value calculated if a normal distribution is in fact the underlying distribution.

If OSL $\leq$ 0.05, one may conclude (at five percent risk of being in error) that the population from which the sample was drawn is not normally distributed. Otherwise, the hypothesis that the population is normally distributed is not rejected. For further information on this test procedure, see Reference 8.5.6.

8.6.4.2 Anderson-Darling test for lognormality. The lognormal distribution is a positively skewed parametric distribution which is related to the normal distribution. If a variable is lognormally distributed, then the logarithm of that variable is normally distributed. Thus, if a sample of data has a lognormal distribution, taking the logarithm of the observations will enable the analyst to

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use normal analysis procedures. The natural (base e) logarithm is used in MIL-HDBK-17.

In order to test the goodness-of-fit of the lognormal distribution, take the log of the data and perform the Anderson-Darling test for normality from Section 8.6.4.1. Using the natural logarithm, let

$$Z_{(i)} = \frac{\ell n\ (X_{(i)}) - \bar{X}_L}{S_L} \qquad 8.6.4.2(a)$$

where $X_{(i)}$ is the ith smallest sample observation and X_L and S_L are the mean and standard deviation of the $\ell n\ (X_i)$ values defined by

$$\bar{X}_L = \frac{\sum_{i=1}^{n} \ell n\ (X_i)}{n} \qquad 8.6.4.2(b)$$

$$S_L = \left[\frac{\sum_{i=1}^{n} (\ell n\ (X_i) - \bar{X}_L)^2}{n - 1} \right]^{0.5} \qquad 8.6.4.2(c)$$

The Anderson-Darling statistic is computed using equation 8.6.4.1(b) and the observed significance level (OSL) is computed as in Equation 8.6.4.1(c).

8.6.4.3 Anderson-Darling test for Weibullness. The Anderson-Darling test for two-parameter Weibullness is used to test the hypothesis that the population from which a given sample of data was drawn has a two-parameter Weibull distribution. See Section 8.5.1.1 for a definition of the two-parameter Weibull distribution.

The test compares the cumulative Weibull distribution function that fits the observed data best with the cumulative distribution function of the observed data. The first step is to compute the parameter estimates $\hat{\alpha}$ and $\hat{\beta}$. A procedure for computing these estimates is given in Section 8.5.6.1. Then let

$$Z_{(i)} = [X_{(i)}/\hat{\alpha}]^{\hat{\beta}} \qquad i = 1, \ldots, n \qquad 8.6.4.3(a)$$

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The Anderson-Darling test statistic is then

$$AD = \sum_{i=1}^{n} \frac{1 - 2i}{n} \{\ell n [1 - \exp(-Z_{(i)})] - Z_{(n + 1 - i)}\} - n \qquad 8.6.4.3(b)$$

and the observed significance level is calculated as

$$OSL = 1/\{1+\exp [-0.10 + 1.24\ \ell n(AD^*) + 4.48\ (AD^*)]\} \qquad 8.6.4.3(c)$$

where

$$AD^* = (1 + 0.2/\sqrt{n})AD \qquad 8.6.4.3(d)$$

The OSL is a measure of the goodness-of-fit of the two-parameter Weibull distribution to the data. Specifically, the OSL measures the probability of observing an Anderson-Darling statistic at least as extreme as the value calculated if the two-parameter Weibull distribution is in fact the underlying distribution.

8.6.4.4 Testing for normality in a regression setting. This section contains a test for normality of the dependent variable in a regression setting. The test is an Anderson-Darling test for normality performed on the residuals

$$e_i = y_i - (a + b\ x_i) \qquad i = 1,\ldots,n \qquad 8.6.4.4(a)$$

from the regression (see Section 8.6.6) assuming equality of variance of the residuals over the range of the independent variable. Letting

$$Z_{(i)} = e_{(i)} / S_y \qquad i = 1,\ldots,n \qquad 8.6.4.4(b)$$

where $e_{(1)},\ldots,e_{(n)}$ are the ordered residuals from smallest to largest and S_y is the root-mean-square error of the regression defined in Section 8.6.6. The Anderson-Darling test statistic is computed using Equation 8.6.4.1(b), and the observed significance level (OSL) is computed as in Equation 8.6.4.1(c).

If OSL $\leq$ 0.05, conclude that the dependent variable is not normally distributed. Otherwise, the assumption that the dependent variable is normally distributed is

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Similar calculations for the remaining batches identify no additional potential outliers in this set of data. Visual inspection of the data also does not identify any additional potential outliers.

No cause could be determined for the potential outlier, 86.4, and it is, therefore, retained in the data set.

Problem 1 - Step 2. The k-sample Anderson-Darling test described in Section 8.6.3.1 will be employed next to determine whether or not the data from the nine batches should be combined. The first step is to order the pooled sample. Table 8.7.1 lists the 47 sorted, distinct values in the column labeled $Z_{(j)}$. The remaining columns show the h_j, H_j, and F_{1j} values used in calculating the terms in the statistic arising from the first batch (i=1). The column labeled f_{1j} shows the number of times that $Z_{(j)}$ is represented in the first batch and is used in calculating F_{1j}. From these numbers, it follows that

$$\frac{1}{n_i}\sum_{j=1}^{L} h_j \frac{(NF_{ij} - n_iH_j)^2}{H_j(N - H_j) - Nh_j/4} = \frac{1}{7}\sum_{j=1}^{47} h_j \frac{(53F_{1j} - 7H_j)^2}{H_j(53 - H_j) - 53h_j/4} = 363.33$$

When these calculations are repeated for the remaining eight batches, the k-sample Anderson-Darling statistic is computed as

$$ADK = \frac{N-1}{N^2(k-1)}\sum_{i=1}^{k}\left[\frac{1}{n_i}\sum_{j=1}^{L} h_j \frac{(NF_{ij} - n_iH_j)^2}{H_j(N - H_j) - Nh_j/4}\right]$$

$$= \frac{52}{53^2(9 - 1)}\,(363.33 + \ldots + 324.11)$$

$$= 2.67$$

The computed value of the statistic is compared to the critical value from Equation 8.6.3.1(j), which is 1.44. Since the computed value of 2.67 is greater than the critical value of 1.44, the hypothesis that the populations from which these groups were drawn are identical is rejected.

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TABLE 8.7.1 Illustration of k-sample Anderson-Darling statistic calculations for the first batch (i=1).

j	$Z_{(j)}$	h_j	H_j	f_{1j}	F_{1j}
1	61.0	1	0.5	1	0.5
2	61.2	1	1.5	0	1.0
3	61.3	1	2.5	1	1.5
4	61.6	1	3.5	1	2.5
5	62.7	1	4.5	0	3.0
6	63.4	1	5.5	1	3.5
7	63.6	2	7.0	1	4.5
8	64.0	1	8.5	0	5.0
9	64.1	1	9.5	0	5.0
10	64.6	1	10.5	0	5.0
11	64.7	2	12.0	0	5.0
12	64.8	1	13.5	0	5.0
13	65.4	3	15.5	2	6.0
14	65.5	1	17.5	0	7.0
15	65.6	1	18.5	0	7.0
16	66.4	1	19.5	0	7.0
17	66.8	2	21.0	0	7.0
18	67.4	1	22.5	0	7.0
19	67.9	1	23.5	0	7.0
20	68.0	1	24.5	0	7.0
21	68.5	1	25.5	0	7.0
22	68.4	1	26.5	0	7.0
23	68.6	1	27.5	0	7.0
24	68.8	1	28.5	0	7.0
25	68.9	1	29.5	0	7.0
26	69.3	1	30.5	0	7.0
27	69.4	1	31.5	0	7.0
28	69.9	1	32.5	0	7.0
29	70.0	1	33.5	0	7.0
30	70.2	1	34.5	0	7.0
31	70.7	1	35.5	0	7.0
32	71.1	1	36.5	0	7.0
33	71.5	1	37.5	0	7.0
34	73.3	1	38.5	0	7.0
35	74.0	1	39.5	0	7.0
36	74.2	1	40.5	0	7.0
37	74.6	2	42.0	0	7.0
38	74.9	1	43.5	0	7.0
39	76.3	1	44.5	0	7.0
40	78.5	1	45.5	0	7.0
41	79.2	1	46.5	0	7.0
42	79.7	1	47.5	0	7.0
43	80.6	1	48.5	0	7.0
44	81.1	1	49.5	0	7.0
45	82.3	1	50.5	0	7.0
46	86.4	1	51.5	0	7.0
47	87.0	1	52.5	0	7.0

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Problem 1 - Step 3. The equality of variance test described in Section 8.6.3.2 is used to determine if the within-batch variances are significantly different. Relevant sample statistics for the nine batches are given in the table below.

Batch	n_i	S_i^2	Y_i	$Z_i = \ell n\ (S_i^2)$
1	7	3.46	6	1.24
2	5	3.61	4	1.28
3	5	5.34	4	1.68
4	7	6.25	6	1.83
5	6	22.37	5	3.11
6	5	19.01	4	2.95
7	5	65.29	4	4.18
8	6	33.76	5	3.52
9	7	44.09	6	3.79

The test statistic is then calculated as follows.

$$EV = \frac{1}{2} \sum_{i=1}^{k} \gamma_i Z_i^2 - \frac{1}{2} (n-k)\bar{Z}^2$$

$$= \frac{1}{2} \sum_{i=1}^{9} \gamma_i Z_i - \frac{1}{2} (53-9)(2.61)$$

$$= 24.63$$

The critical chi-square value from Table 8.8.8 corresponding to k - 1 = 8 degrees of freedom is 15.51. Since the test statistic is greater than the critical value, the hypothesis that the variances are equal is rejected. Since the equality of variance test is a diagnostic test, a B-basis value can be calculated using the ANOVA method. However, a nonconservative B-basis value can result in some instances. Potential problems with consistency in fabrication or processing of the different batches should be examined.

Problem 1 - Step 4. The ANOVA method of Section 8.5.4 is used to calculate a B-basis value. (See Problem 2 for the details of the computations using the ANOVA method.) The B-basis value is calculated as

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$$B = \bar{X} - T\,(S_b^2 + S_e^2)^{0.5}$$

$$= 69.78 - 2.077\,(21.93 + 22.18)^{0.5}$$

$$= 55.99$$

For presentation in MIL-HDBK-17, this B-basis value would be rounded to 56.

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8.7.2 Problem 2 - Outlier detection and ANOVA method. The data set for this problem consists of tensile strength measurements from nine batches of material. This problem illustrates the outlier detection procedure and the calculation of B-basis values by the ANOVA method.

Problem 2 - Step 1. The first step is to screen the data for outliers using the MNR procedure described in Sections 8.5.3.1 and 8.6.2. The screening procedure is performed separately on each batch. (See Problem 1 for details of the outlier detection computations.)

A single observation, the value 109.9 in the ninth batch, is identified as a potential outlier. Assume that the potential outlier was the result of a clerical error, and its value should have been 69.6. The remaining calculations assume that this correction has been made. Upon repeating the entire outlier screening procedure, no potential outliers or extreme outliers are identified.

Problem 2 - Step 2. The k-sample Anderson-Darling test statistic for the data is ADK = 1.81. (See Problem 1 for a detailed computation of the k-sample statistic.) Since 1.81 is greater than the critical value of 1.44, conclude that the batches are not from the same population.

Problem 2 - Step 3. The test for equality of variances between the batches gives a calculated statistic of EV = 5.60, which is less than the critical value of 15.51 from Table 8.8.8. (See Problem 1 for a detailed computation of this test statistic.) Thus, one should proceed as if the data are normally distributed and the variances are equal, and use the ANOVA method of Section 8.5.4 to calculate a B-basis value.

Problem 2 - Step 4. Summary statistics for the data are given in the table below. Preliminary ANOVA calculations covered in Section 8.6.7.1.2 are:

$$\bar{X} = \sum_{i=1}^{k} n_i \bar{X}_i / N = [7(65.60) + \ldots + 7(71.80)]/53 = 68.84$$

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Batch	n_i	$\bar{X}_i$	S_i
1	7	65.60	2.99
2	5	66.32	2.33
3	5	67.84	2.84
4	7	67.33	4.17
5	6	66.93	2.45
6	5	71.64	4.03
7	5	71.10	3.33
8	6	71.52	1.96
9	7	71.80	3.88

$$SSB = \sum_{i=1}^{k} n_i \bar{X}_i^2 - N\bar{X}^2 = [7(65.60)^2 + \ldots + 7(71.80)^2] - 53(68.84)^2$$

$$= 317.255$$

$$SSE = \sum_{i=1}^{k} (n_i - 1)S_i^2 = [6(2.99)^2 + \ldots + 6(3.88)^2] = 460.680$$

$$MSB = SSB/(k-1) = 317.255/(9-1) = 39.66$$

$$MSE = SSE/(N-k) = 460.680/(53-9) = 10.47$$

Preliminary calculations covered in Section 8.5.4 are:

$$n^* = \sum_{i=1}^{k} n_i^2/N = (7^2 + \ldots + 7^2)/53 = 6.0189$$

$$n' = (N - n^*)/(k-1) = (53-6.0189)/(9-1) = 5.8726$$

$$k^* = N/n^* = 53/6.0189 = 8.8056$$

$$S_e^2 = MSE = 10.47$$

$$S_b^2 = (MSB - MSE)/n' = (39.66-10.47)/5.8726 = 4.97$$

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and

$$F_{0.2,k-1,N-k} = F_{0.2,8,44} = 0.56 \quad \text{(from Table 8.8.4)}$$

The upper bound for the ratio of between-batch to within-batch variability is

$$\hat{R} = [(MSB/MSE)/ F_{0.2} - 1]/n'$$

$$= [(39.66/10.47)/0.56 - 1]/5.8726$$

$$= 0.98$$

The tolerance limit factor T should be obtained from the approximation in Section 8.5.4 rather than from Table 8.8.4 because $\hat{R}$, k^* and n^* are not integers. The noncentrality parameter is calculated as

$$\delta = 1.282 \left[\frac{N(\hat{R} + 1)}{n^*\hat{R} + 1} \right]^{0.5} = 1.282 \left[\frac{53\ (0.98 + 1)}{(6.0189)\ (0.98) + 1} \right]^{0.5} = 5.15$$

The degrees of freedom parameter is calculated as

$$\gamma = \frac{(\hat{R} + 1)^2}{\frac{(\hat{R} + \frac{1}{n^*})^2}{k^* - 1} + \frac{(\frac{n^*-1}{n^*})^2}{k^*(n^* - 1)}}$$

$$= \frac{(0.98 + 1)^2}{\frac{\left[0.98 + \frac{1}{6.0189} \right]^2}{8.8056 - 1} + \frac{\left[\frac{6.0189 - 1}{6.0189} \right]^2}{8.8056(6.0189-1)}}$$

$$\gamma = 21.40$$

and

$$t_{\gamma,0.95} = t_{21.4,0.95} = 1.72$$

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The tolerance limit factor is calculated as follows:

$$T = 1.282 + 1.282\,(t_{\gamma,0.95}/\delta) + 1.80\,(1/\gamma) - 1.85\,(1/\gamma^2) + 0.567\,(\delta/\gamma) + 5.24\,(\delta/\gamma^2) - 1.08\,(\delta^2/\gamma^2) + 0.0166\,(\delta^3/\gamma^2) + 7.79\,(1/\gamma^4)$$

$$= 1.282 + 1.282\,(1.72/5.15) + 1.80\,(1/21.4) - 1.85\,(1/21.4^2) + 0.567\,(5.15/21.4) + 5.24\,(5.15/21.4^2) - 1.08\,(5.15^2/21.4^2) + 0.0166\,(5.15^3/21.4^2) + 7.79\,(1/21.4^4)$$

$$= 1.94$$

Thus, a B-basis value is calculated as

$$B = \bar{X} - T\,(S_b^2 + S_e^2)^{0.5} = 68.84 - 1.94\,(4.97 + 10.47)^{0.5} = 61.22$$

For presentation in MIL-HDBK-17, this B-basis value would be rounded to 61.

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8.7.3 Problem 3 - Weibull distribution. The data set for this problem consists of tensile strength measurements from nine batches of material. This problem illustrates the two-parameter Weibull goodness-of-fit test and the calculation of B-basis values by the Weibull method.

Problem 3 - Step 1. There are no detected outliers in this set of data. (See Problem 1 for details of the outlier detection computations.)

Problem 3 - Step 2. The k-sample Anderson-Darling test statistic is ADK = 0.99. (See Problem 1 for details of the computation of the k-sample statistic.) Since this is less than the critical value of 1.44, conclude that the data from the batches may be combined into a single sample.

Problem 3 - Step 3. The maximum normed residual (MNR) test is performed on the pooled data. No potential outliers are detected in the pooled data. (see Problem 1 - Step 1 for details of the outlier detection procedure.)

Problem 3 - Step 4. In order to perform the two-parameter Weibull goodness-of-fit test described in Section 8.6.4.3, it is necessary to compute estimates of the scale and shape parameters, $\hat{\alpha}$ and $\hat{\beta}$. A procedure for doing this is described in Section 8.5.6.1. The geometric mean of the data is computed as

$$\bar{X}_G = \exp\left[\frac{1}{N}\sum_{i=1}^{N} \ell n(X_i)\right]$$

$$= \exp\left[\frac{1}{53}\sum_{i=1}^{53} \ell n(X_i)\right]$$

$$= 67.501$$

For a given value of $\hat{\beta}$, $\hat{\alpha}$ is calculated as

$$\hat{\alpha} = \bar{X}_G \left[\frac{1}{N}\sum_{i=1}^{N}\left(\frac{X_i}{\bar{X}_G}\right)^{\hat{\beta}}\right]^{\frac{1}{\hat{\beta}}}$$

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$$\alpha = (67.501)\left[\frac{1}{53}\sum_{i=1}^{53}\left(\frac{X_i}{67.501}\right)^{\hat{\beta}}\right]^{\frac{1}{\hat{\beta}}}$$

In order to calculate $\hat{\beta}$, define the function $G(\hat{\beta})$ by

$$G(\hat{\beta}) = \frac{1}{N}\sum_{i=1}^{N} \ell n(X_i)\left[\left[\frac{X_i}{\hat{\alpha}}\right]^{\hat{\beta}} - 1\right] - \frac{1}{\hat{\beta}}$$

$$= \frac{1}{53}\sum_{i=1}^{53} \ell n(X_i)\left[\left[\frac{X_i}{\hat{\alpha}}\right]^{\hat{\beta}} - 1\right] - \frac{1}{\hat{\beta}}$$

where $\hat{\alpha}$ is calculated as above. The estimate, $\hat{\beta}$, is the solution to the Equation $G(\hat{\beta}) = 0$. An iterative technique for solving this equation is given in Section 8.5.6.2, and begins by setting

$$\hat{\beta} = \frac{1.28}{S_y} = \frac{1.28}{0.1044} = 12.2605$$

The solution is $\hat{\beta} = 12.27$, which in turn gives $\hat{\alpha} = 70.77$.

The first five ordered observations are listed below with the transformations necessary to compute the goodness-of-fit test statistic.

$X_{(i)}$	$Z_{(i)} = \left[\frac{X_{(i)}}{\hat{\alpha}}\right]^{\hat{\beta}} = \left[\frac{X_{(i)}}{70.77}\right]^{12.27}$
47.1	0.0068
51.6	0.0207
56.0	0.0566
59.4	0.1166
60.4	0.1431
.	.
.	.
.	.

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$$\hat{\beta} = 12.27$$

$$\hat{Q} = \hat{\alpha}(0.10536)^{1/\hat{\beta}} = (70.77)(0.10536)^{1/12.27} = 58.91$$

The B-basis value is calculated as

$$B = \hat{Q} \exp\{-V/(\hat{\beta}\sqrt{n})\}$$

$$= 58.91 \exp\{-4.670/[(12.27)\sqrt{53}]\}$$

$$= 55.91$$

For presentation in MIL-HDBK-17, this B-basis value would be rounded to 56.

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8.7.4 Problem 4 - Normal distribution. The data set for this problem consists of compression test measurements from nine batches of material. This problem illustrates the normal goodness-of-fit test and the calculation of B-basis values by the normal method.

Problem 4 - Step 1. There are no detected outliers in this set of data. (See Problem 1 for details of the outlier detection calculations.)

Problem 4 - Step 2. The k-sample Anderson-Darling test statistic is ADK=1.05 (see Problem 1 for a detailed computation of the k-sample statistic). Since this is less than the critical value of 1.44, conclude that the data from the batches may be combined and treated as a single sample. The next step is to investigate the form of the distribution.

Problem 4 - Step 3. The maximum normed residual (MNR) test is performed on the pooled data. No potential outliers are detected in the pooled data. (see Problem 1 - Step 1 for details of the outlier detection procedure.)

Problem 4 - Step 4. The Weibull goodness-of-fit test yields an observed significance level of 0.025. (See Problem 3 for details of the computation for the Weibull goodness-of-fit test.) Since this is less than 0.05, the normal goodness-of-fit test described in Section 8.6.4.1 is performed.

Problem 4 - Step 5. The average and standard deviation of the sample are 30.25 and 1.495, respectively. The first five ordered observations are listed below with the Z-values and the values of the standard normal distribution necessary for calculation of the normal Anderson-Darling statistic.

$X_{(i)}$	$Z_{(i)} = \frac{X_{(i)} - \bar{X}}{S} = \frac{X_{(i)} - 30.25}{1.49}$	$F_o\left(Z_{(i)}\right)$
26.9	-2.25	0.0122
27.7	-1.71	0.0436
27.9	-1.58	0.0571
27.9	-1.58	0.0571
28.0	-1.51	0.0655
.	.	.
.	.	.

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$$AD = \sum_{i=1}^{N} \frac{1 - 2i}{N} \left[\ell n[F_o(Z_{(i)})] + \ell n[1 - F_o(Z_{(N+1-i)})] \right] - N$$

$$= \sum_{i=1}^{53} \frac{1 - 2i}{53} \left[\ell n[F_o(Z_{(i)})] + \ell n[1 - F_o(Z_{(54-i)})] \right] - 53$$

$$= 0.363$$

$$AD^* = [1 + 4/N - 25/N^2]AD = [1 + 4/53 - 25/(53)^2](0.363) = 0.3872$$

$$OSL = 1/\{1 + \exp[-0.48 + 0.78\ \ell n(AD^*) + 4.58\ (AD^*)]\}$$

$$= 1/\{1 + \exp[-0.48 + 0.78\ \ell n(0.3872) + 4.58\ (0.3872)]\}$$

$$= 0.365$$

Since the normal goodness-of-fit test yields an OSL value (0.365) greater than 0.05, there is insufficient evidence to contradict the assumption that the data are normally distributed. Hence, the normal method in Section 8.5.5 is used to compute a B-basis value.

Problem 4 - Step 6. From Table 8.8.1, the one-sided tolerance limit factor, k_B, is 1.634. The B-basis value for a normally distributed sample is computed as

$$B = \bar{X} - k_B\ S = 30.25 - (1.634)(1.495) = 27.81$$

For presentation in MIL-HDBK-17, this B-basis value would be rounded to 28.

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8.7.5 Problem 5 - Lognormal distribution. The data set for this problem consists of compression test measurements from nine batches of material. This problem illustrates the lognormal goodness-of-fit test and the calculation of B-basis values by the lognormal method.

Problem 5 - Step 1. There are no detected outliers in this set of data. (See Problem 1 for details of the outlier detection calculations.)

Problem 5 - Step 2. The k-sample Anderson-Darling test statistic is ADK = 1.24. (See Problem 1 for details of the computation of the k-sample statistic.) Since this is less than the critical value of 1.44, conclude that the data from the batches may be combined into a single sample.

Problem 5 - Step 3. The maximum normed residual (MNR) test is performed on the pooled data. No potential outliers are detected in the pooled data. (see Problem 1 - Step 1 for details of the outlier detection procedure.)

Problem 5 - Step 4. The observed significance levels (OSL) for the two-parameter Weibull and the normal goodness-of-fit tests are given below:

Distribution	OSL
Two-parameter Weibull	0.000
Normal	0.034

(See Problems 3 and 4 for details of the computations for these tests.) Since the OSLs are both less than 0.05, neither of the distributions adequately describe the data. Thus, the lognormal goodness-of-fit test is performed.

Problem 5 - Step 5. In order to perform the lognormal goodness-of-fit test described in Section 8.6.4.2, the natural logarithms of the data are used. The average and standard deviation of the transformed data are

$$\bar{X}_L = 3.41$$

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$$S_L = 0.0548$$

The first five ordered observations are listed below with the transformations necessary to compute the goodness-of-fit statistic.

$X_{(i)}$	$\ell n(X_{(i)})$	$Z_{(i)} = \frac{\ell n(X_{(i)}) - \bar{X}_L}{S_L} = \frac{\ell n(X_{(i)}) - 3.41}{0.0548}$	$F_o[Z_{(i)}]$
27.3	3.31	-1.82	0.0344
27.3	3.31	-1.82	0.0344
27.5	3.31	-1.82	0.0344
27.9	3.33	-1.45	0.0735
28.4	3.35	-1.09	0.1379
.	.	.	.
.	.	.	.
.	.	.	.

The goodness-of-fit statistic and observed significance level are calculated as:

$$AD = \sum_{i=1}^{N} \frac{1 - 2i}{N} \left[\ell n[F_o(Z_{(i)})] + \ell n[1 - F_o(Z_{(N+1-i)})] \right] - N$$

$$= \sum_{i=1}^{53} \frac{1 - 2i}{53} \left[\ell n[F_o(Z_{(i)})] + \ell n[1 - F_o(Z_{(54-i)})] \right] - 53$$

$$= 0.650$$

$$AD^* = [1 + 4/N - 25/N^2]AD = [1 + 4/53 - 25/(53)^2](0.650) = 0.6933$$

$$OSL = 1/\{1 + \exp[-0.48 + 0.78\ \ell n(AD^*) + 4.58\ (AD^*)]\}$$

$$= 1/\{1 + \exp[-0.48 + 0.78\ \ell n(0.6933) + 4.58\ (0.6933)]\}$$

$$= 0.083$$

Since the lognormal goodness-of-fit test results in an OSL value greater than 0.05, there is insufficient evidence to contradict the assumption that the data

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are lognormally distributed. Hence, the lognormal method in Section 8.5.5.2 is used to compute a B-basis value.

Problem 5 - Step 6. The B-basis value for lognormally distributed data is computed as

$$B = \exp\{\bar{X}_L - k_B S_L\} = \exp\{3.41 - 1.634(0.0548)\} = 27.68$$

For presentation in MIL-HDBK-17, this B-basis value would be rounded to 27.

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8.7.6 Problem 6 - Nonparametric method. The data set for this problem consists of tensile strength measurements for nine batches of material. This problem illustrates the calculation of B-basis values by the nonparametric method.

Problem 6 - Step 1. There are no detected outliers in this set of data. (See Problem 1 for details of the outlier detection calculations.)

Problem 6 - Step 2. The k-sample Anderson-Darling test statistic is ADK = 0.89. (See Problem 1 for details of the computation of the k-sample statistic.) Since this is less than the critical value of 1.44, conclude that the data from the batches may be combined into a single sample.

Problem 6 - Step 3. The maximum normed residual (MNR) test is performed on the pooled data. No potential outliers are detected in the pooled data. (see Problem 1 - Step 1 for details of the outlier detection procedure.)

Problem 6 - Step 4. The results of the goodness-of-fit tests for the three distributions are:

Distribution	OSL
Two-parameter Weibull	0.013
Normal	0.008
Lognormal	0.007

(See problems 3, 4, and 5 for details of the computations for each of these tests.)

Since all of the observed significance levels are less than 0.05, it is concluded that the data do not follow any of the three distributions. Thus, the nonparametric method described in Section 8.5.7 must be used to calculate the B-basis value.

Problem 6 - Step 5. The first step in computing a B-basis value by the nonparametric method is to order the data values from smallest to largest. The five

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smallest values are 64.3, 64.9, 65.1, 65.3, and 65.6. The next step is to obtain the appropriate rank from Table 8.8.3 corresponding to the sample of size n. With an n of 53, the rank of the observation to be used as a B-basis value is r = 2. Thus, the second observation, or 64.9, is the B-basis value for this sample. For presentation in MIL-HDBK-17, this B-basis value would be rounded to 65.

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8.7.7 Problem 7 - Insufficient data. The data set for this problem consists of tensile strength measurements for five batches of material. This problem illustrates the situation where none of the standard distributions adequately fit the data, and there is insufficient data to perform the nonparametric method.

Problem 7 - Step 1. There are no detected outliers in this set of data. (See Problem 2 for details of the outlier detection calculations.)

Problem 7 - Step 2. The k-sample Anderson-Darling test statistic is ADK = 0.59. (See Problem 1 for details of the computation of the k-sample statistic.) Since this is less than the critical value of 1.63, conclude that the data from the batches may be combined into a single sample.

Problem 7 - Step 3. The maximum normed residual (MNR) test is performed on the pooled data. No potential outliers are detected in the pooled data. (see Problem 1 - Step 1 for details of the outlier detection procedure.)

Problem 7 - Step 4. The results of the goodness-of-fit tests for the three distributions are:

Distribution	OSL
Two-parameter Weibull	0.031
Normal	0.011
Lognormal	0.006

(See problems 3, 4, and 5 for details of the computations for each of these tests.)

Since all of the observed significance levels are less than 0.05, it is concluded that the data do not follow any of the three distributions. The nonparametric method cannot be used, however, because there are only 25 data values in the sample. The Hanson-Koopmans method should be used to calculate a B-basis value for these data.

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Problem 7 - Step 5. Following the procedure described in Section 8.5.7.2, a B-basis value can be estimated. For n = 25, from Table 8.8.11 it is determined that r = 11 and k = 1.087. After ranking the data in ascending order, the first and eleventh values are found.

$$X_{(1)} = 55.9 \qquad X_{(11)} = 72.2$$

$$B = X_{(r)} - k\,[X_{(r)} - X_{(1)}]$$

$$= 72.2 - 1.087\ (72.2 - 55.9)$$

$$= 54.48$$

These data can be included in MIL-HDBK-17 as interim data, but the B-value would not be reported in the handbook.

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where X_i is a Weibull random variable with unknown shape and scale parameters β and α and $\hat{\alpha}$ and $\hat{\beta}$ are the maximum likelihood estimators (MLE's) of β and α given by Equations 8.5.6.1(a) and 8.5.6.1(c). For a particular n, the V value is the 0.95th quantile of the conditional distribution of the random variable

$$V = \frac{\sqrt{n}[\ell n(\hat{Q}) - \ell n(Q)]}{1/\hat{\beta}} \qquad 8.8.2(b)$$

given that

$$A_i = \frac{\ell n(x_i') - \ell n(\hat{\alpha}')}{1/\hat{\beta}'} \qquad 8.8.2(c)$$

where

$$x_i' = -\ell n\left(1 - \frac{i - 0.5}{n + 0.25}\right) \quad i = 1, \ldots, n \qquad 8.8.2(d)$$

$$\hat{Q} = \hat{\alpha}(0.10536)^{1/\beta} \qquad 8.8.2(e)$$

$$Q = \alpha(0.10536)^{1/\beta} \qquad 8.8.2(f)$$

and $\hat{\alpha}'$ and $\hat{\beta}'$ are the MLE's of the two-parameter Weibull scale and shape parameters for the sample $x_1', \ldots, x_n'$. The conditional distribution of V is determined by the relationship

$$V = \sqrt{n}\,[Z + \ell n(0.10536)] \qquad 8.8.2(g)$$

where the distribution of Z is given in Theorem 4.1.3 on page 150 of reference 8.6.4.1. Numerical integration was used to determine the V values in Table 8.8.2 based on these results.

An approximation to the V values in Table 8.8.2 is:

$$V \approx 3.803 + \exp\{1.79 - 0.516\,\ell n(n) + 5.1/n\} \qquad 8.8.2(h)$$

This approximation is accurate to within 0.5% of the tabulated values for n greater than or equal to 16.

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8.8.3 Ranks, r, of observations for determining B-values for an unknown distribution. For $n > 29$, an approximation to the ranks for B-basis values in Table 8.8.3 is

$$r = n/10 - 1.645\sqrt{9n/100} + 0.23 \qquad 8.8.3$$

rounded to the nearest integer. This approximation is exact for all but 12 values of n in the range of the table ($29 \le n \le 10499$). For this small percentage of n values (0.1%), the approximation errs by one rank on the conservative side.

8.8.4 0.20 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. An approximation to the $F_{0.20}$ values in Table 8.8.4 is

$$F_{0.20} = \exp\left(2\delta \left[1 + \frac{z^2-1}{3} - \frac{4\sigma^2}{3} \right] + 2\sigma z \left[1 + \frac{\sigma^2(z^2-3)}{6} \right]^{0.5} \right) \qquad 8.8.4(a)$$

where

$$z = -0.84$$

$$\delta = 0.5(1/(\gamma_2-1) - 1/(\gamma_1-1)) \qquad 8.8.4(b)$$

$$\sigma^2 = 0.5(1/(\gamma_2-1) + 1/(\gamma_1-1)) \qquad 8.8.4(c)$$

γ_1 = numerator degrees of freedom

γ_2 = denominator degrees of freedom.

(See Reference 8.8.4.)

Equations 8.8.4(a-d) are not valid when either γ_1 or γ_2 equals one. The following equations are to be used for these special cases:

For $\gamma_1 = 1$

$$F_{0.20} = \left[0.25334720 + 0.06740204/\gamma_2 + 0.01068163/\gamma_2^2 - 0.00912432/\gamma_2^3 - 0.00291922/\gamma_2^4 \right]^2 \qquad 8.8.4(e)$$

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For $\gamma_2 = 1$

$$F_{0.20} = \left[1.28155100 + 0.84658450/\gamma_1 + 0.57089040/\gamma_1^2 + 0.25851690/\gamma_1^3 + 0.05439575/\gamma_1^4 \right]^{-2} \qquad 8.8.4(f)$$

8.8.5 0.95 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. An approximation to the $F_{0.95}$ values in Table 8.8.5 is

$$F_{0.95} = \exp\left(2\delta \left[1 + \frac{z^2-1}{3} - \frac{4\sigma^2}{3} \right] + 2\sigma z \left[1 + \frac{\sigma^2(z^2-3)}{6} \right]^{0.5} \right) \qquad 8.8.5(a)$$

where

$$z = 1.645$$

$$\delta = 0.5(1/(\gamma_2-1) - 1/(\gamma_1-1)) \qquad 8.8.5(b)$$

$$\sigma^2 = 0.5(1/(\gamma_2-1) + 1/(\gamma_1-1)) \qquad 8.8.5(c)$$

γ_1 = numerator degrees of freedom

γ_2 = denominator degrees of freedom.

(See Reference 8.8.4.)

Equations 8.8.5(a-d) are not valid when either γ_1 or γ_2 equals one. The following equations are to be used for these special cases:

For $\gamma_1 = 1$

$$F_{0.95} = \left[1.95996400 + 2.37227200/\gamma_2 + 2.82250000/\gamma_2^2 + 2.555585200/\gamma_2^3 + 1.58953600/\gamma_2^4 \right]^{2} \qquad 8.8.5(e)$$

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For $\gamma_2 = 1$

$$F_{0.95} = \left[\begin{array}{l} 0.06270671 + 0.01573832/\gamma_1 + 0.00200073/\gamma_1^2 \\ - 0.00243852/\gamma_1^3 - 0.00064811/\gamma_1^4 \end{array} \right]^{-2} \qquad 8.8.5(f)$$

8.8.6 <u>One-sided tolerance factors, T, for the one-way random effects ANOVA, 0.95 confidence</u>. The tolerance limit factor T is defined in as

$$T = \{(n^*\hat{R}+1)/[N(\hat{R}+1)]\}^{0.5} \, t_{\gamma,0.95}(\delta) \qquad 8.8.6(a)$$

where $t_{\gamma,0.95}(\delta)$ is the 0.95 quantile of the non-central t-distribution with noncentrality parameter

$$\delta = 1.282 \{ N(\hat{R}+1)/(n^*\hat{R}+1)\}^{0.5} \qquad 8.8.6(b)$$

degrees of freedom approximated by

$$\gamma = \frac{(\hat{R} + 1)^2}{\frac{(\hat{R} + \frac{1}{n^*})^2}{k^* - 1} + \frac{(\frac{n^*-1}{n^*})^2}{k^*(n^* - 1)}} \qquad 8.8.6(c)$$

Tolerance limit factors may also be computed from the following approximate formula:

$$T = 1.282 + 1.282 \, (t_{\gamma,0.95}/\delta) + 1.80 \, (1/\gamma) \qquad 8.8.6(d)$$

$$- 1.85 \, (1/\gamma^2) + 0.567 \, (\delta/\gamma) + 5.24 \, (\delta/\gamma^2)$$

$$- 1.08 \, (\delta^2/\gamma^2) + 0.0166 \, (\delta^3/\gamma^2) + 7.79 \, (1/\gamma^4)$$

This approximation is accurate to within 0.45% of the tabulated values. This approximation may be used for values of n^*, k^* and $\hat{R}$ both within and beyond the range of the table.

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8.8.7 Critical values for the k-sample Anderson-Darling test at the 0.05 significance level. The k-sample Anderson-Darling test critical values in Table 8.8.7 were calculated using Equation 8.6.3.1(j) for the case of samples of equal size n.

8.8.8 0.95 Quantiles of the chi-squared distribution associated with γ degrees of freedom. An approximation to the chi-squared quantiles ($\chi^2_{0.95}$) in Table 8.8.8 is:

$$\chi^2_{0.95} = \gamma \left[1 - \frac{2}{9\gamma} + 1.645\left(\frac{2}{9\gamma}\right)^{0.5}\right]^3 + \frac{9}{100\gamma} \qquad 8.8.8$$

where γ is the degrees of freedom. This approximation is accurate to within 0.2% of the tabulated values. (See Reference 8.8.8.)

8.8.9 Critical values for the MNR outlier detection procedure ($\alpha = 0.01$). The critical values in Table 8.8.9 are computed by the following formula:

$$C.V. = \frac{n - 1}{\sqrt{n}} \left[\frac{t^2}{n - 2 + t^2}\right]^{0.5} \qquad 8.8.9$$

where t is the $[1 - \alpha/(2n)]$ quantile of the t-distribution with n - 2 degrees of freedom, α is the significance level of the test, and n is the sample size. Numbers in Table 8.8.9 are computed with a significance level of $\alpha = 0.01$. (See Reference 8.6.2(b))

8.8.10 Critical values for the MNR outlier detection procedure ($\alpha = 0.05$). The critical values in Table 8.8.10 are computed according to Equation 8.8.9 with significance level $\alpha = 0.05$.

8.8.11 Nonparametric B-basis material properties for small samples. The values in Table 8.8.11 are based on Reference 8.5.7.2.

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TABLE 8.8.1 One-sided tolerance limit factors, k_B, for the normal distribution, 0.95 confidence.

n = 2 - 136

n	k_B	n	k_B	n	k_B
2	20.581	47	1.660	92	1.539
3	6.157	48	1.655	93	1.537
4	4.163	49	1.650	94	1.536
5	3.408	50	1.646	95	1.534
6	3.007	51	1.642	96	1.533
7	2.756	52	1.638	97	1.531
8	2.583	53	1.634	98	1.530
9	2.454	54	1.630	99	1.529
10	2.355	55	1.626	100	1.527
11	2.276	56	1.623	101	1.526
12	2.211	57	1.619	102	1.525
13	2.156	58	1.616	103	1.523
14	2.109	59	1.613	104	1.522
15	2.069	60	1.609	105	1.521
16	2.034	61	1.606	106	1.519
17	2.002	62	1.603	107	1.518
18	1.974	63	1.600	108	1.517
19	1.949	64	1.597	109	1.516
20	1.927	65	1.595	110	1.515
21	1.906	66	1.592	111	1.513
22	1.887	67	1.589	112	1.512
23	1.870	68	1.587	113	1.511
24	1.854	69	1.584	114	1.510
25	1.839	70	1.582	115	1.509
26	1.825	71	1.579	116	1.508
27	1.812	72	1.577	117	1.507
28	1.800	73	1.575	118	1.506
29	1.789	74	1.572	119	1.505
30	1.778	75	1.570	120	1.504
31	1.768	76	1.568	121	1.503
32	1.758	77	1.566	122	1.502
33	1.749	78	1.564	123	1.501
34	1.741	79	1.562	124	1.500
35	1.733	80	1.560	125	1.499
36	1.725	81	1.558	126	1.498
37	1.718	82	1.556	127	1.497
38	1.711	83	1.554	128	1.496
39	1.704	84	1.552	129	1.495
40	1.698	85	1.551	130	1.494
41	1.692	86	1.549	131	1.493
42	1.686	87	1.547	132	1.492
43	1.680	88	1.545	133	1.492
44	1.675	89	1.544	134	1.491
45	1.669	90	1.542	135	1.490
46	1.664	91	1.540	136	1.489

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TABLE 8.6.4 0.20 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom.

		γ_1 Numerator degrees of freedom								
		1	2	3	4	5	6	7	8	9
	1	0.106	0.281	0.373	0.425	0.459	0.482	0.500	0.513	0.523
	2	0.0833	0.250	0.347	0.405	0.443	0.470	0.489	0.505	0.517
	3	0.0765	0.241	0.341	0.403	0.444	0.473	0.495	0.513	0.526
	4	0.0733	0.236	0.338	0.403	0.447	0.478	0.502	0.520	0.535
	5	0.0714	0.233	0.337	0.404	0.449	0.482	0.507	0.526	0.542
	6	0.0701	0.232	0.337	0.404	0.451	0.485	0.511	0.531	0.548
	7	0.0693	0.230	0.336	0.405	0.453	0.488	0.514	0.535	0.552
	8	0.0686	0.230	0.336	0.406	0.454	0.490	0.517	0.539	0.556
	9	0.0681	0.229	0.336	0.406	0.455	0.492	0.520	0.542	0.560
	10	0.0677	0.228	0.336	0.407	0.456	0.493	0.522	0.544	0.562
	11	0.0674	0.228	0.336	0.407	0.457	0.495	0.523	0.546	0.565
	12	0.0671	0.227	0.336	0.407	0.458	0.496	0.525	0.548	0.567
	13	0.0669	0.227	0.335	0.408	0.459	0.497	0.526	0.550	0.569
	14	0.0667	0.227	0.335	0.408	0.459	0.498	0.527	0.551	0.570
	15	0.0665	0.227	0.335	0.408	0.460	0.498	0.528	0.552	0.572
	16	0.0664	0.226	0.335	0.408	0.460	0.499	0.529	0.553	0.573
	17	0.0662	0.226	0.335	0.409	0.461	0.500	0.530	0.554	0.574
	18	0.0661	0.226	0.335	0.409	0.461	0.500	0.531	0.555	0.575
	19	0.0660	0.226	0.335	0.409	0.461	0.501	0.532	0.556	0.576
	20	0.0659	0.226	0.335	0.409	0.462	0.501	0.532	0.557	0.577
	21	0.0658	0.226	0.335	0.409	0.462	0.502	0.533	0.558	0.578
	22	0.0656	0.225	0.335	0.409	0.462	0.502	0.533	0.558	0.579
	23	0.0657	0.225	0.335	0.409	0.463	0.503	0.534	0.559	0.580
	24	0.0656	0.225	0.335	0.410	0.463	0.503	0.534	0.559	0.580
	25	0.0656	0.225	0.335	0.410	0.463	0.503	0.535	0.560	0.581
	26	0.0655	0.225	0.335	0.410	0.463	0.503	0.535	0.560	0.582
	27	0.0655	0.225	0.335	0.410	0.463	0.504	0.535	0.561	0.582
	28	0.0654	0.225	0.335	0.410	0.464	0.504	0.536	0.561	0.583
	29	0.0654	0.225	0.335	0.410	0.464	0.504	0.536	0.562	0.583
	30	0.0653	0.225	0.335	0.410	0.464	0.504	0.536	0.562	0.583
	40	0.0650	0.224	0.335	0.411	0.465	0.506	0.539	0.565	0.587
	60	0.0648	0.224	0.335	0.411	0.466	0.508	0.541	0.568	0.590
	120	0.0645	0.224	0.335	0.412	0.467	0.510	0.543	0.571	0.594
	∞	0.0643	0.223	0.335	0.412	0.468	0.512	0.546	0.574	0.597

γ_2 denominator degrees of freedom

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TABLE 8.8.4 0.20 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom - Continued.

γ_2 denominator degrees of freedom	γ_1 numerator degrees of freedom									
	10	12	15	20	24	30	40	60	120	∞
1	0.531	0.544	0.556	0.569	0.576	0.582	0.589	0.596	0.602	0.607
2	0.527	0.542	0.557	0.573	0.581	0.589	0.597	0.605	0.613	0.621
3	0.537	0.554	0.572	0.590	0.599	0.608	0.618	0.627	0.637	0.645
4	0.547	0.566	0.585	0.605	0.615	0.625	0.635	0.646	0.657	0.663
5	0.555	0.575	0.595	0.617	0.628	0.639	0.650	0.662	0.674	0.685
6	0.561	0.582	0.604	0.627	0.638	0.650	0.663	0.675	0.688	0.699
7	0.566	0.588	0.611	0.635	0.647	0.660	0.673	0.686	0.700	0.712
8	0.571	0.593	0.617	0.642	0.655	0.668	0.682	0.696	0.710	0.724
9	0.574	0.598	0.622	0.648	0.661	0.675	0.689	0.704	0.719	0.735
10	0.578	0.601	0.626	0.653	0.667	0.681	0.696	0.712	0.727	0.744
11	0.580	0.605	0.630	0.658	0.672	0.687	0.702	0.718	0.735	0.752
12	0.583	0.607	0.634	0.662	0.676	0.692	0.707	0.724	0.741	0.757
13	0.585	0.610	0.637	0.665	0.680	0.696	0.712	0.729	0.747	0.762
14	0.587	0.612	0.639	0.668	0.684	0.700	0.716	0.734	0.752	0.771
15	0.588	0.614	0.642	0.671	0.687	0.703	0.720	0.738	0.757	0.776
16	0.590	0.616	0.644	0.674	0.690	0.707	0.724	0.742	0.761	0.781
17	0.591	0.618	0.646	0.676	0.693	0.709	0.727	0.746	0.766	0.786
18	0.592	0.619	0.648	0.679	0.695	0.712	0.730	0.749	0.769	0.789
19	0.593	0.620	0.649	0.681	0.697	0.715	0.733	0.752	0.773	0.793
20	0.594	0.622	0.651	0.682	0.699	0.717	0.736	0.755	0.776	0.796
21	0.595	0.623	0.652	0.684	0.701	0.719	0.738	0.758	0.779	0.802
22	0.596	0.624	0.653	0.686	0.703	0.721	0.740	0.761	0.782	0.805
23	0.597	0.625	0.655	0.687	0.705	0.723	0.742	0.763	0.785	0.806
24	0.598	0.626	0.656	0.689	0.706	0.725	0.744	0.765	0.788	0.810
25	0.599	0.627	0.657	0.690	0.708	0.726	0.746	0.767	0.790	0.812
26	0.599	0.627	0.658	0.691	0.709	0.728	0.748	0.769	0.793	0.816
27	0.600	0.628	0.659	0.692	0.710	0.729	0.750	0.771	0.795	0.820
28	0.600	0.629	0.660	0.693	0.711	0.731	0.751	0.773	0.797	0.821
29	0.601	0.629	0.660	0.694	0.713	0.732	0.753	0.775	0.799	0.823
30	0.601	0.630	0.661	0.695	0.714	0.733	0.754	0.776	0.801	0.827
40	0.605	0.635	0.667	0.702	0.722	0.743	0.765	0.789	0.816	0.845
60	0.609	0.640	0.673	0.710	0.731	0.753	0.777	0.804	0.834	0.869
120	0.613	0.645	0.680	0.719	0.741	0.765	0.791	0.822	0.857	0.902
∞	0.618	0.651	0.688	0.729	0.753	0.779	0.809	0.843	0.889	0.977

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TABLE 8.6.5 0.95 Quantiles of the F-distribution associated with r_1 and r_2 degrees of freedom.

r_2 denominator degrees of freedom	r_1 numerator degrees of freedom								
	1	2	3	4	5	6	7	8	9
1	161.	199.	216.	225.	230.	234.	237.	239.	241.
2	18.5	19.0	19.2	19.2	19.3	19.3	19.4	19.4	19.4
3	10.1	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96
∞	3.84	3.00	2.61	2.38	2.22	2.10	2.01	1.94	1.88

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TABLE 8.8.5 0.95 Quantiles of the F-distribution associated with ν_1 and ν_2 degrees of freedom - Continued.

ν_2 denominator degrees of freedom	ν_1 numerator degrees of freedom									
	10	12	15	20	24	30	40	60	120	∞
1	242.	244.	246.	248.	249.	250.	251.	252.	253.	256.
2	19.4	19.4	19.4	19.4	19.5	19.5	19.5	19.5	19.5	19.5
3	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.56
4	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.68
5	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.38
6	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.68
7	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.24
8	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.22
14	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.85
21	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.79
23	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.74
25	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.72
26	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.66
29	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.26
∞	1.83	1.75	1.67	1.57	1.52	1.46	1.40	1.32	1.22	1.05

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TABLE 8.8.6 One-sided tolerance limit factors, T, for the one-way random effects ANOVA, 0.95 confidence - Continued.

$\bar{R}$ = 2.00

n \ k	2	3	4	5	6	7	8	9	10	15	20	∞
1	20.588	6.157	4.163	3.408	3.007	2.756	2.583	2.454	2.355	2.069	1.927	1.282
2	10.092	4.419	3.337	2.875	2.613	2.441	2.319	2.226	2.153	1.936	1.824	1.282
3	8.129	3.979	3.105	2.716	2.491	2.342	2.234	2.152	2.087	1.891	1.789	1.282
4	7.325	3.779	2.995	2.640	2.432	2.293	2.192	2.115	2.054	1.869	1.771	1.282
5	6.890	3.664	2.931	2.595	2.397	2.264	2.167	2.093	2.034	1.855	1.760	1.282
6	6.617	3.591	2.890	2.566	2.374	2.245	2.151	2.079	2.021	1.846	1.753	1.282
7	6.431	3.539	2.860	2.545	2.358	2.231	2.139	2.068	2.012	1.839	1.748	1.282
8	6.296	3.501	2.838	2.529	2.345	2.221	2.130	2.061	2.005	1.835	1.744	1.282
9	6.193	3.471	2.821	2.517	2.336	2.213	2.123	2.054	1.999	1.831	1.741	1.282
10	6.112	3.448	2.808	2.508	2.328	2.207	2.118	2.050	1.995	1.828	1.739	1.282
11	6.047	3.429	2.797	2.500	2.322	2.202	2.113	2.046	1.991	1.825	1.737	1.282
12	5.994	3.413	2.788	2.493	2.317	2.197	2.110	2.042	1.988	1.823	1.736	1.282
13	5.949	3.400	2.780	2.488	2.312	2.194	2.107	2.039	1.986	1.821	1.734	1.282
14	5.911	3.389	2.773	2.483	2.309	2.190	2.104	2.037	1.984	1.820	1.733	1.282
15	5.878	3.379	2.768	2.479	2.305	2.188	2.101	2.035	1.982	1.818	1.731	1.282
∞	5.442	3.246	2.689	2.422	2.260	2.150	2.069	2.006	1.955	1.800	1.717	1.282

$\bar{R}$ = 10.00

n \ k	2	3	4	5	6	7	8	9	10	15	20	∞
1	20.580	6.157	4.163	3.408	3.007	2.756	2.583	2.454	2.355	2.069	1.927	1.282
2	16.565	5.564	3.893	3.237	2.883	2.658	2.501	2.384	2.294	2.029	1.896	1.282
3	15.433	5.382	3.808	3.163	2.843	2.626	2.474	2.361	2.273	2.016	1.886	1.282
4	14.901	5.294	3.766	3.156	2.823	2.610	2.461	2.350	2.263	2.009	1.881	1.282
5	14.592	5.242	3.741	3.133	2.811	2.600	2.453	2.343	2.257	2.005	1.878	1.282
6	14.390	5.207	3.724	3.123	2.803	2.594	2.448	2.338	2.252	2.002	1.876	1.282
7	14.248	5.183	3.713	3.121	2.797	2.590	2.444	2.335	2.250	2.000	1.874	1.282
8	14.142	5.165	3.704	3.116	2.793	2.586	2.441	2.332	2.248	1.999	1.873	1.282
9	14.061	5.151	3.697	3.111	2.790	2.584	2.439	2.330	2.246	1.998	1.872	1.282
10	13.996	5.139	3.692	3.108	2.787	2.581	2.437	2.329	2.245	1.997	1.871	1.282
11	13.943	5.130	3.687	3.105	2.785	2.580	2.436	2.328	2.244	1.996	1.871	1.282
12	13.899	5.123	3.684	3.102	2.783	2.578	2.434	2.327	2.243	1.996	1.870	1.282
13	13.863	5.116	3.680	3.100	2.782	2.577	2.433	2.326	2.242	1.995	1.870	1.282
14	13.831	5.111	3.678	3.099	2.780	2.576	2.432	2.325	2.241	1.995	1.870	1.282
15	13.804	5.106	3.676	3.097	2.779	2.575	2.432	2.324	2.241	1.994	1.869	1.282
∞	13.429	5.040	3.643	3.076	2.763	2.563	2.421	2.315	2.233	1.989	1.866	1.282

$\bar{R}$ = ∞

All n	20.588	6.157	4.163	3.408	3.007	2.756	2.583	2.454	2.355	2.069	1.927	1.282

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TABLE 8.8.7 Critical values for the k-sample Andereon-Darling test at the 0.05 significance level, (samples of equal size n).

	k													
n	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	2.11	1.80	1.65	1.56	1.50	1.46	1.42	1.39	1.37	1.35	1.33	1.32	1.31	1.29
4	2.20	1.86	1.70	1.60	1.54	1.49	1.45	1.42	1.39	1.37	1.36	1.34	1.33	1.31
5	2.25	1.89	1.73	1.63	1.56	1.51	1.47	1.43	1.41	1.39	1.37	1.35	1.34	1.32
6	2.29	1.92	1.74	1.64	1.57	1.52	1.48	1.45	1.42	1.40	1.38	1.36	1.34	1.33
7	2.32	1.94	1.76	1.65	1.58	1.53	1.49	1.45	1.43	1.40	1.38	1.36	1.35	1.34
8	2.34	1.95	1.77	1.66	1.59	1.53	1.49	1.46	1.43	1.41	1.39	1.37	1.35	1.34
9	2.35	1.96	1.78	1.67	1.59	1.54	1.50	1.46	1.43	1.41	1.39	1.37	1.36	1.34
10	2.37	1.97	1.78	1.67	1.60	1.54	1.50	1.47	1.44	1.41	1.39	1.37	1.36	1.35
11	2.38	1.97	1.79	1.68	1.60	1.55	1.50	1.47	1.44	1.42	1.39	1.38	1.36	1.35
12	2.39	1.98	1.79	1.68	1.60	1.55	1.51	1.47	1.44	1.42	1.40	1.38	1.36	1.35
13	2.39	1.98	1.80	1.68	1.61	1.55	1.51	1.47	1.44	1.42	1.40	1.38	1.36	1.35
14	2.40	1.99	1.80	1.69	1.61	1.55	1.51	1.47	1.44	1.42	1.40	1.38	1.37	1.35
15	2.41	1.99	1.80	1.69	1.61	1.55	1.51	1.48	1.45	1.42	1.40	1.38	1.37	1.35
16	2.41	2.00	1.80	1.69	1.61	1.56	1.51	1.48	1.45	1.42	1.40	1.38	1.37	1.35
17	2.42	2.00	1.81	1.69	1.61	1.56	1.51	1.48	1.45	1.42	1.40	1.38	1.37	1.35
18	2.42	2.00	1.81	1.69	1.62	1.56	1.51	1.48	1.45	1.42	1.40	1.39	1.37	1.35
19	2.42	2.00	1.81	1.70	1.62	1.56	1.52	1.48	1.45	1.43	1.40	1.39	1.37	1.36
20	2.43	2.01	1.81	1.70	1.62	1.56	1.52	1.48	1.45	1.43	1.40	1.39	1.37	1.36
∞	2.49	2.05	1.84	1.72	1.64	1.58	1.53	1.50	1.46	1.44	1.42	1.40	1.38	1.37

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